Feasibility of using the ADS1299

Dmytro Stavskyi, Luis Wong 10/14/2025

Objective

- Investigate the feasibility of acquiring and processing electromyography (EMG) signals using the ADS1299 analog front-end and MSPM0C1104 microcontroller.
- Specifically:
    1. Understand key characteristics of EMG signals (frequency range, amplitude, bandwidth, etc.).
    2. Evaluate if the ADS1299 is suitable for EMG signal acquisition.
    3. Design a connection and data flow plan between ADS1299 and MSPM0C1104.

Methods

Literature Review:

- Analyzed Rashid et al. [1] and Li et al. [2] to assess the ADS1299's performance in acquiring biopotential signals under experimental conditions.

Chip Evaluation Approach:

- Extracted and analyzed ADS1299 specifications (gain, noise, sampling rate) from the TI datasheet.

Interface Design Methodology:

- Designed a high-level connection diagram based on ADS1299 and MSPM0C1104 documentation.

Results

Rashit et al. [1] demonstrated that the ADS1299, and 8-channel, 24-bit ADC, can record biopotentials with laboratory grade precision.

Quote: “The main findings of this research are that the ADS1299 is comparable with respect to power across EEG bands, power ratio, pre-movement noise of the EEG, and the signal-to-noise ratio, the amplitude and time of the negative peak, and cosine similarity of the MRCPs” [1].

In Rashid et al.’s [1] work, MRCPs are used as a benchmark signal to evaluate the quality of EEG acquisition by the ADS1299 compared to a laboratory-grade amplifier. Because MRCPs are:

- Low-frequency (~0.1–5 Hz)
- Very low amplitude (5–50 μV)
- Time-locked to movement

ADC1299 Specs:

- Input-referred noise: ~1 μV RMS
- Programmable gain: 1× – 24×
- Sampling range: 250 SPS – 16 kSPS
- Digital filter: Built-in decimation filter provides strong low-pass response.

Note: This assumes that due to EEG being a harder signal to capture this would imply that the ADS is more than sufficient for EMG.

Results

- Li et al. [2] designed and implemented an 8-channel surface EMG (sEMG) acquisition system using the ADS1299 analog front-end.
- The system integrates a high-density electrode array, bias-drive circuitry, and an STM32 microcontroller for signal processing.
- They focused on back-of-hand EMG acquisition during voluntary fist-clenching movements.
- Provides direct experimental proof that the ADS1299 is effective for EMG acquisition, not just EEG [2].

Connecting ADS1299 to MSPM0C1104

- Communication: SPI bus used for configuration and data readout, as described in [1].
- Signal flow:
  1. MSPM0C1104 configures ADS1299 registers over SPI.
  2. ADS1299 digitizes EMG signals and signals data readiness (GPIO).
  3. MSPM0C1104 retrieves and processes 24-bit samples.

Example Layout

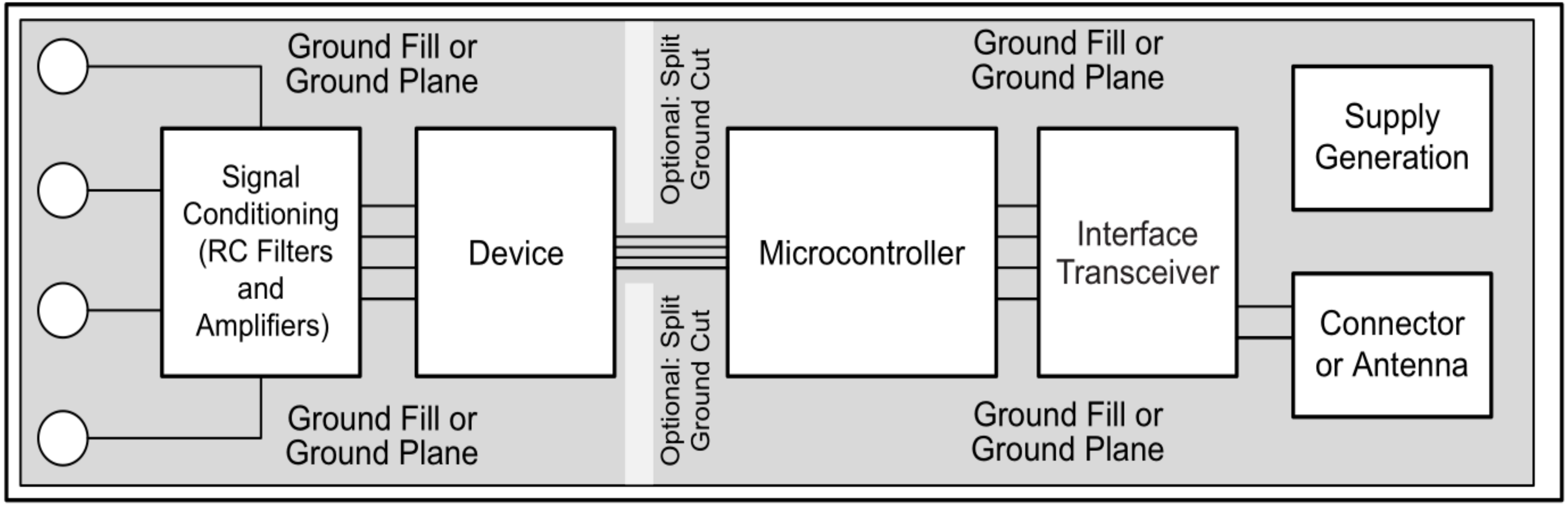


Figure 79. System Component Placement

Example Application

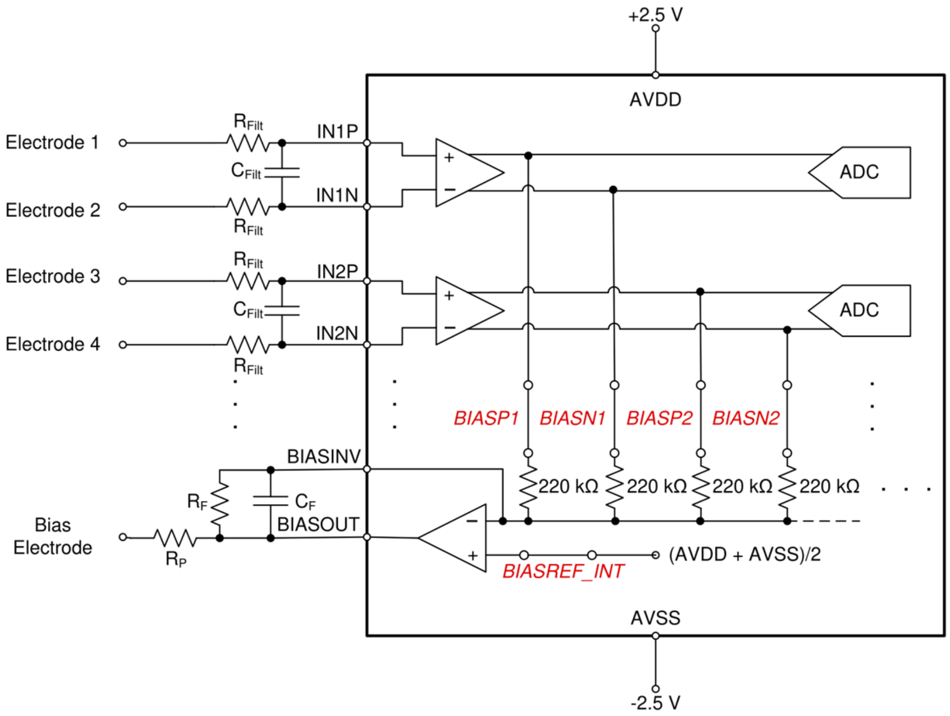


Figure 72. Example Schematic Using the ADS1299 in an EEG Data Acquisition Application, Sequential Montage

Observation/Conclusion:

- Options for the proof of concept:
  - ADS1299 PDK (Performance Demonstration Kit)
    - Fast, but potentially harder to integrate with main circuit
  - ADS1299 on a separate PCB
    - Need to properly design first, but would be easier to integrate with the main PCB

EMG Signals (Questions to ask)

- What level of noise should we expect in real EMG measurements, and how much filtering is typically required?
- Should we consider active filtering (analog) before the ADS1299, or is digital filtering sufficient post-acquisition?
- Record EMG as in store the signal data in a file or just read the EMG signals and provide it as an output?
    - For how long and how much?

Next steps:

Determine the option for the proof of concept

Help Jeremiah plan how integrate the ADS1299 into the current design.

Reference

[1] U. Rashid, I. Niazi, N. Signal, and D. Taylor, “An EEG experimental study evaluating the performance of Texas Instruments ADS1299,” *Frontiers in Neuroscience*, vol. 12, p. 627, 2018, doi: 10.3390/s18113721

[2] Y. Li, H. Pan, and Q. Song, “ADS1299-Based Array Surface Electromyography Signal Acquisition System,” J. Phys.: Conf. Ser., vol. 2383, 012054, 2022. [Online]. Available: https://doi.org/10.1088/1742-6596/2383/1/012054

https://www.ti.com/product/ADS1299

https://www.ti.com/tool/ADS1299EEGFE-PDK

Initial Power Amplifier Design

Luis Wong 10/21/2025

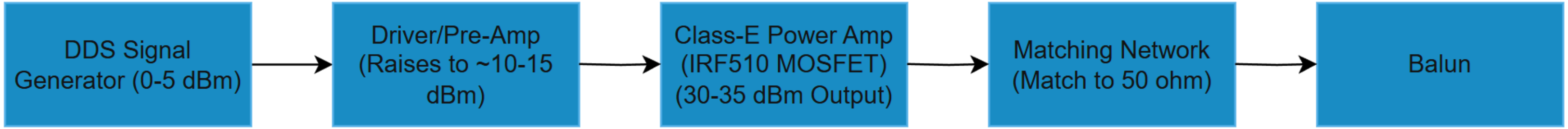
Objective

- To fully understand and attempt design an RF Power Amplifier for use in device.
    - We need 30 - 35 dBm at output, DDS can give 0-5 dBm.
    - 50-ohm resistance

Methods

- Read through various sources to better understand the basics of RF PAs due to being unfamiliar with the subject.
- Model a Class-E power amplifier to efficiently convert low-power DDS output into the required output power.
    - Will make use of PA used in TI Reference Design Guide [1] and also designs found in [3],[5] as a guide.

Results



High-Level View of the PAs design.

- **Power Switch (MP5000ADQ-LF-P):** Disconnects PA stage when not transmitting.

Results

- Different classes of power amplifiers [2],[6],[7]:

Class	Conduction Angle	Power Efficiency	Linearity
A	360° (always on)	~25-35%	Highest
B	180°	~65%	Moderate
AB	180-360°	~50-70%	Better than AB
C	>360°	~60-80%	Poor
D	Switch	~80-90%	Poor
E	Switch	~80-90%	Moderate
F	Switch + harmonic tuning	~85-90%	Moderate

Results

Marki microwave

Power to Voltage Conversion Table

P (dBm)	P (mW)	V_{RMS} (V)	Vp (V)[1]	Vpp (V)	P (dBm)	P (mW)	V_{RMS} (V)	Vp (V)[1]	Vpp (V)
-30	0.001	0.007	0.010	0.020	0	1.000	0.224	0.316	0.632
-29	0.001	0.008	0.011	0.022	1	1.259	0.251	0.355	0.710
-28	0.002	0.009	0.013	0.025	2	1.585	0.282	0.398	0.796
-27	0.002	0.010	0.014	0.028	3	1.995	0.316	0.447	0.893
-26	0.003	0.011	0.016	0.032	4	2.512	0.354	0.501	1.002
-25	0.003	0.013	0.018	0.036	5	3.162	0.398	0.562	1.125
-24	0.004	0.014	0.020	0.040	6	3.981	0.446	0.631	1.262
-23	0.005	0.016	0.022	0.045	7	5.012	0.501	0.708	1.416
-22	0.006	0.018	0.025	0.050	8	6.310	0.562	0.794	1.589
-21	0.008	0.020	0.028	0.056	9	7.943	0.630	0.891	1.783
-20	0.010	0.022	0.032	0.063	10	10.000	0.707	1.000	2.000
-19	0.013	0.025	0.035	0.071	11	12.589	0.793	1.122	2.244
-18	0.016	0.028	0.040	0.080	12	15.849	0.890	1.259	2.518
-17	0.020	0.032	0.045	0.089	13	19.953	0.999	1.413	2.825
-16	0.025	0.035	0.050	0.100	14	25.119	1.121	1.585	3.170
-15	0.032	0.040	0.056	0.112	15	31.623	1.257	1.778	3.557
-14	0.040	0.045	0.063	0.126	16	39.811	1.411	1.995	3.991
-13	0.050	0.050	0.071	0.142	17	50.119	1.583	2.239	4.477
-12	0.063	0.056	0.079	0.159	18	63.096	1.776	2.512	5.024
-11	0.079	0.063	0.089	0.178	19	79.433	1.993	2.818	5.637
-10	0.100	0.071	0.100	0.200	20	100.000	2.236	3.162	6.325
-9	0.126	0.079	0.112	0.224	21	125.893	2.509	3.548	7.096
-8	0.158	0.089	0.126	0.252	22	158.489	2.815	3.981	7.962
-7	0.200	0.100	0.141	0.283	23	199.526	3.159	4.467	8.934
-6	0.251	0.112	0.158	0.317	24	251.189	3.544	5.012	10.024
-5	0.316	0.126	0.178	0.356	25	316.228	3.976	5.623	11.247
-4	0.398	0.141	0.200	0.399	26	398.107	4.462	6.310	12.619
-3	0.501	0.158	0.224	0.448	27	501.187	5.006	7.079	14.159
-2	0.631	0.178	0.251	0.502	28	630.957	5.617	7.943	15.887
-1	0.794	0.199	0.282	0.564	29	794.328	6.302	8.913	17.825
0	1.000	0.224	0.316	0.632	30	1000.000	7.071	10.000	20.000

[1]For Square wave signal, $V_P = V_{RMS}$

Note: The converted voltages in the above table are for R = 50 Ω

$P_{(dBm)} = 10 \log_{10} P_{(mW)}$

$P_{(mW)} = 10^{(P_{(dBm)}/10)}$

$P_{(mW)} = [V_{RMS\,(V)}]^2 * 10^3/R$

$V_{RMS\,(V)} = \sqrt{(P_{(mW)} \cdot R / 10^3)}$

$V_P = \sqrt{2} * V_{RMS}$, for sinusoid – Fig. a.

$V_P = V_{RMS}$, for square wave – Fig. b.

$V_{P\text{-}P} = 2 * V_P$

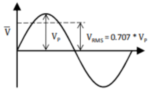


Fig. a. Sinusoidal signal

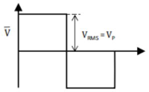


Fig. b. Square wave signal

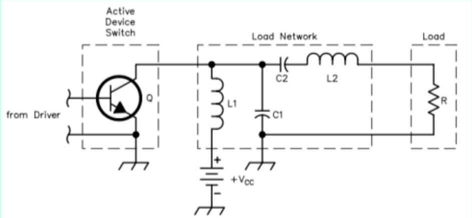
345 Digital Drive, Morgan Hill, CA 95037 | Ph: 408.778.4200 | Fax 408.778.4300
info@markimicrowave.com www.markimicrowave.com

https://markimicrowave.com/tools/power-to-voltage.pdf

Results

Enter Vcc: 12 V
Enter Vo (transistor/FET saturation voltage): 0 V
Enter Coss (transistor/FET output capacitance): 10p F
Enter frequency: 25M Hz
Enter the desired power output: 1 W
Enter L1: 50u H
Calculate

R is: 74.39 Ω (Impedance transformer to 50Ω: secondary (50Ω) to primary (amplifier) turns ratio: 819.81 m)
C1 is: 8.4 p F (XC1 is: 757.78 Ω)
C2 is: 22.86 p F (XC2 is: 278.46 Ω)
L1min is: 41.27 μ Ω (XL1 is: 7.85 k Ω)
L2 is: 2.37 μ H (XL2 is: 371.97 Ω)
XL2 - XC2 = 93.52 Ω
Expect Icc of at least: 83.33 m I
Select a FET/transistor with a max Vds/Vce of at least: 53.4 V (Includes a safety factor of 0.8)
1 / fr1 = 21.63 M Hz. When Q1 closed, should be less than the frequency of operation.
1 / fr2 = 32.39 M Hz. When Q1 open, should be greater than the frequency of operation.
1 / (fr1 + f2 half period) = 27.01 M Hz. Should be about equal to the frequency of operation.



Schematic from Nathan Sokal's paper titled "Class-E RF power amplifiers", published in QEX Jan/Feb 2001.

For more info on designing Class-E amplifiers for LF/MF, see VK1SV's Class-E for beginners

L1min and Fr1/2 calculations are from here.

Many thanks to Daniel O'Connor (doconnor AT gsoft.com.au) for his contribution of the SI prefix capability, the L1min and Fr1/2 calculation as well as a general tidy up of the javascript code!

Back to calculators page or main page.

$$R = \left(\frac{(V_{CC} - V_o)^2}{P}\right) 0.576801 \left(1.0000086 - \frac{0.414395}{Q_L} - \frac{0.577501}{Q_L^2} + \frac{0.205967}{Q_L^3}\right)$$

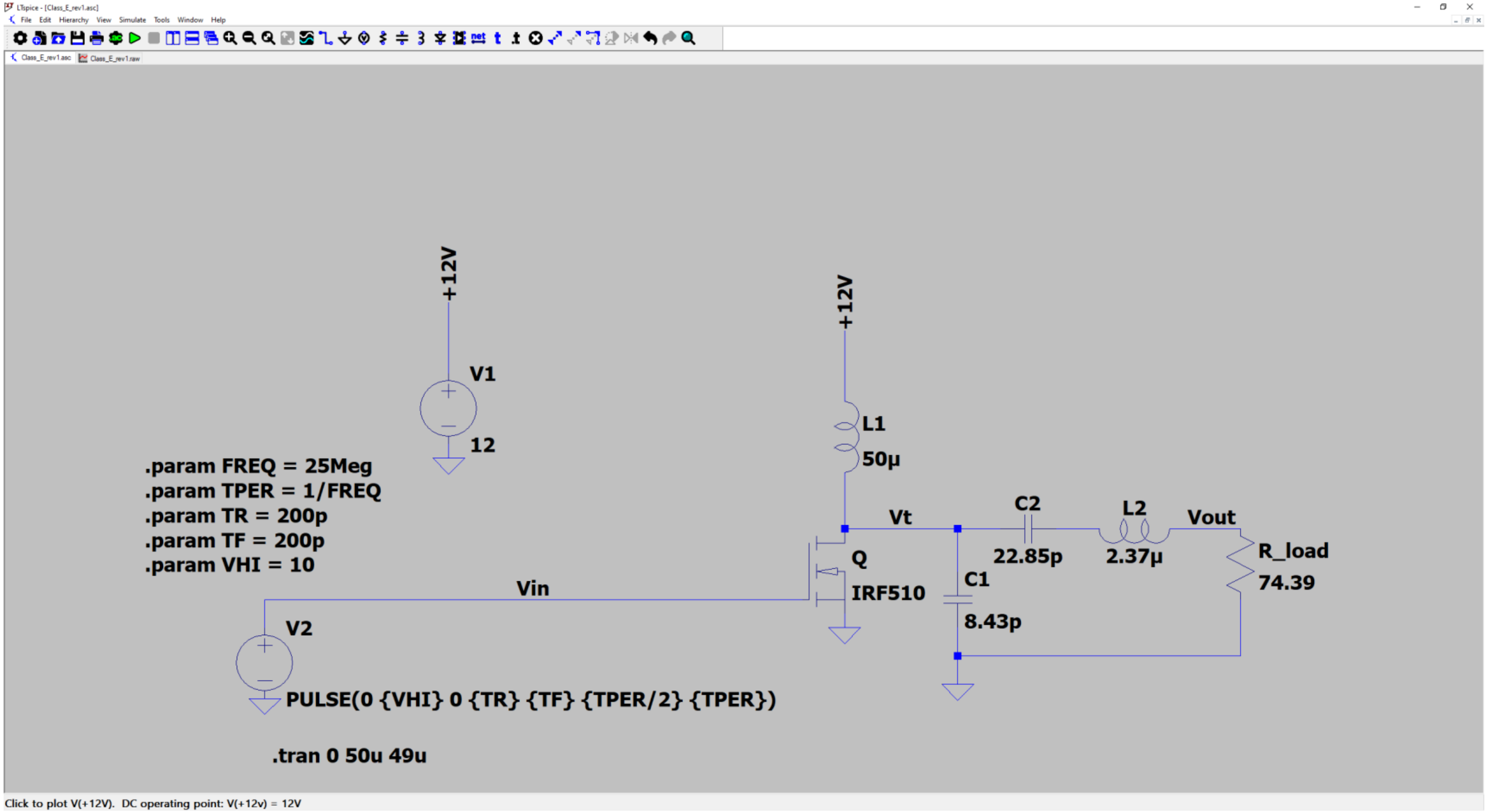
$$C1 = \frac{1}{34.2219\, f\, R}\left(0.99866 + \frac{0.91424}{Q_L} - \frac{1.03175}{Q_L^2}\right) + \frac{0.6}{(2\pi f)^2 L1}$$

$$C2 = \frac{1}{2\pi f R}\left(\frac{1}{Q_L - 0.104823}\right)\left(1.00121 + \frac{1.01468}{Q_L - 1.7879}\right) - \frac{0.2}{(2\pi f)^2 L1}$$

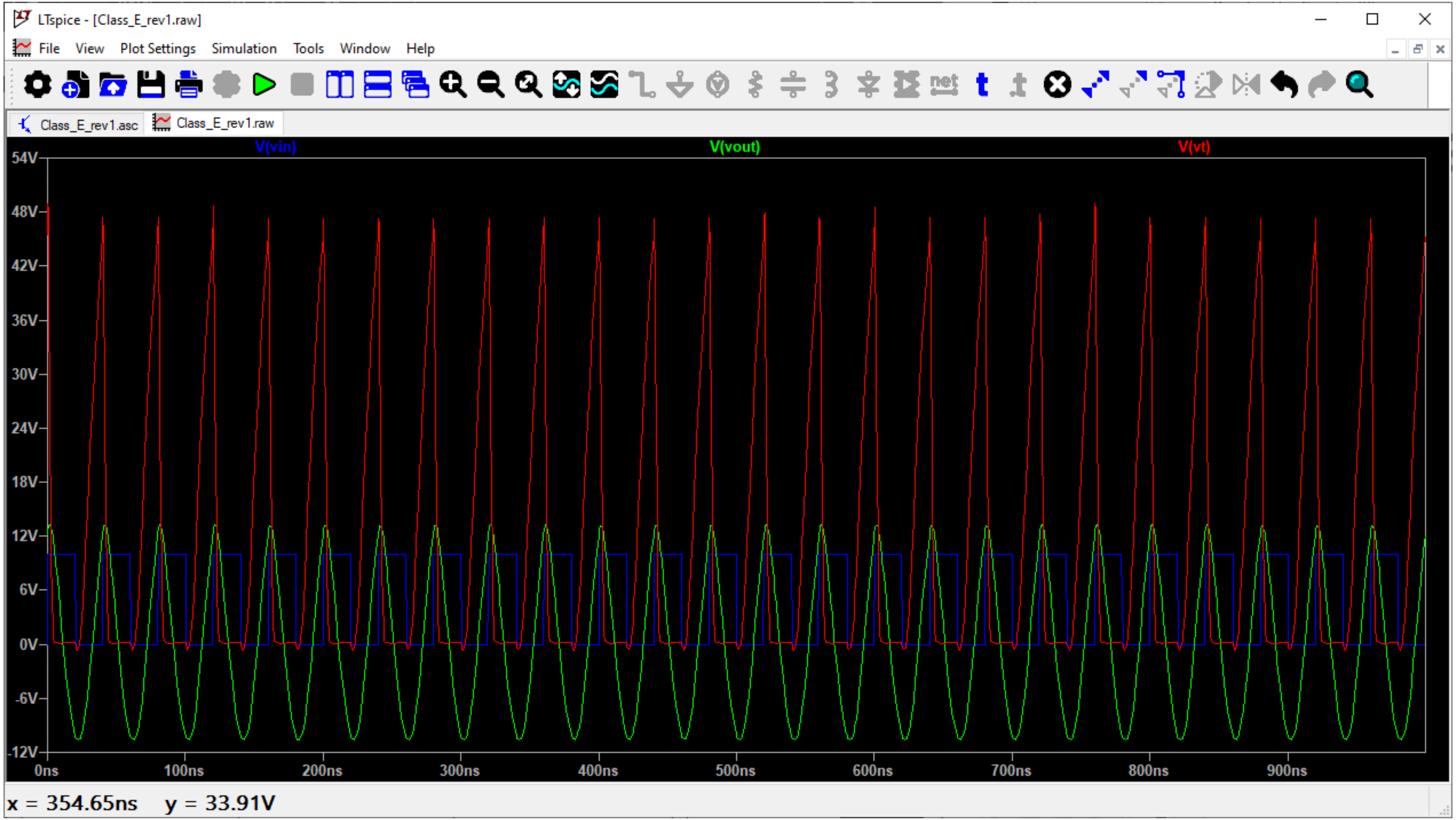
$$L2 = \frac{Q_L R}{2\pi f}$$

https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php

Results


LTspice - [Class_E_rev1.asc]
File Edit Hierarchy View Simulate Tools Window Help
Class_E_rev1.asc
Class_E_rev1.raw
+12V
V1
12
+12V
L1
50µ
.param FREQ = 25Meg
.param TPER = 1/FREQ
.param TR = 200p
.param TF = 200p
.param VHI = 10
Vt
C2
22.85p
L2
2.37µ
Vout
R_load
74.39
Q
IRF510
C1
8.43p
Vin
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
.tran 0 50u 49u
Click to plot V(+12V). DC operating point: V(+12v) = 12V

Results


LTspice - [Class_E_rev1.raw]
File View Plot Settings Simulation Tools Window Help
Class_E_rev1.asc
Class_E_rev1.raw
V(vin)
V(vout)
V(vt)
54V
48V
42V
36V
30V
24V
18V
12V
6V
0V
-6V
-12V
0ns
100ns
200ns
300ns
400ns
500ns
600ns
700ns
800ns
900ns
x = 354.65ns y = 33.91V

Observation/Conclusion

- Class-E PAs provide the best trade-off between high efficiency (~80–90%), simplicity, and output power for powering of implants [1], [3], [4], [7].
- The current Class-E PA isn’t designed to see 50 Ω directly so we could use:
  - Use a transformer with ~1.22:1 (from the calculator) voltage ratio (secondary to primary)
  - Use a simple L-network to transform 50 Ω → 74.39 Ω instead of a physical transformer.
- For wideband operation, push-pull Class-AB could be explored.

Next Steps

Refine design and integrate it into the overall device with the help of Jerimiah through the use of Altium.

References

[1] N. R., V. A., B. Chokkalingam, S. Padmanaban, and Z. Leonowicz, “Class E Power Amplifier Design and Optimization for the Capacitive Coupled Wireless Power Transfer System in Biomedical Implants,” Energies, vol. 10, no. 9, p. 1409, Sep. 2017, doi: https://doi.org/10.3390/en10091409.

[2] M. Ludens, “RF Power Amplifier Design,” ludens.cl, 2020. https://ludens.cl/Electron/RFamps/RFamps.html

[3] Y. Ben Fadhel, S. Ktata, K. Sedraoui, S. Rahmani, and K. Al-Haddad, “A Modified Wireless Power Transfer System for Medical Implants,” Energies, vol. 12, no. 10, p. 1890, May 2019, doi: https://doi.org/10.3390/en12101890.

[4] S. R. Khan, S. K. Pavuluri, G. Cummins, and M. P. Y. Desmulliez, “Wireless Power Transfer Techniques for Implantable Medical Devices: a Review,” Sensors, vol. 20, no. 12, p. 3487, Jun. 2020, doi: https://doi.org/10.3390/s20123487.

[5] Texas Instruments, “HF Power Amplifier (Reference Design Guide) RFID Systems / ASP 1.) Scope,” 2004. Accessed: Oct. 21, 2025. [Online]. Available: https://e2e.ti.com/cfs-file/__key/telligent-evolution-components-attachments/00-667-00-00-00-11-91-53/appnote_5F00_trf796x_5F00_pwramp_5F00_4w.pdf

[6] S. C. Cripps, *RF Power Amplifiers for Wireless Communications*, 2nd ed., Norwood, MA, USA: Artech House, 2006.

[7] F. H. Raab *et al.*, “Power amplifiers and transmitters for RF and microwave,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 3, pp. 814–826, Mar. 2002. doi: 10.1109/22.989965.

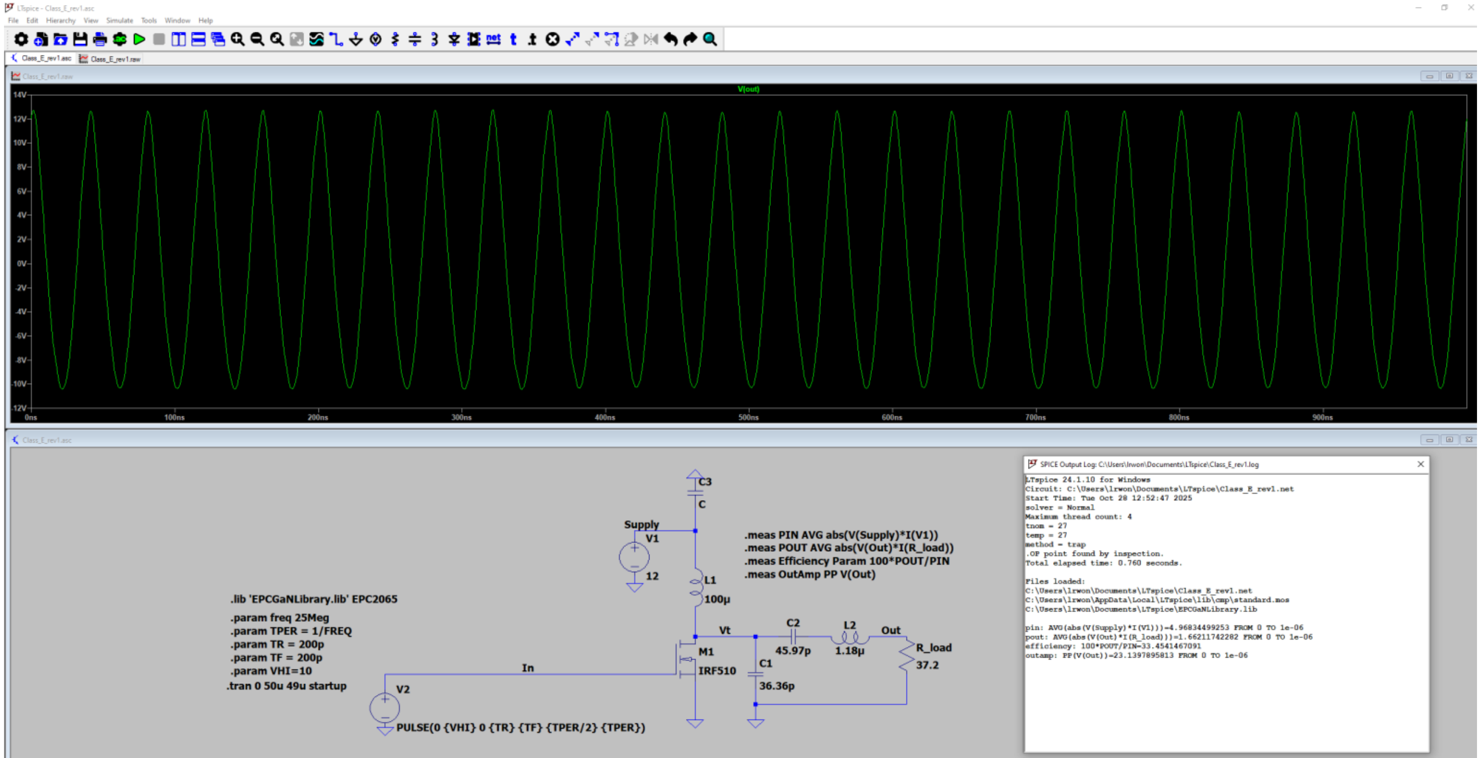
Power Amplifier Design Continued…

Luis Wong 10/28/2025

Methods

- Made use of LTspice to graph different aspects of the design and modify accordingly.
  - Compared power efficiency from 20-25MHz
- Some further research on Class E implementations.
- Used Altium in order to make schematic (at the request of Jerimiah).

Results



LTspice - Class_E_rev1.asc
File Edit Hierarchy View Simulate Tools Window Help
Class_E_rev1.asc
Class_E_rev1.raw
V(out)
14V
12V
10V
8V
6V
4V
2V
0V
-2V
-4V
-6V
-8V
-10V
-12V
0ns
100ns
200ns
300ns
400ns
500ns
600ns
700ns
800ns
900ns
C3
C
Supply
V1
12
L1
100µ
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
.lib 'EPCGaNLibrary.lib' EPC2065
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200p
.param TF = 200p
.param VHI=10
.tran 0 50u 49u startup
Vt
C2
45.97p
L2
1.18µ
Out
R_load
37.2
M1
IRF510
C1
36.36p
In
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
SPICE Output Log: C:\Users\lrwon\Documents\LTspice\Class_E_rev1.log
LTspice 24.1.10 for Windows
Circuit: C:\Users\lrwon\Documents\LTspice\Class_E_rev1.net
Start Time: Tue Oct 28 12:52:47 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = trap
.OP point found by inspection.
Total elapsed time: 0.760 seconds.
Files loaded:
C:\Users\lrwon\Documents\LTspice\Class_E_rev1.net
C:\Users\lrwon\AppData\Local\LTspice\lib\cmp\standard.mos
C:\Users\lrwon\Documents\LTspice\EPCGaNLibrary.lib
pin: AVG(abs(V(Supply)*I(V1)))=4.96834499253 FROM 0 TO 1e-06
pout: AVG(abs(V(Out)*I(R_load)))=1.66211742282 FROM 0 TO 1e-06
efficiency: 100*POUT/PIN=33.4541467091
outamp: PP(V(Out))=23.1397895813 FROM 0 TO 1e-06

Results

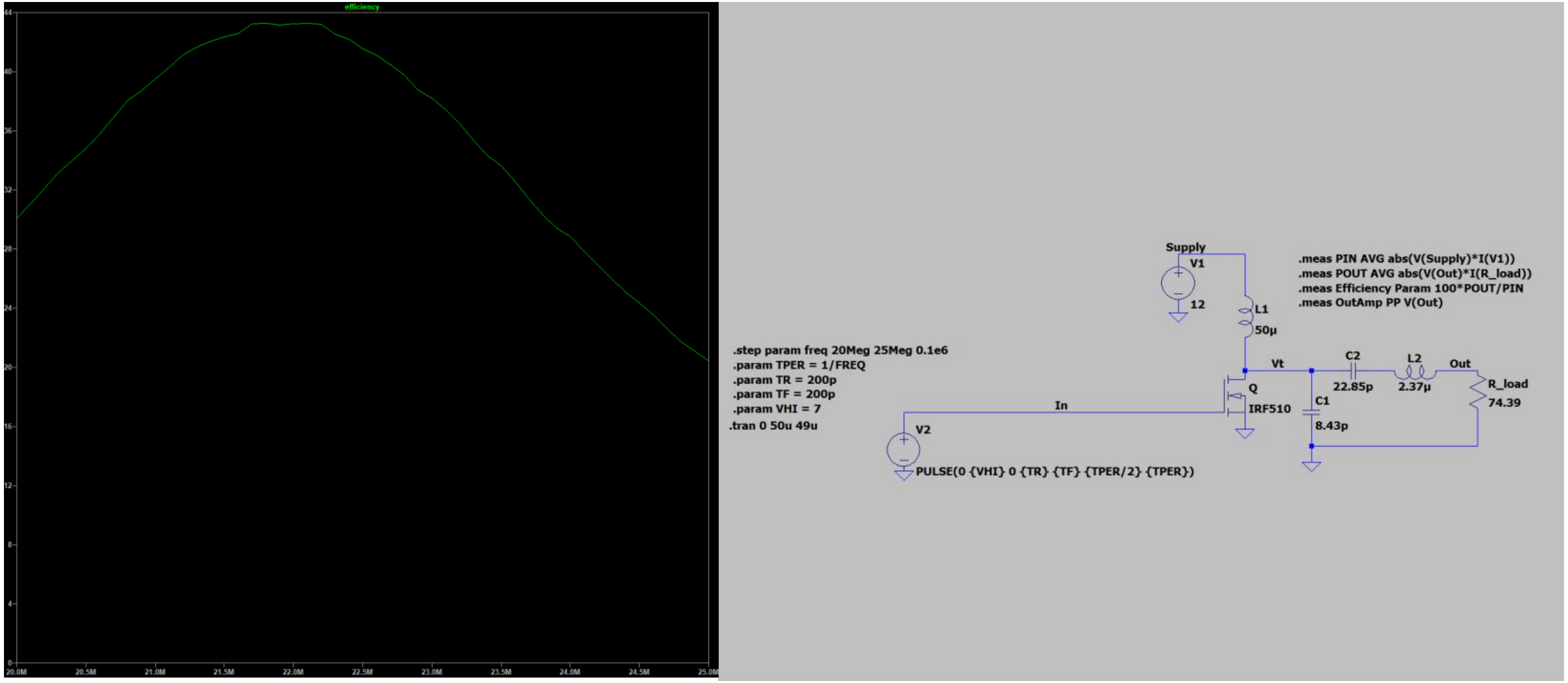


Figure 1: Efficiency graph of IRF510 (left) made using the following schematic (right) and calculated as described in the spice directives on the top-right of the design.

Results

Dynamic							
Input capacitance	C_{iss}	V_{GS} = 0 V, V_{DS} = 25 V, f = 1.0 MHz, see fig. 5		-	180	-	pF
Output capacitance	C_{oss}			-	81	-	
Reverse transfer capacitance	C_{rss}			-	15	-	
Total gate charge	Q_g	V_{GS} = 10 V	I_D = 5.6 A, V_{DS} = 80 V V_{DS} = 10 V, see fig. 6 and fig. 13 [b]	-	-	8.3	nC
Gate-source charge	Q_{gs}			-	-	2.3	
Gate-drain charge	Q_{gd}			-	-	3.8	

Figure 2: From Datasheet for IRF510. Shows Coss of 81pF

- If the calculations result in a shunt capacitance lower than Coss this would eliminate the need of the capacitor entirely.
- Adding any external capacitance would further slow down the drain transition and ruin ZVS.
- The device is too “capacitively heavy” for the target frequency and load.
- Will lead to the lower than expected efficiency (expecting ~90%).

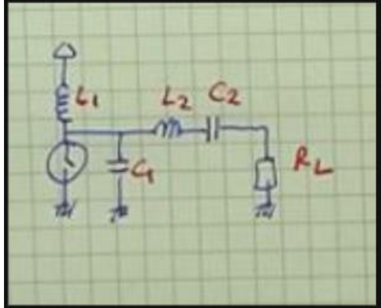
Results

- Redid calculations with Coss in mind.

Enter Q (around 5 is a good start): 5
Enter Vcc: 12 V
Enter Vo (transistor/FET saturation voltage): 0 V
Enter Coss (transistor/FET output capacitance): 84p F
Enter frequency: 25M Hz
Enter the desired power output: 1 W
Enter L1: 50u H
Calculate

R is: 74.39 Ω (Impedance transformer to 50Ω: secondary (50Ω) to primary (amplifier) turns ratio: 819.81 m)
C1 is: -65.6 p F (XC1 is: -97.05 Ω)
C2 is: 22.86 p F (XC2 is: 278.46 Ω)
L1min is: 41.27 μ Ω (XL1 is: 7.85 k Ω)
L2 is: 2.37 μ H (XL2 is: 371.97 Ω)
XL2 - XC2 = 93.52 Ω
Expect Icc of at least: 83.33 m I
Select a FET/transistor with a max Vds/Vce of at least: 53.4 V (Includes a safety factor of 0.8)
1 / fr1 = 21.63 M Hz. When Q1 closed, should be less than the frequency of operation.
1 / fr2 = 32.39 M Hz. When Q1 open, should be greater than the frequency of operation.
1 / (fr1 + f2 half period) = 27.01 M Hz. Should be about equal to the frequency of operation.

Figure 3: https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php




```
V_supply = 12;
Pout = 1;
Zout = 50;
f = 25*10^7;
Q = 5; %ranges from 0-10
w = 2*pi*f;
Coss = 9.5e-12;
```

```
Coss = 9.5000e-12
```

```
R_L = 1.154*((V_supply^2)/(2*Pout))
```

```
R_L = 83.0880
```

Figure 4: R_L calculation based on this video: https://www.youtube.com/watch?v=Tgrakttus3c&t=1s

Results

Dynamic Characteristics# ($T_J = 25°C$ unless otherwise stated)						
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
C_{ISS}	Input Capacitance	$V_{GS} = 0$ V, $V_{DS} = 50$ V		14	17	pF
C_{RSS}	Reverse Transfer Capacitance			0.1		
C_{OSS}	Output Capacitance			6.5	10	
$C_{OSS(ER)}$	Effective Output Capacitance, Energy Related (Note 2)	$V_{GS} = 0$ V, $V_{DS} = 0$ to 50 V		9.5		
$C_{OSS(TR)}$	Effective Output Capacitance, Time Related (Note 3)			12		
R_G	Gate Resistance			0.5		Ω
Q_G	Total Gate Charge	$V_{GS} = 5$ V, $V_{DS} = 50$ V, $I_D = 0.1$ A		115	145	pC
Q_{GS}	Gate-to-Source Charge	$V_{DS} = 50$ V, $I_D = 0.1$ A		32		
Q_{GD}	Gate-to-Drain Charge			25		
$Q_{G(TH)}$	Gate Charge at Threshold			24		
Q_{OSS}	Output Charge	$V_{GS} = 0$ V, $V_{DS} = 50$ V		600	900	
Q_{RR}	Source-Drain Recovery Charge			0		

Defined by design. Not subject to production test.
All measurements were done with substrate connected to source.
Note 2: $C_{OSS(ER)}$ is a fixed capacitance that gives the same stored energy as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.
Note 3: $C_{OSS(TR)}$ is a fixed capacitance that gives the same charging time as C_{OSS} while V_{DS} is rising from 0 to 50% BV_{DSS}.

Figure 5: EPC2037 datasheet showing 9.5pF Coss. Might make this better for our purposes.

EPC2037 – Enhancement Mode Power Transistor

V_{DS} , 100 V
$R_{DS(on)}$, 550 mΩ
I_D , 1.7 A

EFFICIENT POWER CONVERSION

RoHS Halogen-Free

Revised August 21, 2024

Gallium Nitride's exceptionally high electron mobility and low temperature coefficient allows very low $R_{DS(on)}$, while its lateral device structure and majority carrier diode provide exceptionally low Q_G and zero Q_{RR}. The end result is a device that can handle tasks where very high switching frequency, and low on-time are beneficial as well as those where on-state losses dominate.

Application Notes:

- Easy-to-use and reliable gate, Gate Drive ON = 5 V typical, OFF = 0 V (negative voltage not needed)
- Top of FET is electrically connected to source

Questions: Ask a GaN Expert

Maximum Ratings			
PARAMETER		**VALUE**	**UNIT**
V_{DS}	Drain-to-Source Voltage (Continuous)	100	V
	Drain-to-Source Voltage (up to 10,000 5 ms pulses at 150°C)	120	
I_D	Continuous (T_A = 25°C, $R_{θJA}$ = 44°C/W)	1.7	A
	Pulsed (25°C, T_{PULSE} = 300 µs)	2.4	
V_{GS}	Gate-to-Source Voltage	6	V
	Gate-to-Source Voltage	–4	
	Recommended Gate-to-Source Voltage Operating Range*	4.5 – 5.5	
T_J	Operating Temperature	–40 to 150	°C
T_{STG}	Storage Temperature	–40 to 150	

*Operating at less than 4 V_{GS} is not recommended.

Thermal Characteristics			
PARAMETER		**TYP**	**UNIT**
$R_{θJC}$	Thermal Resistance, Junction-to-Case	14	°C/W
$R_{θJB}$	Thermal Resistance, Junction-to-Board	79	
$R_{θJA}$	Thermal Resistance, Junction-to-Ambient (Note 1)	100	

Note 1: $R_{θJA}$ is determined with the device mounted on one square inch of copper pad, single layer 2 oz copper on FR4 board.
See https://epc-co.com/epc/documents/product-training/Appnote_Thermal_Performance_of_eGaN_FETs.pdf for details

Static Characteristics (T_J = 25°C unless otherwise stated)						
PARAMETER		**TEST CONDITIONS**	**MIN**	**TYP**	**MAX**	**UNIT**
BV_{DSS}	Drain-to-Source Voltage	V_{GS} = 0 V, I_D = 125 µA	100			V
I_{DSS}	Drain-Source Leakage	V_{GS} = 0 V, V_{DS} = 80 V		10	100	µA
I_{GSS}	Gate-to-Source Forward Leakage	V_{GS} = 5 V		0.1	1	mA
	Gate-to-Source Reverse Leakage	V_{GS} = -4 V		10	100	µA
$V_{GS(TH)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 0.08 mA	0.8	1.5	2.5	V
$R_{DS(on)}$	Drain-Source On Resistance	V_{GS} = 5 V, I_D = 0.1 A		400	550	mΩ
V_{SD}	Source-Drain Forward Voltage#	V_{GS} = 0 V, I_S = 0.5 A		2.5		V

Defined by design. Not subject to production test.
All measurements were done with substrate connected to source.

Die size: 0.9 x 0.9 mm

EPC2037 eGaN® FETs are supplied only in passivated die form with solder bumps.

Applications

- High speed DC-DC conversion
- Wireless power transfer
- Lidar/pulsed power applications
- Class-D audio

Benefits

- Ultra high efficiency
- Ultra low $R_{DS(on)}$
- Ultra low Q_G
- Ultra small footprint

Scan QR code or click link below for more information including reliability reports, device models, demo boards!

https://l.ead.me/EPC2037

eGaN® FETs for Low Cost Resonant Wireless Power Applications

EFFICIENT POWER CONVERSION

Yuanzhe Zhang, Ph.D., Director of Applications Engineering, Michael de Rooij, Ph.D., Vice President of Applications Engineering

Resonant wireless power systems use loosely-coupled, highly-resonant coils that are tuned to high frequencies (6.78 MHz or 13.56 MHz). The AirFuel Alliance has developed the standard for resonant wireless power applications. They address convenience-of-use issues such as source to device distance, device orientation on the source, multiple devices on a single source, higher power capability, simplicity of use, and imperfect placement.

eGaN® FETs from EPC offer significantly lower capacitance and inductance with zero reverse recovery charge (Q_{RR}) in a smaller footprint for a given $R_{DS(on)}$ than comparable MOSFETs. This enables a number of applications that require higher switching frequency such as 6.78 MHz highly resonant wireless power transfer.

In this application note, we will present a differential class E amplifier using EPC2037 for 6.78 MHz loosely coupled highly resonant wireless power applications. A photo of the EPC2037 die is shown in figure 1, with its specifications given in table 1. It has very low gate charge and parasitics. As a result, no dedicated gate driver chips are required and it can be driven directly from logic.

Class E amplifier design basics

The schematic of a single-ended class E amplifier is shown in figure 2. It consists of a ground-referenced transistor (Q_1), an RF choke (L_{RFck}), an extra inductor (L_e) and a shunt capacitor (C_{sh}). The load of the amplifier (Z_{Load}) represents the power transmitting coil and is assumed to be inductive. A capacitor (C_s) is connected in series for coil tuning. The resistance of the tuned coil is R_{Load}. The amplifier operates at a fixed frequency with a fixed duty cycle of 50%. The transistor voltage and current waveforms under ideal operation is sketched in figure 3. Because of zero-voltage switching (ZVS) and zero-current switching (ZCS) at the optimum operating point, the class E amplifier has high efficiency (usually well above 90%).

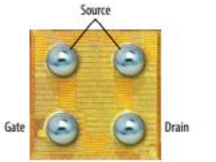


Figure 1: Mounting side of EPC2037 eGaN FET Dimension: 0.9 x 0.9 mm

EPC Part Number	V_{DS} (V)	$R_{DS(on)}$ @5 V (mΩ)	Q_G @5 V Typ (pC)	Q_{GS} Typ (pC)	Q_{GD} Typ (pC)	Q_{OSS} (pC)	I_D (A)	Package (mm)
EPC2037	100	550	115	32	25	600	1.7	BGA 0.9 x 0.9

Table 1. EPC2037 specifications

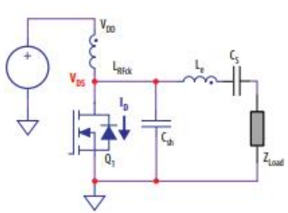


Figure 2. Schematic of a single-ended class E amplifier

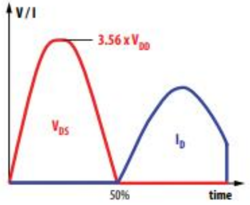


Figure 3. Ideal FET voltage and current waveforms for the class E amplifier

Results

- https://open.library.ubc.ca/soa/cIRcle/collections/ubctheses/24/items/1.0406624
- Paper shows a PA designed closely to our specifications at frequencies of 13.56 MHz and 27.12 MHz.
- Provides a design for how to drive the transistor and mentions need for heatsink.

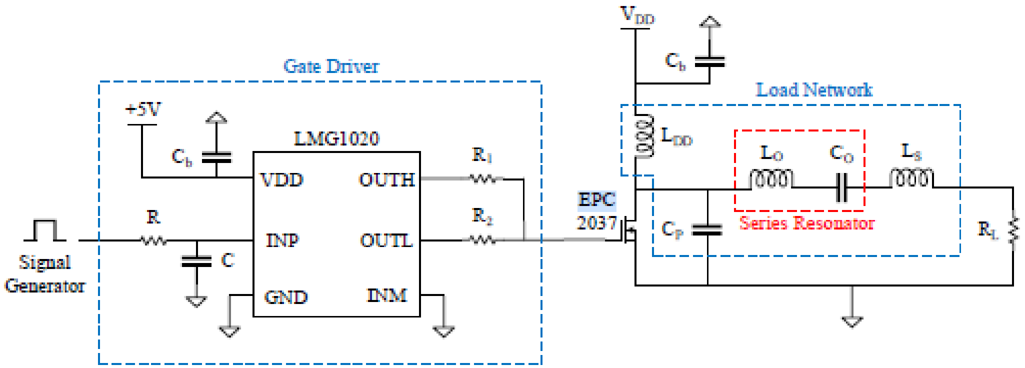


Figure 2.2: A signal generator (clock oscillator) feds a pulse to the LMG1020 gate driver integrated with the class-E power amplifier circuit [2].

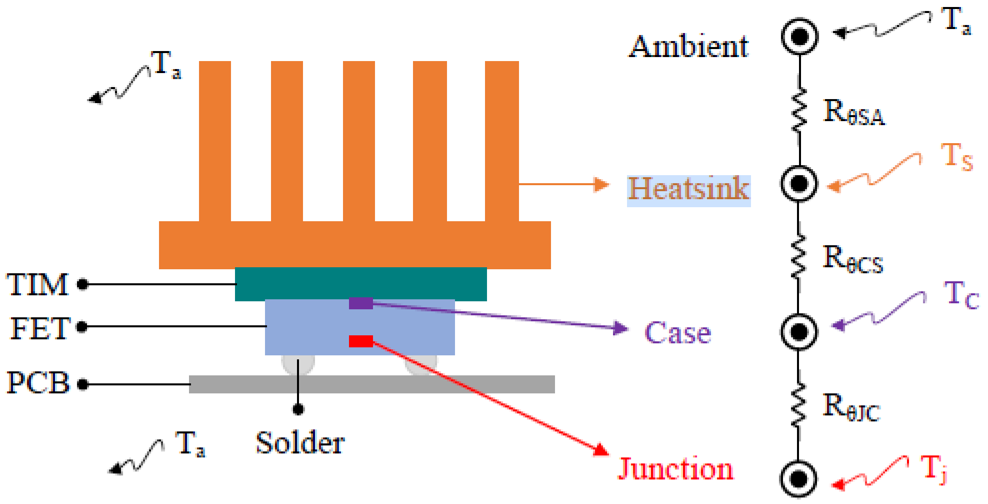


Figure 2.9: Illustration of a heatsink attachment with corresponding thermal resistance dissipation for a typical FET device.

Results

- https://epc-co.com/epc/Portals/0/epc/documents/application-notes/AN021%20FETs%20for%20Low%20Cost%20Class%20E%20WiPo.pdf is a document dedicated to explaining the EPC2037's application demonstrating it's use in multiple different class E PAs.

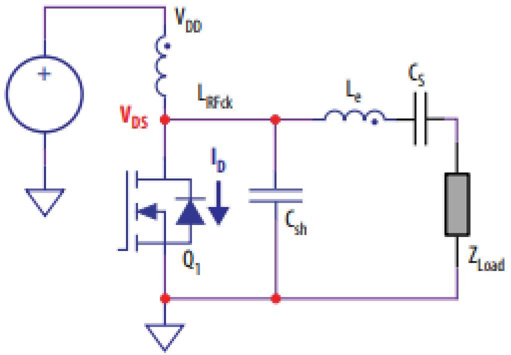


Figure 2. Schematic of a single-ended class E amplifier

Results

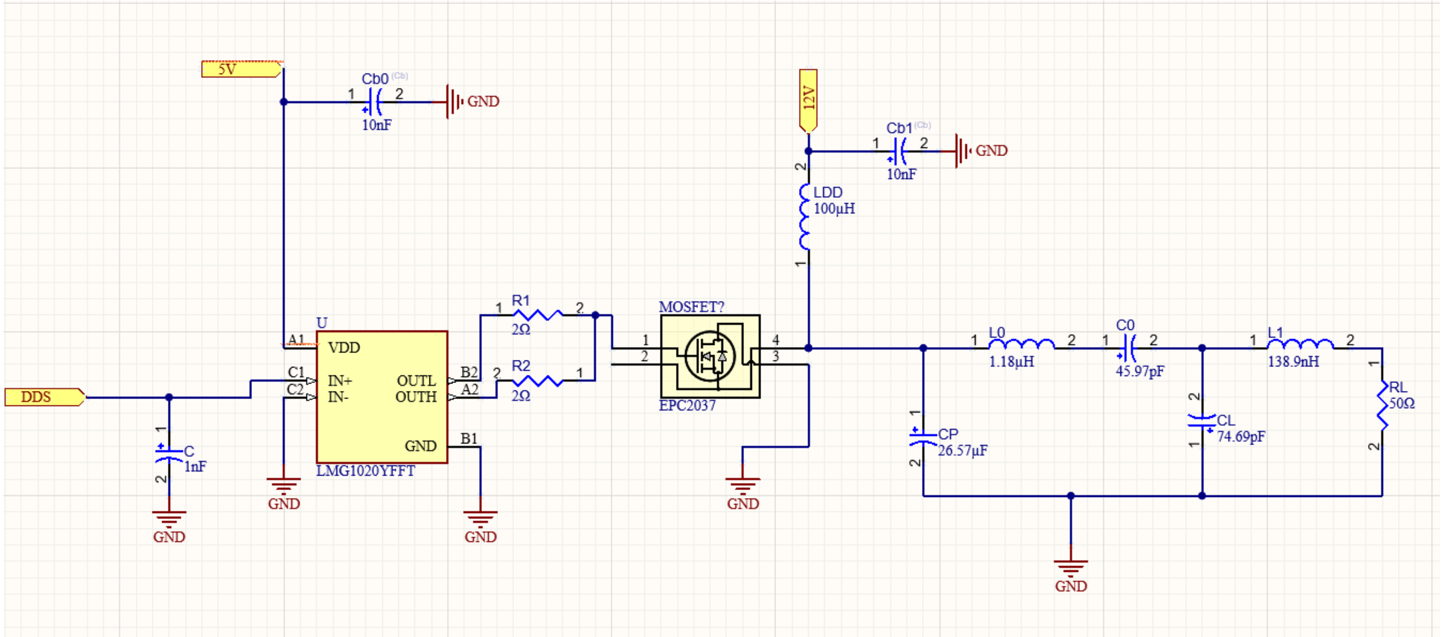


Figure 8: Altium Schematic Block implementing Driver + GaN-FET. Assumes DDS, 5V, and 12V rail are handled my MCU as it is supposed to control whether they are on or off.

Observation/Conclusion

- Previous design was lacking in power efficiency and tuned to the wrong frequency.
- The EPC2037 GaN-FET in conjunction with a driver amplifying a square wave may be preferable to 2 amplifiers in series.
  - The EPC2037 GaN-FET seems preferrable over the IRF510 (due to IRF510s high output capacitance) supported by documentation and paper.
  - PCB implementation needs to take into account the inaccessible pins in the smaller device.
- Will need to collaborate in order to properly design each schematic block.

Next Steps

- Do some more testing in LTspice using the EPC2037 check power efficiency.
- Change Q value of amplifier to allow PA to accept a wider range of frequencies **efficiently** (was told it needed to be 20-25MHz). Confirm range using LTspice.
- Refine design (using real component values) and integrate it into the overall device with the help of Jerimiah and Dmytro through the use of Altium 365.

Power Amplifier Design Continued…

Luis Wong 11/04/2025

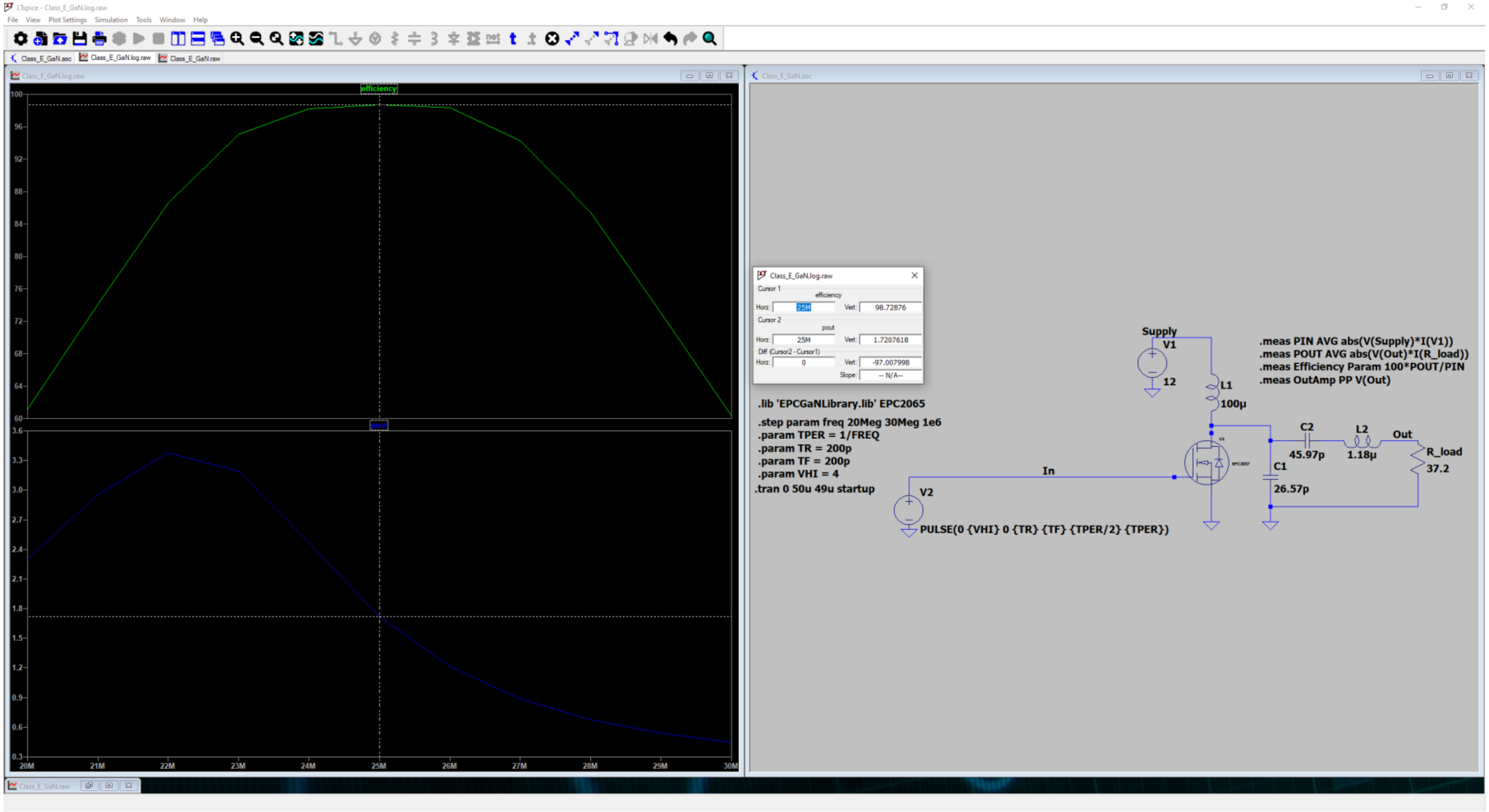
Objective

- Refine class E Design so that it may be implementable in Altium.
  - Simulate the efficiency of an ideal GaN_FET setup.
  - Achieve decent efficiency using IRF510 MOSFET.
  - Simulate 500us of the PA working.
  - User Driver in design.
  - Use capacitor and inductor values that are attainable.

Methods

- Use Altium and LTspice in order to make a working schematic and simulation.
- Use ONLINE SMITH CHART TOOL in order to make matching network.

Results



LTspice - Class_E_GaN.log.raw
efficiency
pout
Class_E_GaN.log.raw
Cursor 1
Horz: 25M Vert: 98.72876
Cursor 2
Horz: 25M Vert: 1.7207618
Diff (Cursor2 - Cursor1)
Horz: 0 Vert: -97.007998
Slope: -- N/A--
.lib 'EPCGaNLibrary.lib' EPC2065
.step param freq 20Meg 30Meg 1e6
.param TPER = 1/FREQ
.param TR = 200p
.param TF = 200p
.param VHI = 4
.tran 0 50u 49u startup
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
Supply
V1
12
L1
100µ
C2
45.97p
L2
1.18µ
Out
R_load
37.2
C1
26.57p
In
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
20M
21M
22M
23M
24M
25M
26M
27M
28M
29M
30M

Results

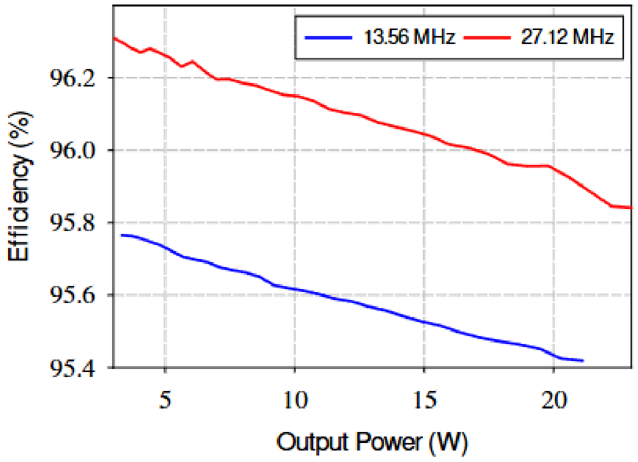
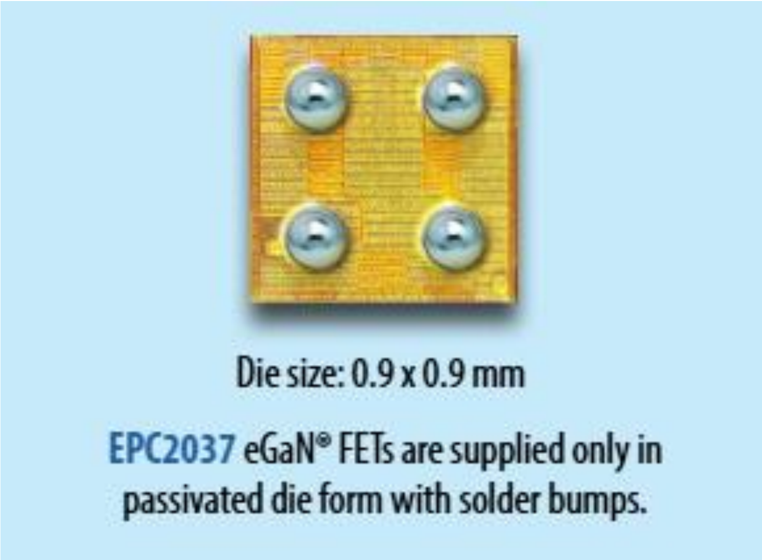
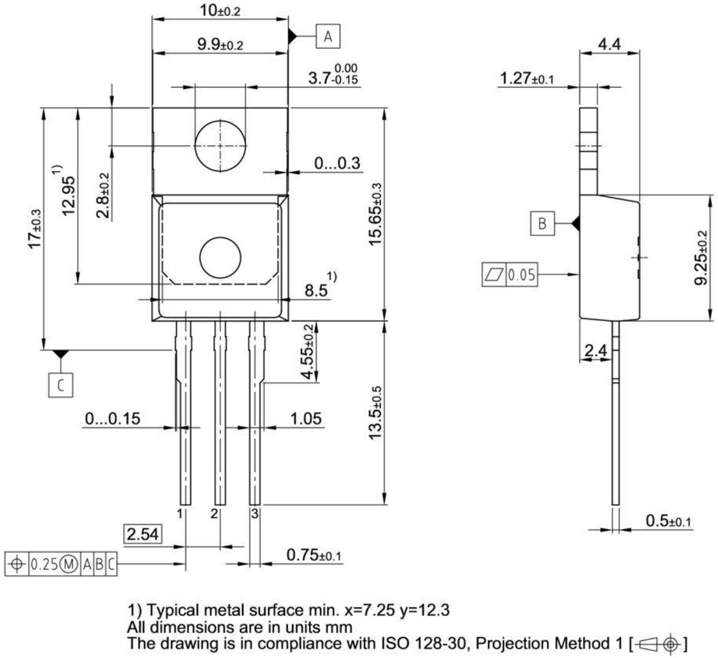


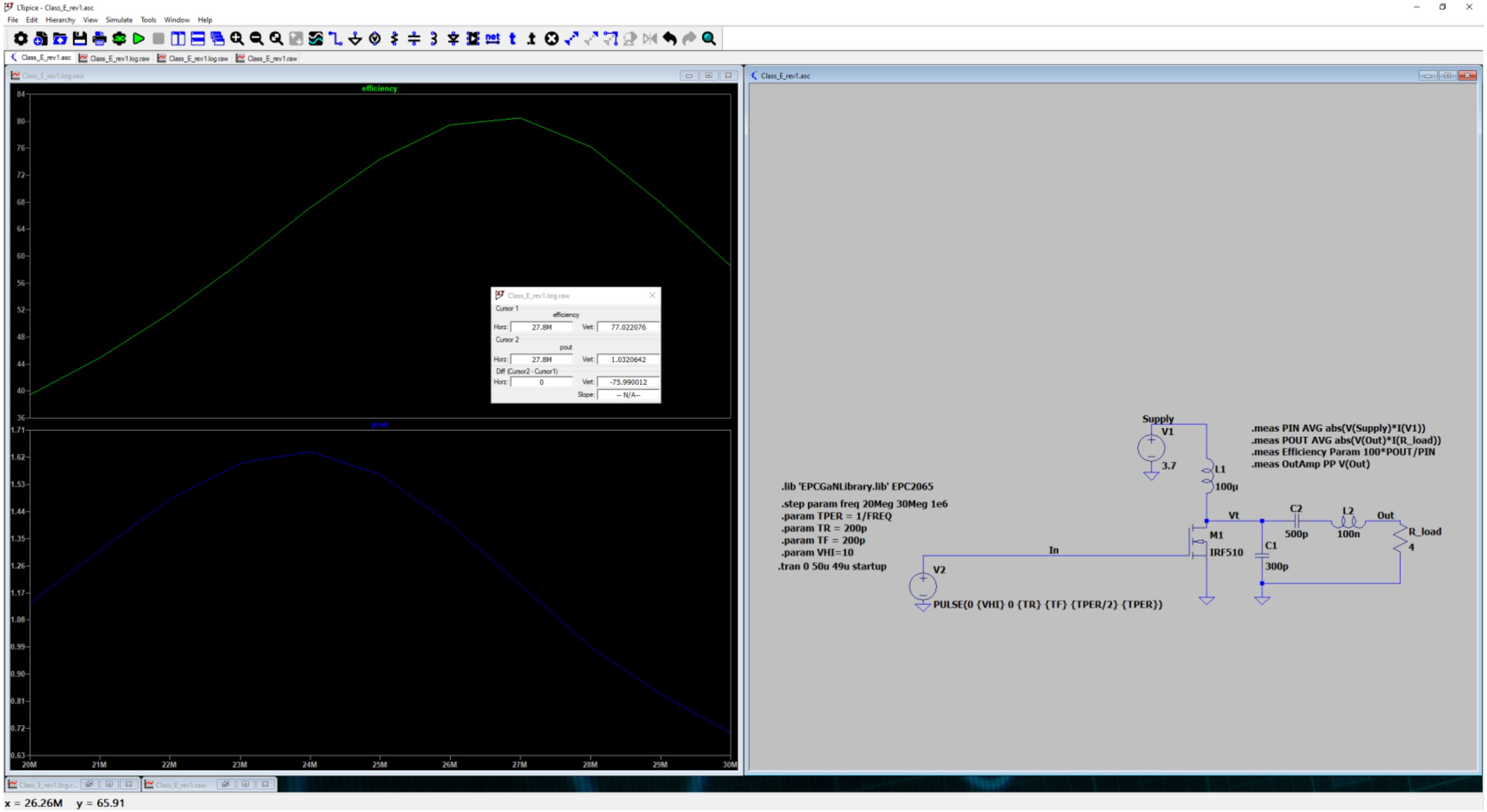
Figure 2.5: Simulated results for efficiency versus power of class-E amplifiers designed for 13.56 MHz and 27.12 MHz frequency bands.

Results


Die size: 0.9 x 0.9 mm
EPC2037 eGaN® FETs are supplied only in passivated die form with solder bumps.


10±0.2
9.9±0.2
A
3.7 0.00 -0.15
4.4
1.27±0.1
0...0.3
17±0.3
12.95 1)
2.8±0.2
15.65±0.3
B
0.05
9.25±0.2
8.5 1)
4.55±0.2
C
2.4
0...0.15
1.05
13.5±0.5
1
2
3
2.54
0.5±0.1
0.25 M A B C
0.75±0.1
1) Typical metal surface min. x=7.25 y=12.3
All dimensions are in units mm
The drawing is in compliance with ISO 128-30, Projection Method 1

Results



LTspice - Class_E_rev1.asc
efficiency
pout
Cursor 1
Horz: 27.8M
Vert: 77.022076
Cursor 2
Horz: 27.8M
Vert: 1.0320642
Diff (Cursor2 - Cursor1)
Horz: 0
Vert: -75.990012
Slope: -- N/A--
Supply
V1
3.7
L1
100µ
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
.lib 'EPCGaNLibrary.lib' EPC2065
.step param freq 20Meg 30Meg 1e6
.param TPER = 1/FREQ
.param TR = 200p
.param TF = 200p
.param VHI=10
.tran 0 50u 49u startup
Vt
C2
500p
L2
100n
Out
R_load
4
M1
IRF510
C1
300p
In
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
x = 26.26M y = 65.91

Results

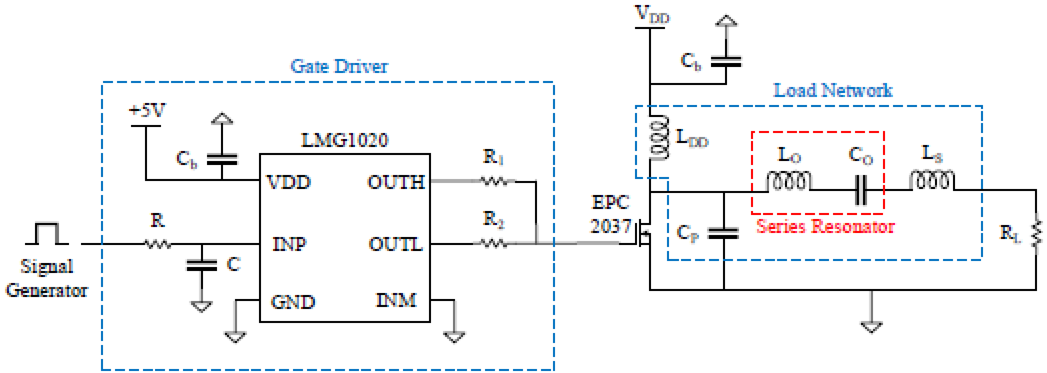
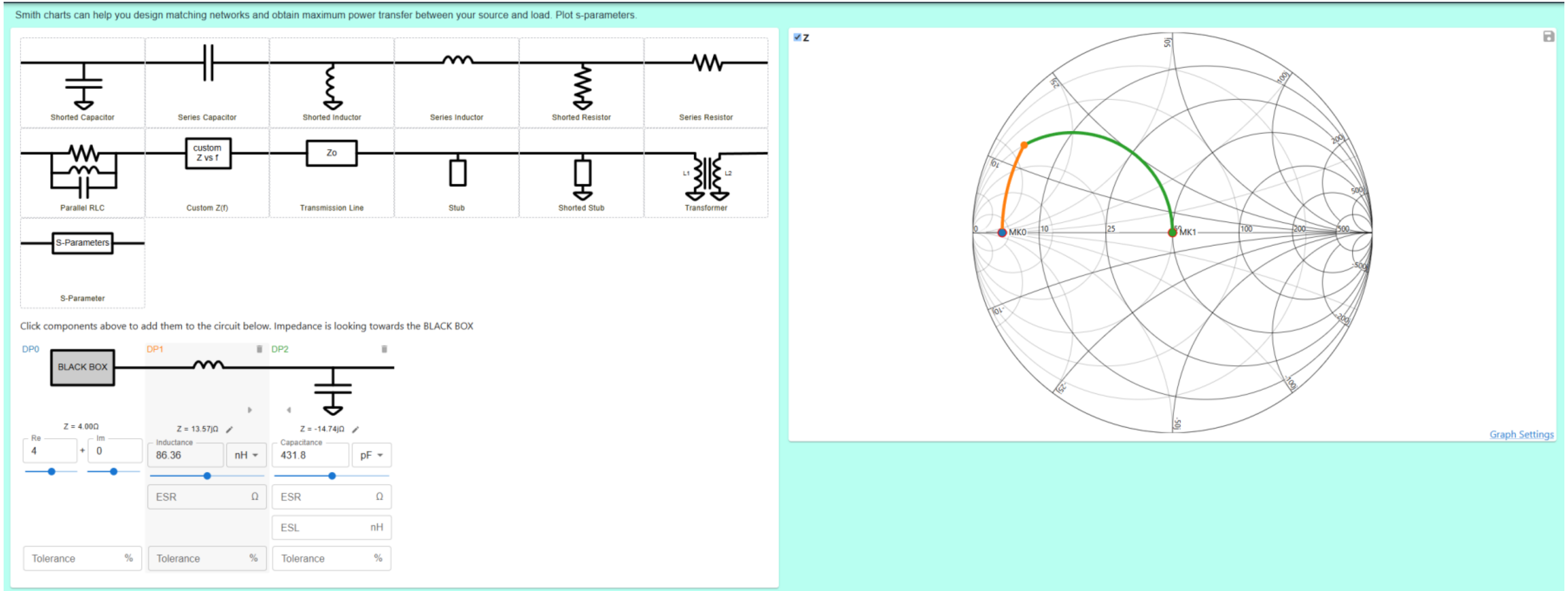


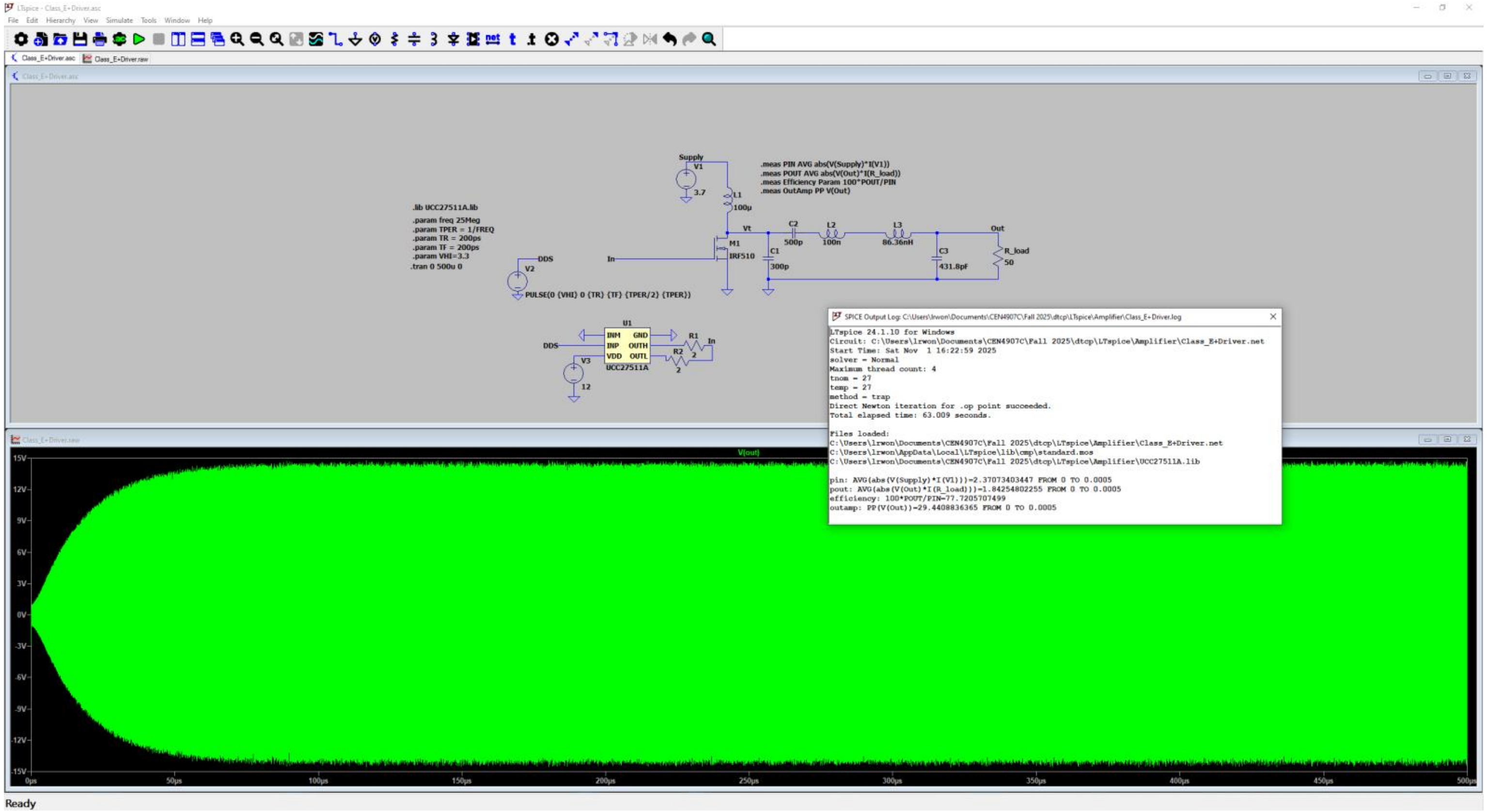
Figure 2.2: A signal generator (clock oscillator) feds a pulse to the LMG1020 gate driver integrated with the class-E power amplifier circuit [2].

Results

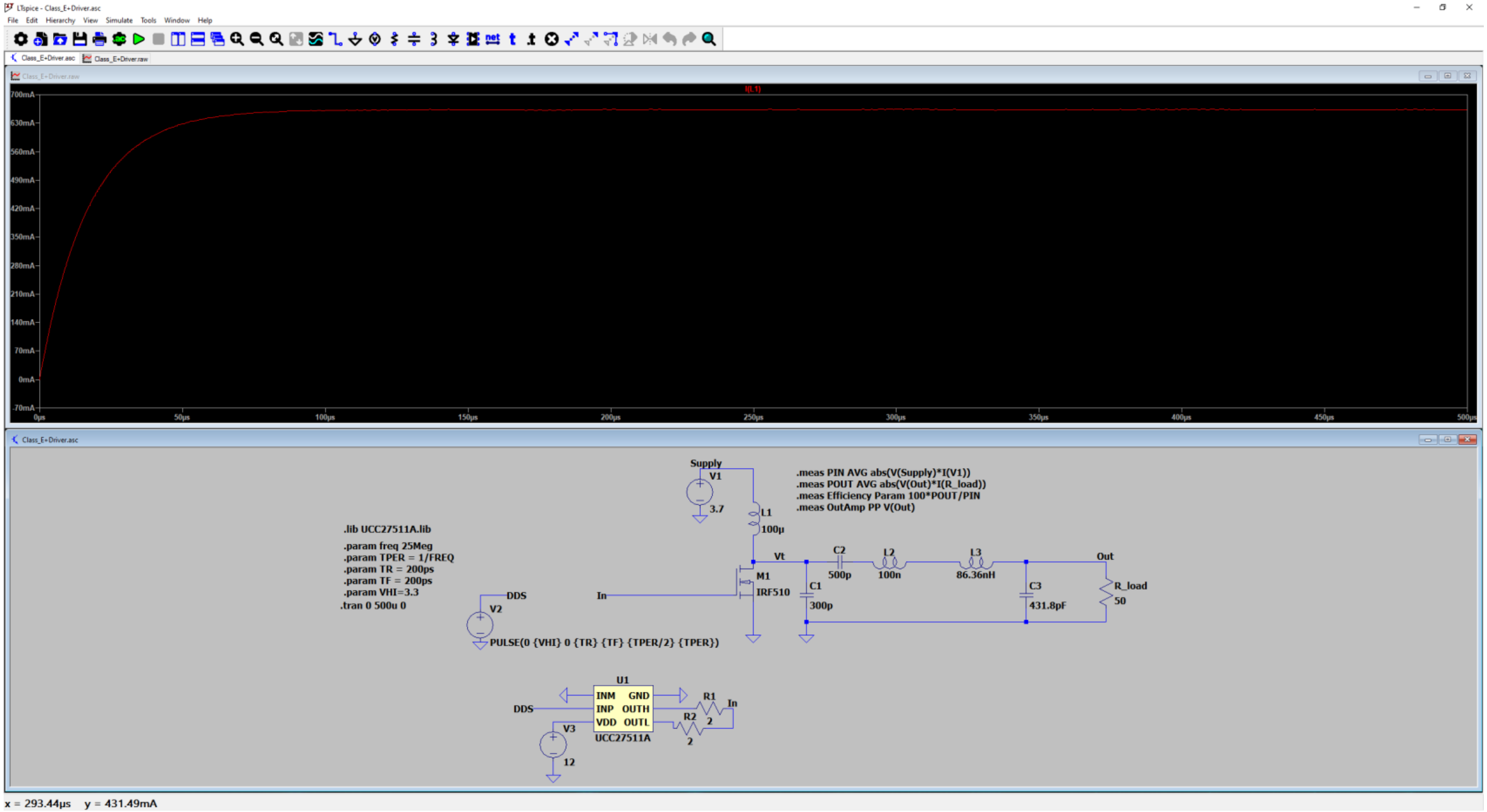


Will use calculator in order to tune matching network (an approx. 4 ohm to 50 ohm impedance).

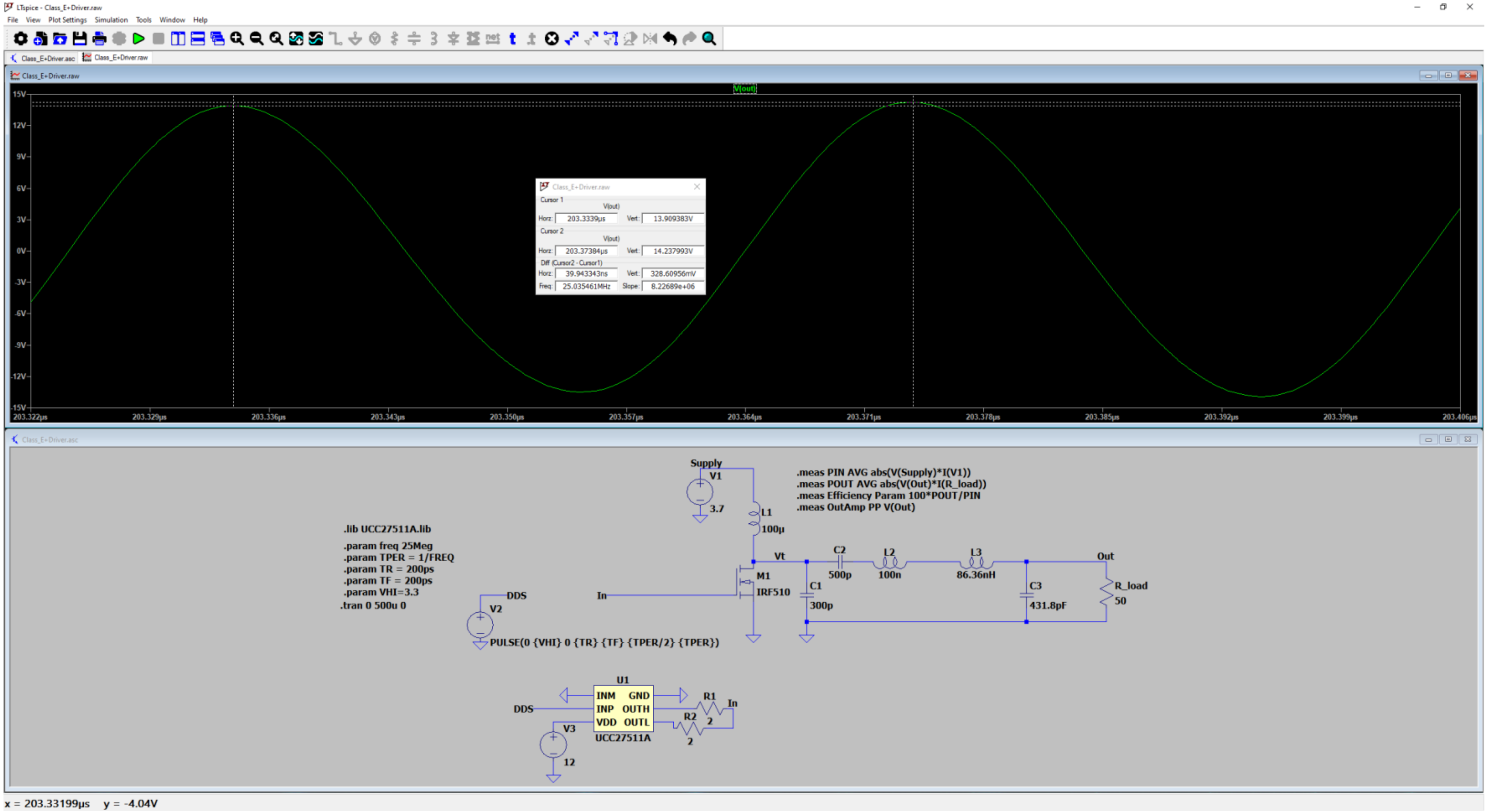
Results (DDS: 3.3V Square Wave)



Results (DDS: 3.3V Square Wave)

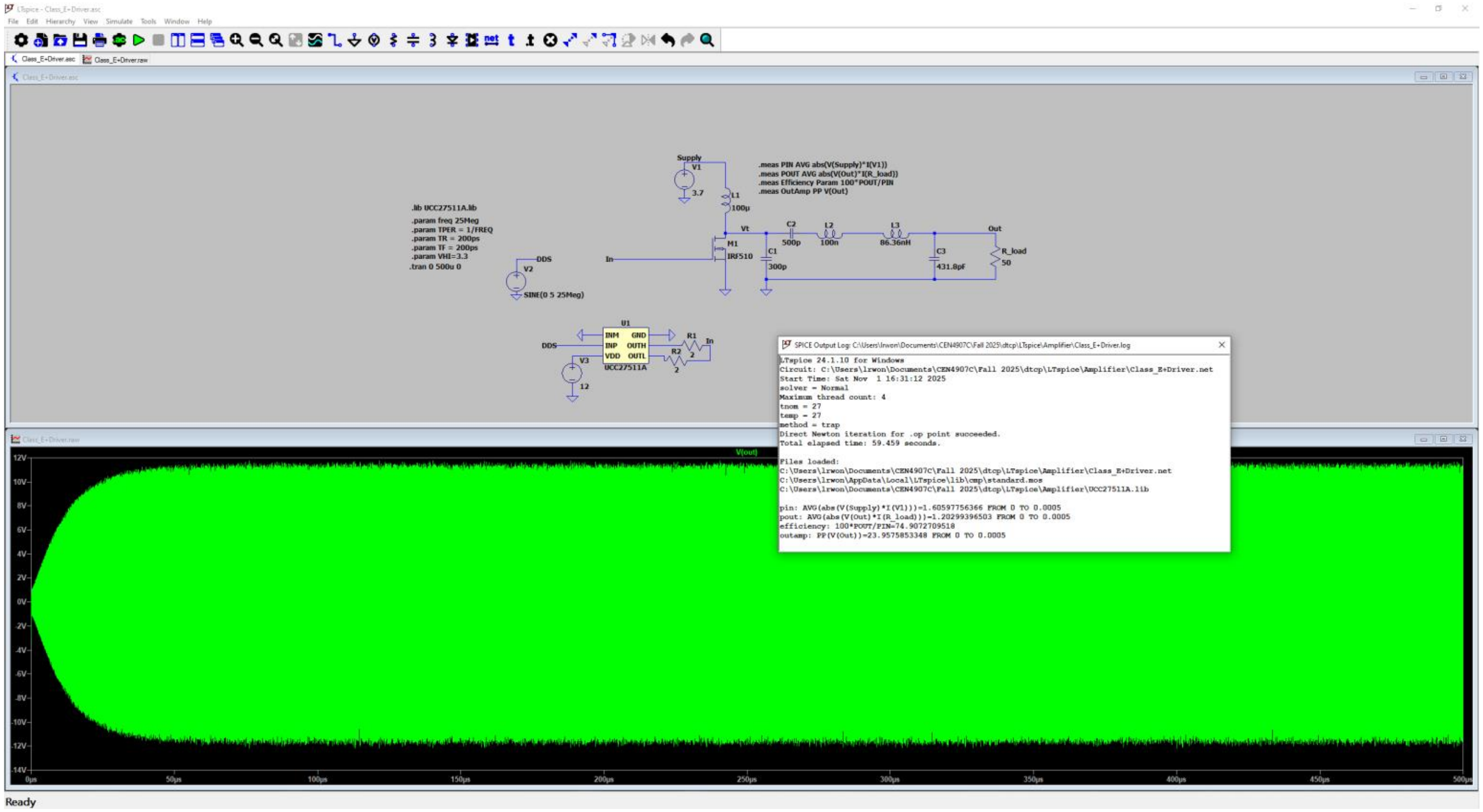


Results (DDS: 3.3V Square Wave)

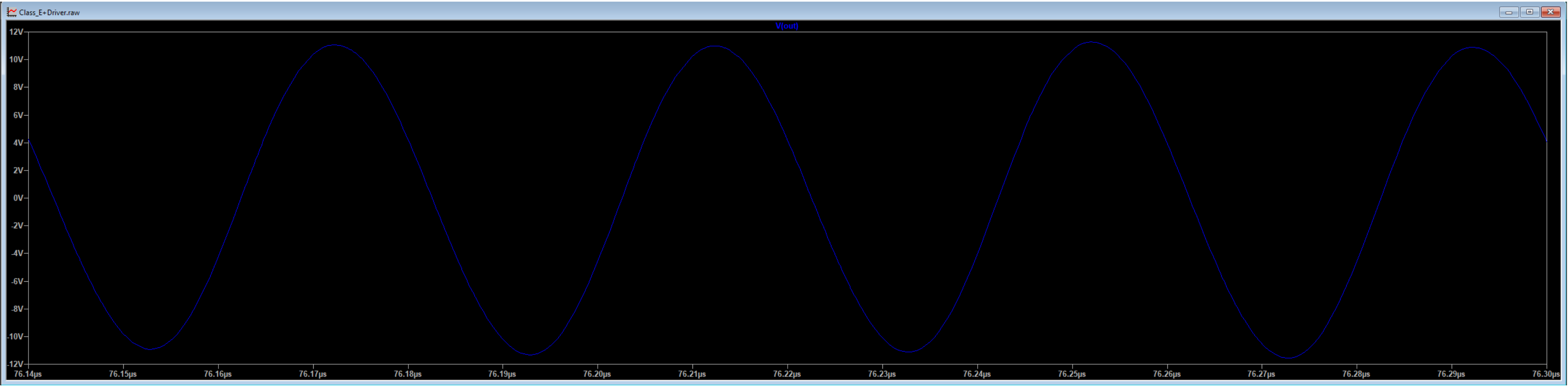


Results (DDS: 5V Sine Wave)

3.3 V makes power out go down to 0.7W

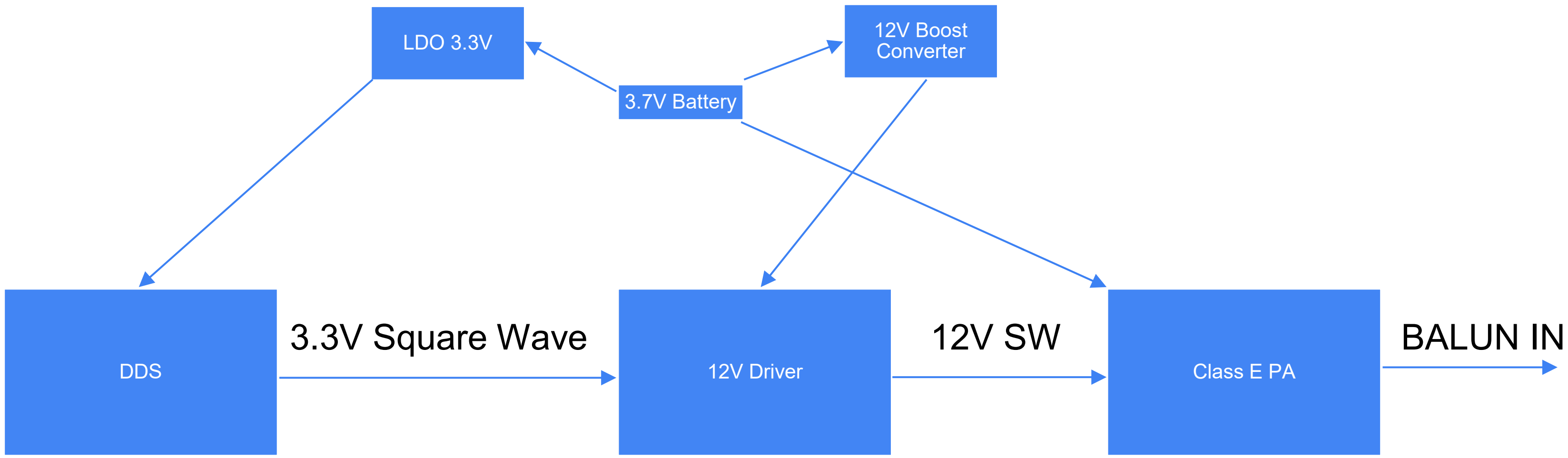


Results

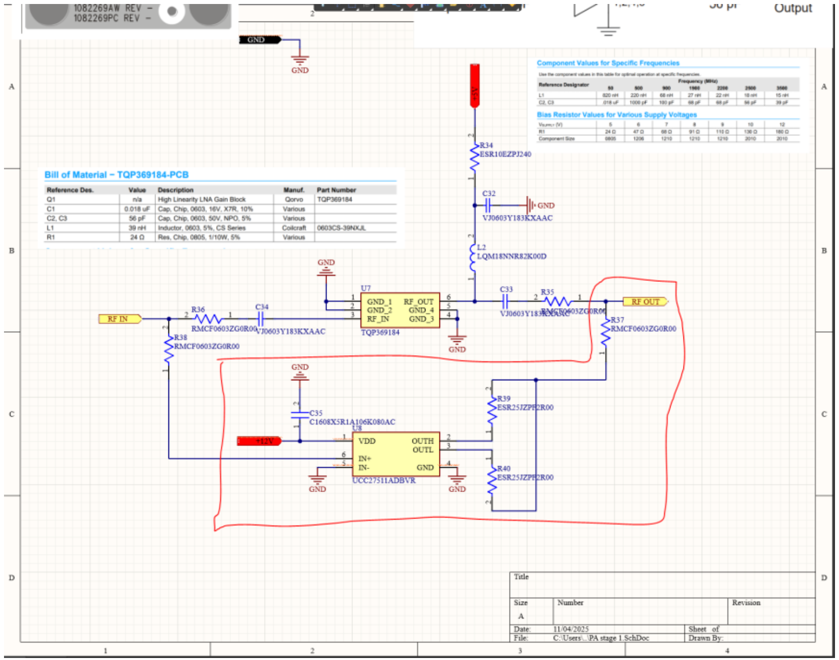


This results in distorted sine wave as output a period of about 50ns instead of the 40ns we expect (Each tick is 10ns).

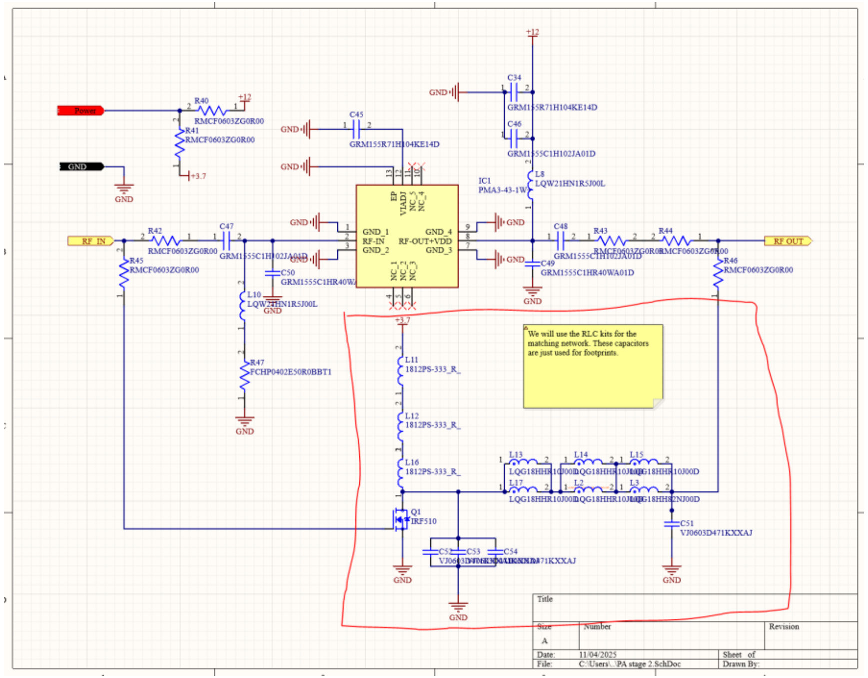
Results


LDO 3.3V
12V Boost Converter
3.7V Battery
DDS
3.3V Square Wave
12V Driver
12V SW
Class E PA
BALUN IN

Results



Output
GND
RF IN
RF OUT
TQP369184
UCC27511ADBVR
Bill of Material – TQP369184-PCB
Component Values for Specific Frequencies
Bias Resistor Values for Various Supply Voltages
Title
Size
Number
Revision
Date: 11/04/2025
File: C:\Users\..\PA stage 1.SchDoc
Sheet of
Drawn By:



Power
GND
RF IN
RF OUT
+12
+3.7
PMA3-43-1W
IRF510
We will use the RLC kits for the matching network. These capacitors are just used for footprints.
Title
Size
Number
Revision
Date: 11/04/2025
File: C:\Users\..\PA stage 2.SchDoc
Sheet of
Drawn By:

Observation/Conclusion

- As we are approaching the end of the semester the IRF510 while not as efficient is easier to prototype with.
  - Manages the >50% efficiency Jerimiah was looking for.
- Altium Schematic was implemented.

Next Steps

- Correct any mistakes in Altium design and adjust based on feedback.
- Adjust matching network design to work with real values.
- Order PCB in order to begin firmware development
- Assemble PCB and test functionality.

Inductors and Simulating New GaN-FET

Luis Wong 11/04/2025

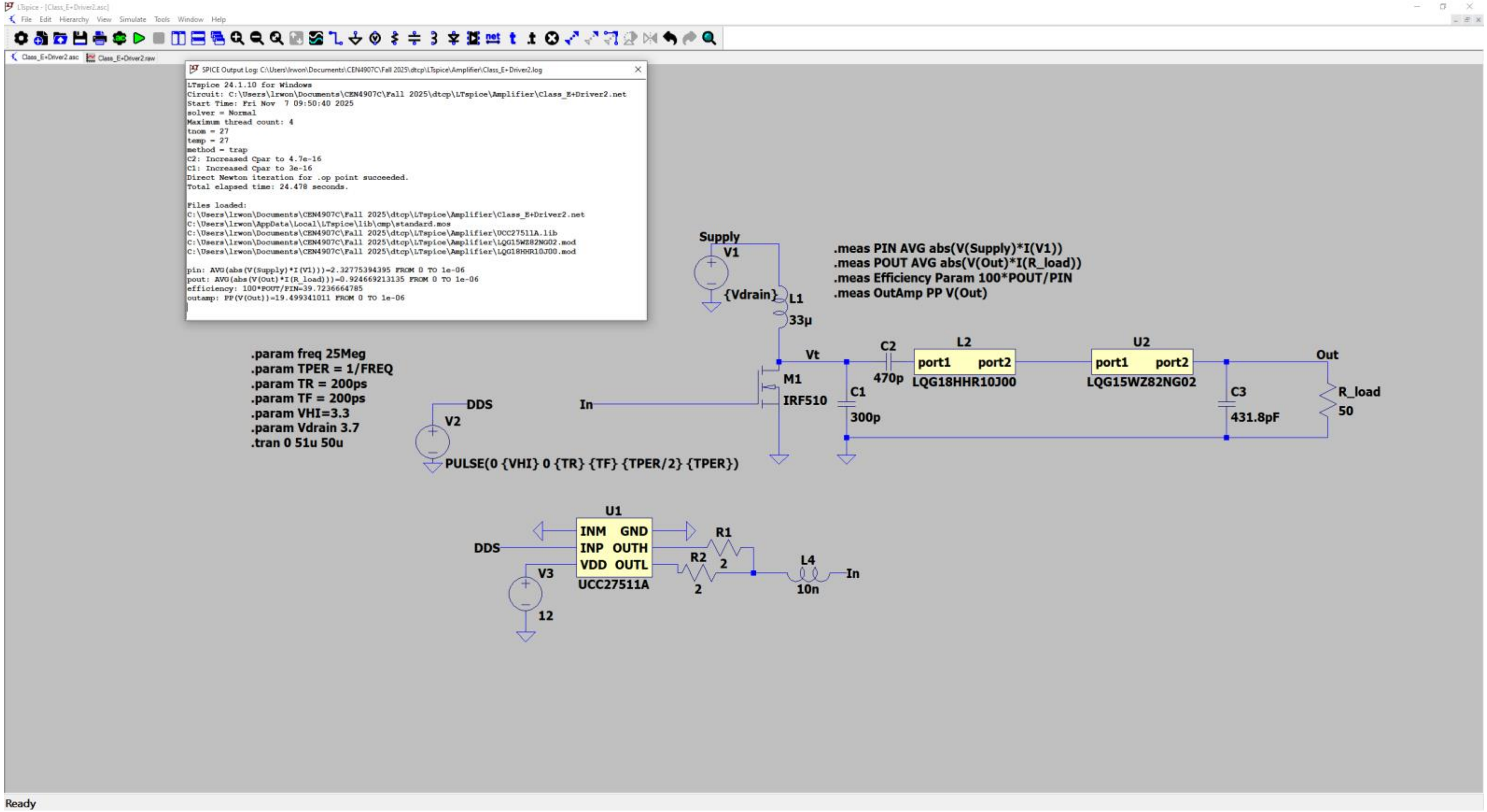
Objective

- Simulate Class-E PA with all real parts.
- If efficiency loss is present find out why.
- Fix it if possible.

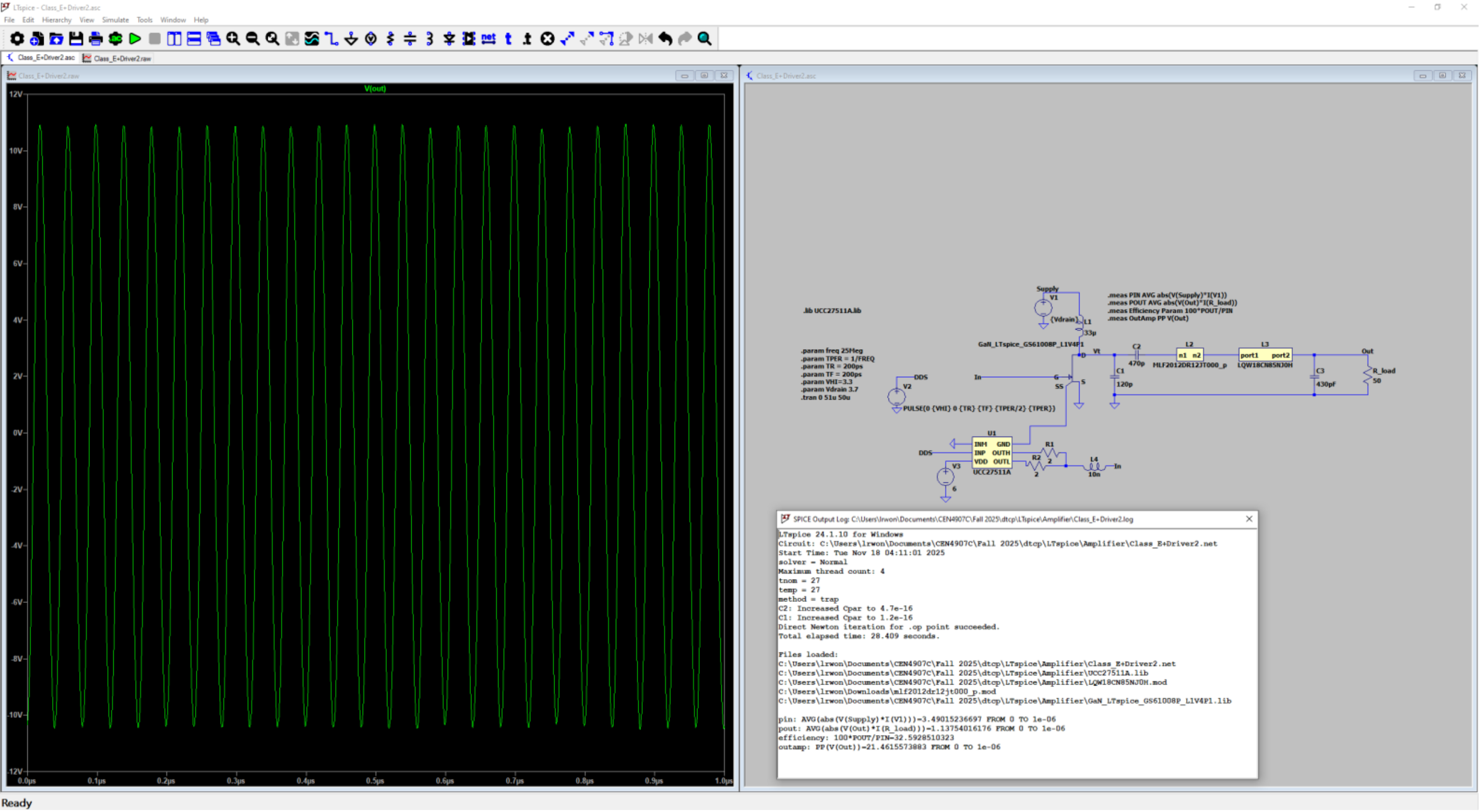
Methods

- Use LTspice and spec sheets.

Results

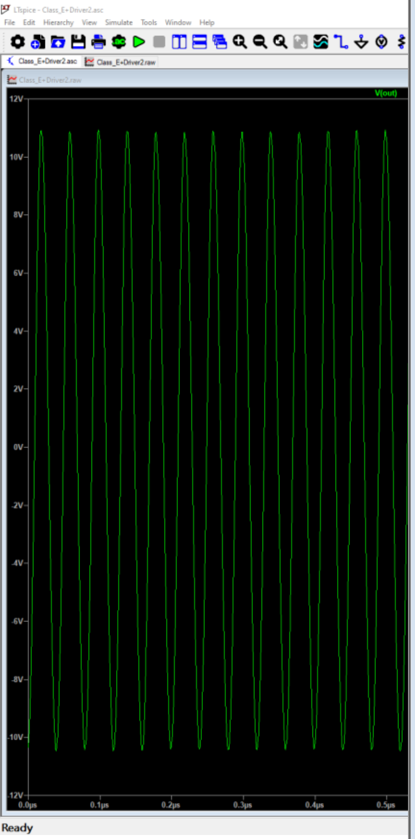

SPICE Output Log: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.log
LTspice 24.1.10 for Windows
Circuit: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
Start Time: Fri Nov 7 09:50:40 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = trap
C2: Increased Cpar to 4.7e-16
C1: Increased Cpar to 3e-16
Direct Newton iteration for .op point succeeded.
Total elapsed time: 24.478 seconds.
Files loaded:
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
C:\Users\lrwon\AppData\Local\LTspice\lib\cmp\standard.mos
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\UCC27511A.lib
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\LQG15WZ82NG02.mod
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\LQG18HHR10J00.mod
pin: AVG(abs(V(Supply)*I(V1)))=2.32775394395 FROM 0 TO 1e-06
pout: AVG(abs(V(Out)*I(R_load)))=0.924669213135 FROM 0 TO 1e-06
efficiency: 100*POUT/PIN=39.7236664785
outamp: PP(V(Out))=19.499341011 FROM 0 TO 1e-06
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
Supply
V1
{Vdrain}
L1
33µ
Vt
C2
470p
L2
port1
port2
LQG18HHR10J00
U2
port1
port2
LQG15WZ82NG02
Out
M1
IRF510
C1
300p
C3
431.8pF
R_load
50
DDS
V2
In
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
U1
INM GND
INP OUTH
VDD OUTL
UCC27511A
DDS
V3
12
R1
2
R2
2
L4
10n
In
Ready

Results

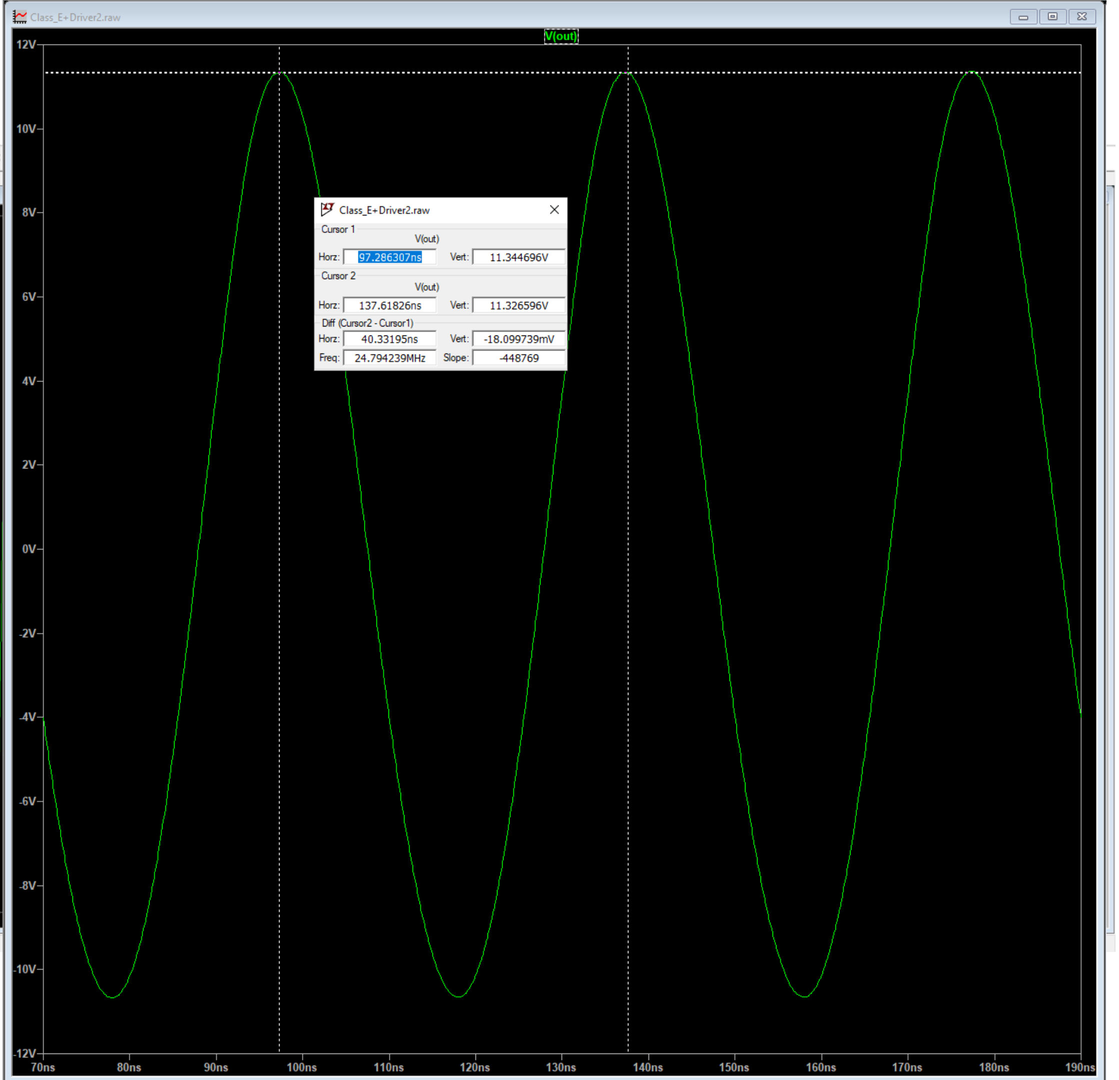


LTspice - Class_E+Driver2.asc
File Edit Hierarchy View Simulate Tools Window Help
Class_E+Driver2.asc
Class_E+Driver2.raw
V(out)
12V
10V
8V
6V
4V
2V
0V
-2V
-4V
-6V
-8V
-10V
-12V
0.0µs
0.1µs
0.2µs
0.3µs
0.4µs
0.5µs
0.6µs
0.7µs
0.8µs
0.9µs
1.0µs
.lib UCC27511A.lib
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
Supply
V1
{Vdrain}
L1
33µ
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
GaN_LTspice_GS61008P_L1V4P1
D
Vt
C2
470p
L2
n1 n2
MLF2012DR12JT000_p
L3
port1 port2
LQW18CN85NJ0H
Out
C1
120p
C3
430pF
R_load
50
DDS
V2
In
G
SS
S
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
U1
INM GND
INP OUTH
VDD OUTL
UCC27511A
DDS
V3
6
R1
2
R2
2
L4
10n
In
SPICE Output Log: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.log
LTspice 24.1.10 for Windows
Circuit: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
Start Time: Tue Nov 18 04:11:01 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = trap
C2: Increased Cpar to 4.7e-16
C1: Increased Cpar to 1.2e-16
Direct Newton iteration for .op point succeeded.
Total elapsed time: 28.409 seconds.
Files loaded:
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\UCC27511A.lib
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\LQW18CN85NJ0H.mod
C:\Users\lrwon\Downloads\mlf2012dr12jt000_p.mod
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\GaN_LTspice_GS61008P_L1V4P1.lib
pin: AVG(abs(V(Supply)*I(V1)))=3.49015236697 FROM 0 TO 1e-06
pout: AVG(abs(V(Out)*I(R_load)))=1.13754016176 FROM 0 TO 1e-06
efficiency: 100*POUT/PIN=32.5928510323
outamp: PP(V(Out))=21.4615573883 FROM 0 TO 1e-06
Ready

Results

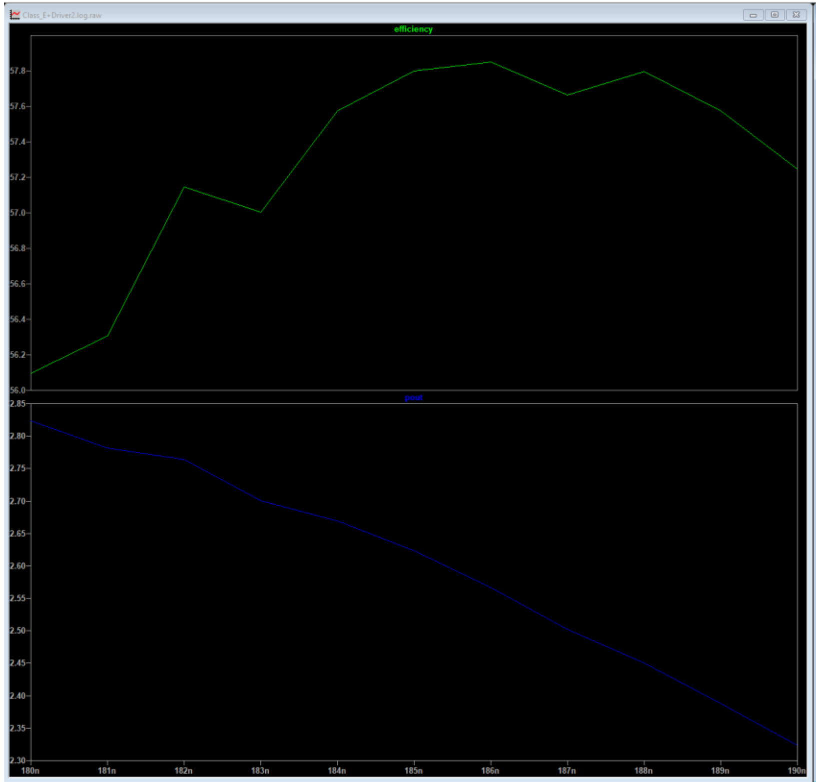


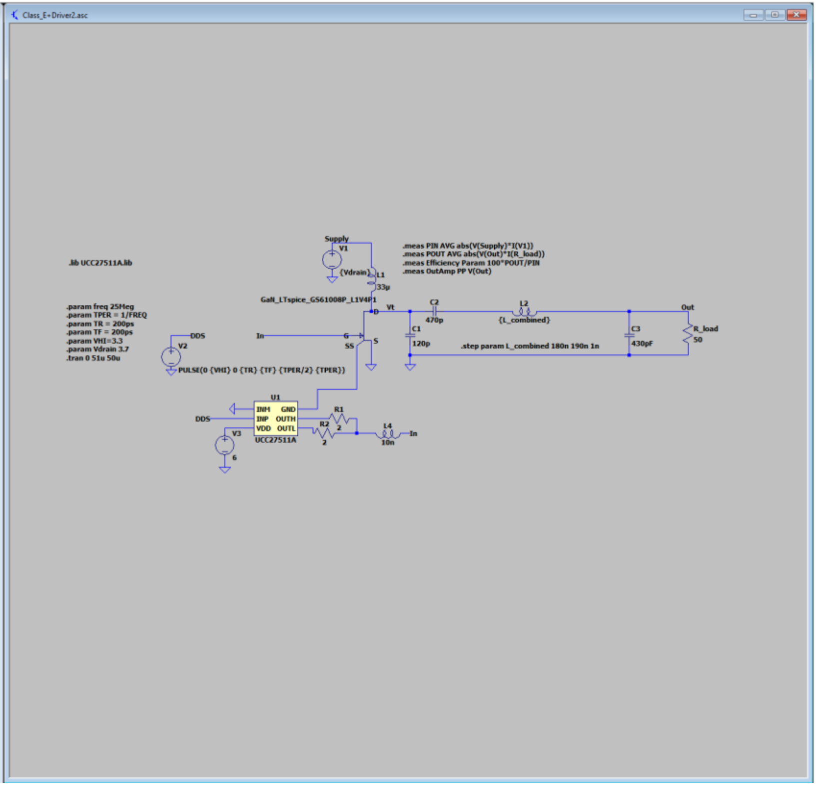
LTspice - Class_E+Driver2.asc
File Edit Hierarchy View Simulate Tools Window Help
Class_E+Driver2.asc
Class_E+Driver2.raw
V(out)
12V
10V
8V
6V
4V
2V
0V
-2V
-4V
-6V
-8V
-10V
-12V
0.0µs
0.1µs
0.2µs
0.3µs
0.4µs
0.5µs
Ready



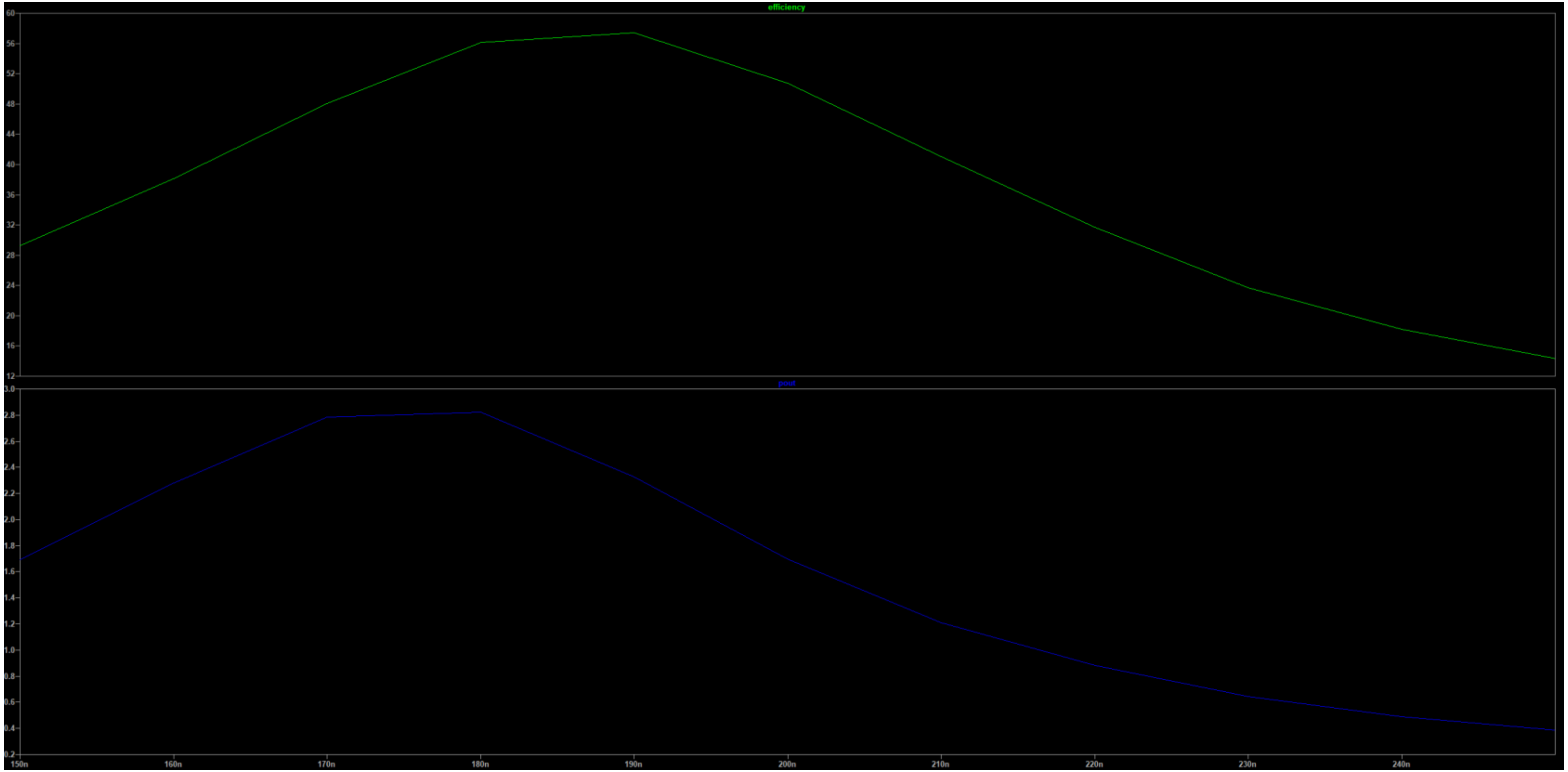
Class_E+Driver2.raw
V(out)
12V
10V
8V
6V
4V
2V
0V
-2V
-4V
-6V
-8V
-10V
-12V
70ns
80ns
90ns
100ns
110ns
120ns
130ns
140ns
150ns
160ns
170ns
180ns
190ns
Class_E+Driver2.raw
Cursor 1
V(out)
Horz: 97.286307ns
Vert: 11.344696V
Cursor 2
V(out)
Horz: 137.61826ns
Vert: 11.326596V
Diff (Cursor2 - Cursor1)
Horz: 40.33195ns
Vert: -18.099739mV
Freq: 24.794239MHz
Slope: -448769

Results

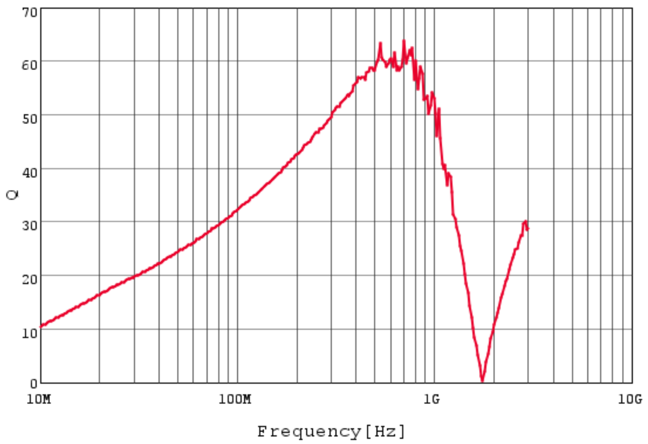

Class_E+Driver2.log.raw
efficiency
57.8
57.6
57.4
57.2
57.0
56.8
56.6
56.4
56.2
56.0
pout
2.85
2.80
2.75
2.70
2.65
2.60
2.55
2.50
2.45
2.40
2.35
2.30
180n
181n
182n
183n
184n
185n
186n
187n
188n
189n
190n


Class_E+Driver2.asc
.lib UCC27511A.lib
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
Supply
V1
{Vdrain}
L1
33µ
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
GaN_LTspice_GS61008P_L1V4P1
D
Vt
C2
470p
L2
{L_combined}
Out
C1
120p
C3
430pF
R_load
50
.step param L_combined 180n 190n 1n
DDS
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
In
G
SS
S
U1
INM
GND
INP
OUTH
VDD
OUTL
UCC27511A
DDS
V3
6
R1
2
R2
2
L4
10n
In

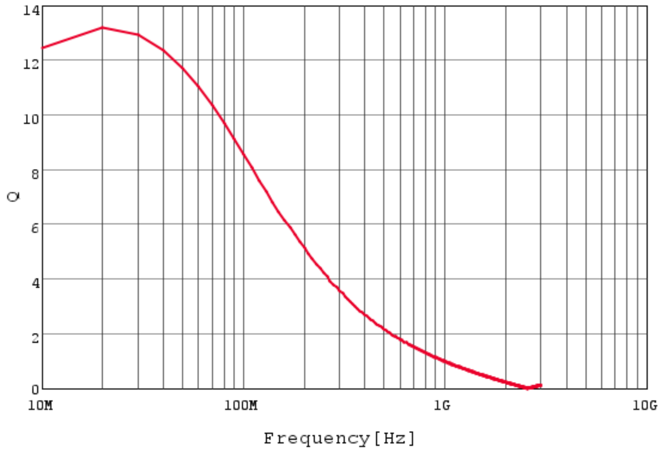
Results


efficiency
60
56
52
48
44
40
36
32
28
24
20
16
12
pout
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
150n
160n
170n
180n
190n
200n
210n
220n
230n
240n

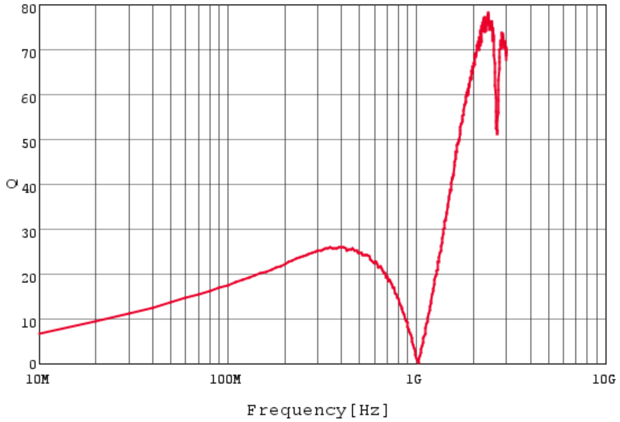
Results



LQW18ANR11G8Z Q by Imp. Analyzer

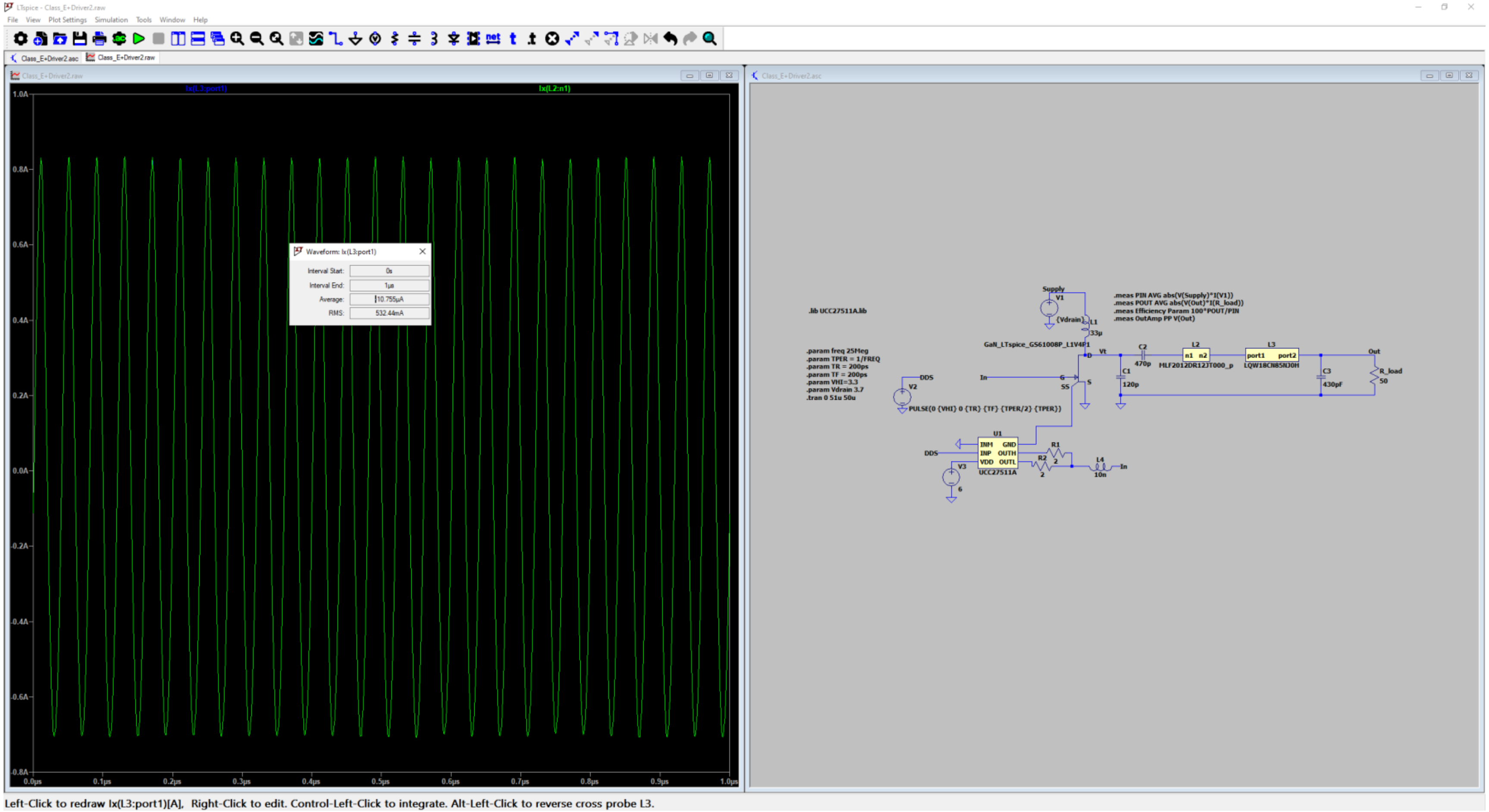


LQW18CN85NJ0H Q



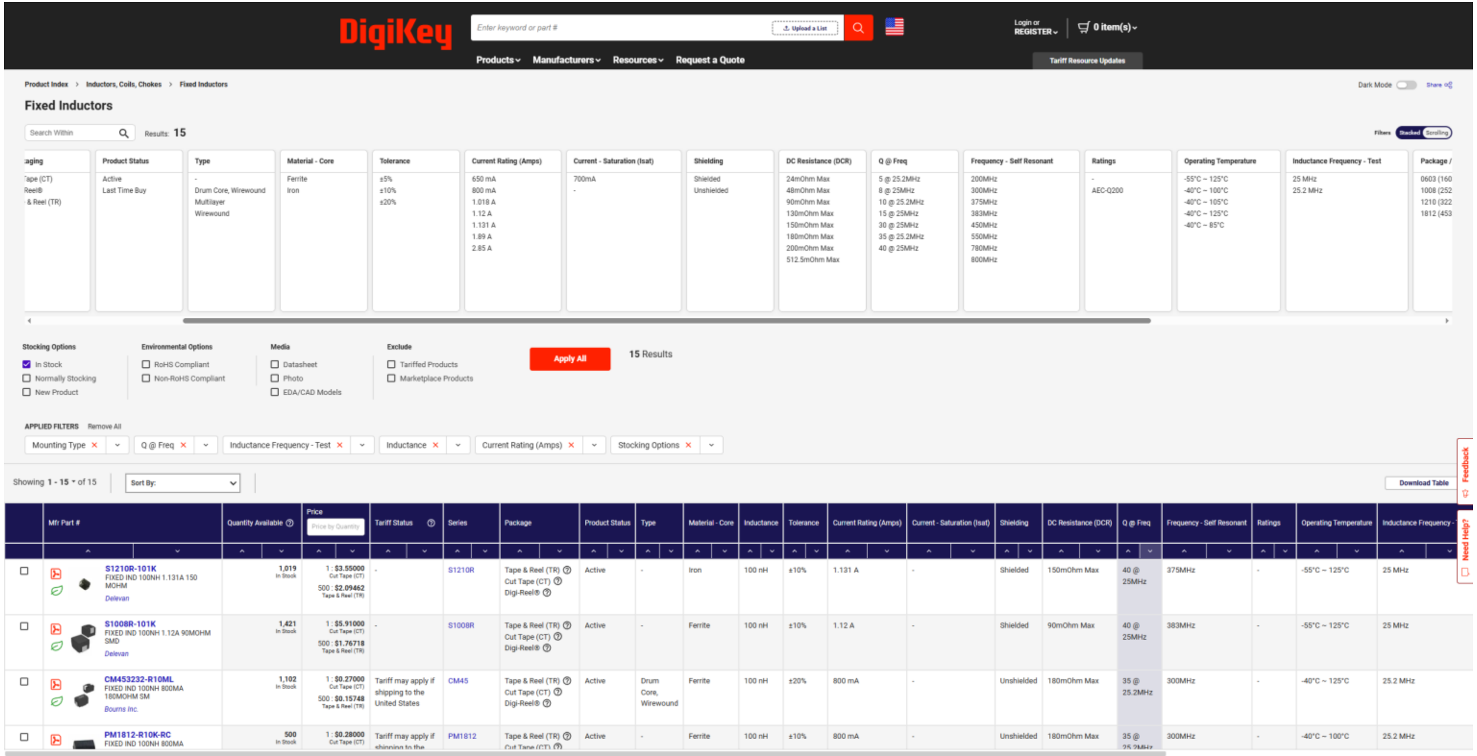
LQG18HHR10J00 Q by Imp. Analyzer

Results



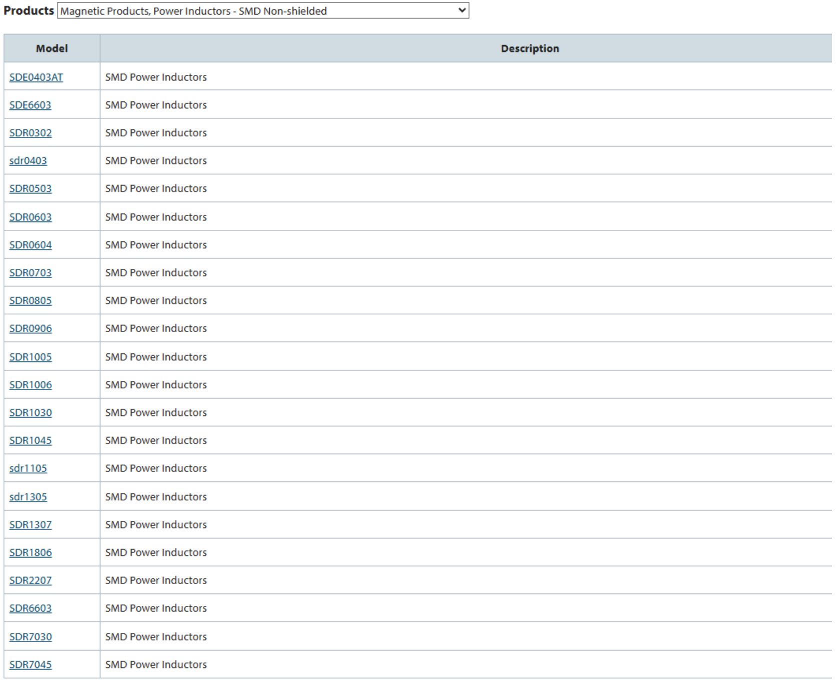
LTspice - Class_E+Driver2.raw
File View Plot Settings Simulation Tools Window Help
Class_E+Driver2.asc
Class_E+Driver2.raw
Ix(L3:port1)
Ix(L2:n1)
Waveform: Ix(L3:port1)
Interval Start: 0s
Interval End: 1µs
Average: 10.755µA
RMS: 532.44mA
.lib UCC27511A.lib
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
GaN_LTspice_GS61008P_L1V4P1
MLF2012DR12JT000_p
LQW18CN85NJ0H
UCC27511A
Left-Click to redraw Ix(L3:port1)[A], Right-Click to edit. Control-Left-Click to integrate. Alt-Left-Click to reverse cross probe L3.

Results



DigiKey
Products
Manufacturers
Resources
Request a Quote
Tariff Resource Updates
Product Index > Inductors, Coils, Chokes > Fixed Inductors
Fixed Inductors
Results: 15
Product Status
Type
Material - Core
Tolerance
Current Rating (Amps)
Current - Saturation (Isat)
Shielding
DC Resistance (DCR)
Q @ Freq
Frequency - Self Resonant
Ratings
Operating Temperature
Inductance Frequency - Test
Stocking Options
Environmental Options
Media
Exclude
Apply All
15 Results
APPLIED FILTERS
Remove All
Mounting Type
Q @ Freq
Inductance Frequency - Test
Inductance
Current Rating (Amps)
Stocking Options
Showing 1 - 15 of 15
Sort By:
Download Table
S1210R-101K
FIXED IND 100NH 1.131A 150 MOHM
Delevan
S1008R-101K
FIXED IND 100NH 1.12A 90MOHM SMD
Delevan
CM453232-R10ML
FIXED IND 100NH 800MA 180MOHM SM
Bourns Inc.
PM1812-R10K-RC
FIXED IND 100NH 800MA

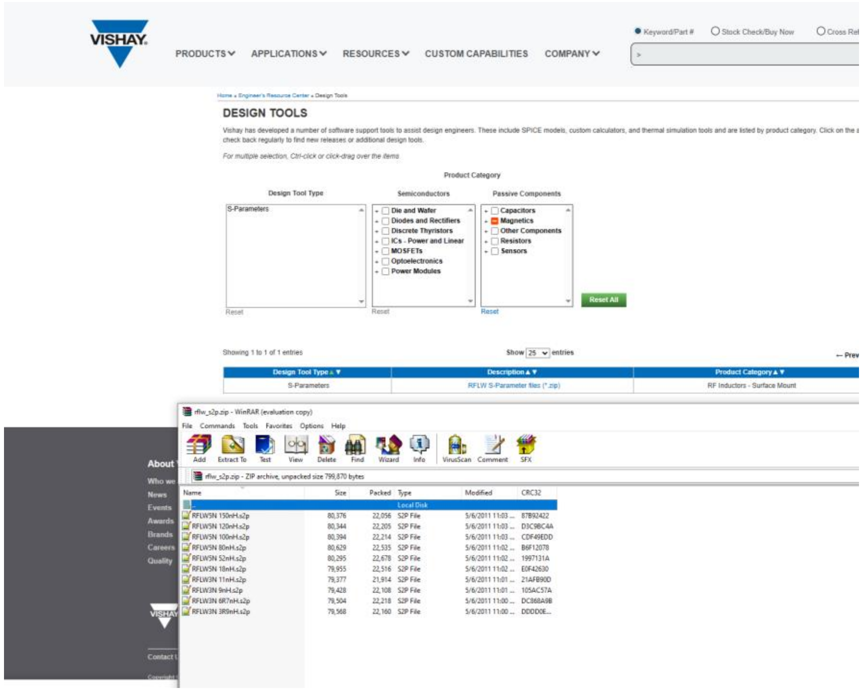
Results



Products Magnetic Products, Power Inductors - SMD Non-shielded

Model	Description
SDE0403AT	SMD Power Inductors
SDE6603	SMD Power Inductors
SDR0302	SMD Power Inductors
sdr0403	SMD Power Inductors
SDR0503	SMD Power Inductors
SDR0603	SMD Power Inductors
SDR0604	SMD Power Inductors
SDR0703	SMD Power Inductors
SDR0805	SMD Power Inductors
SDR0906	SMD Power Inductors
SDR1005	SMD Power Inductors
SDR1006	SMD Power Inductors
SDR1030	SMD Power Inductors
SDR1045	SMD Power Inductors
sdr1105	SMD Power Inductors
sdr1305	SMD Power Inductors
SDR1307	SMD Power Inductors
SDR1806	SMD Power Inductors
SDR2207	SMD Power Inductors
SDR6603	SMD Power Inductors
SDR7030	SMD Power Inductors
SDR7045	SMD Power Inductors

Bourns Website Showing All Design File Listings



Vishay Website Showing All Design File Listings For SMD Inductors

Observation/Conclusion

- The ~30% efficiency we see is probably that low due to low Q of parts at 25MHz.
- Finding inductors that are SMDs, fit our required inductance, are listed with their Q values at 25MHz, and have a readily available spice file is difficult.
    - It seems similar implementations of Class-E Amplifiers make use of hand wound inductors.
- If we would like to test these higher Q inductors, we will need to probably buy them to test them on the physical board.

Assembling PCB

Luis Wong 01/21/2026

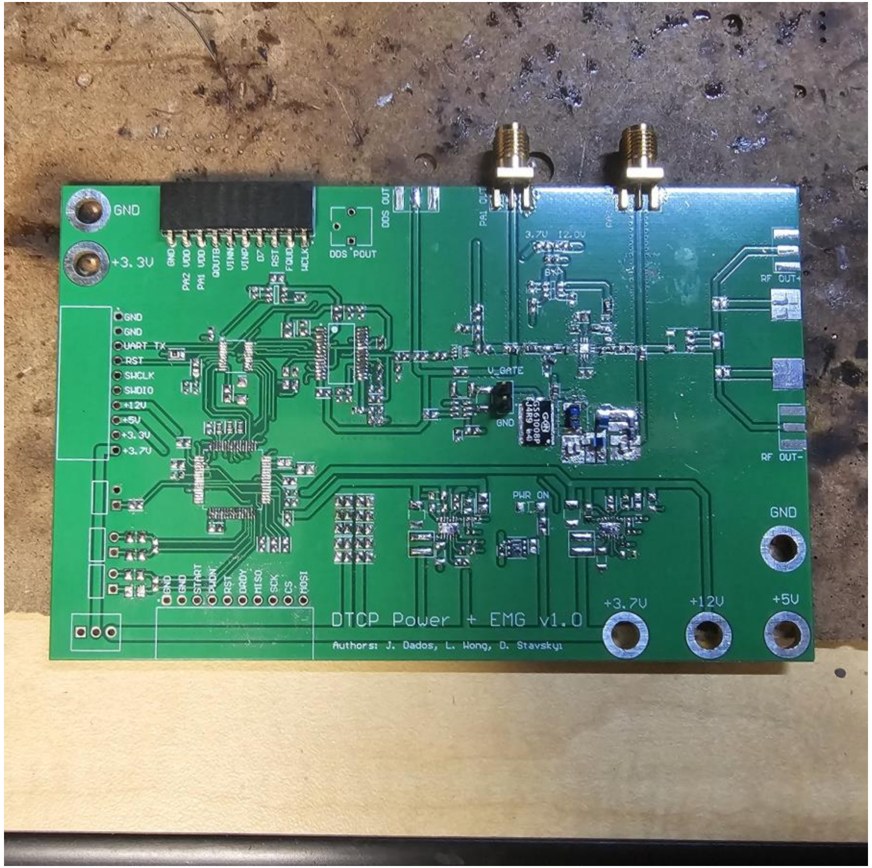
Objective

- Build the Class-E PA
- Test to see if we get the desired output at 30dBm.

Methods

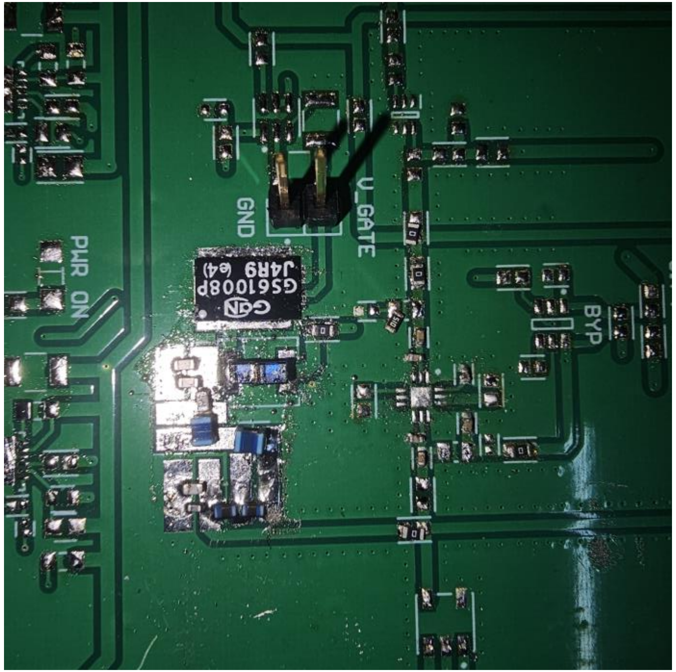
- Use spectrum analyzer, PSU, and assembled PCB to test whether the output is about 30dBm at 25MHz.

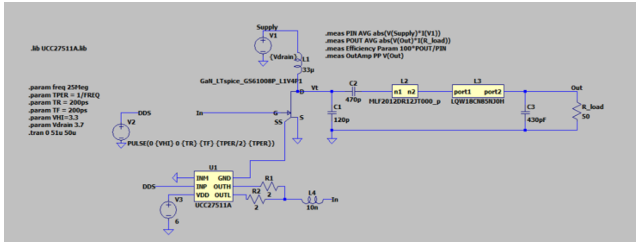
Results



DTCP Power + EMG v1.0
Authors: J. Dados, L. Wong, D. Stavsky
GND
+3.3V
DDS POUT
DDS OUT
PA1 OUT
RF OUT+
RF OUT-
V_GATE
PWR ON
+3.7V
+12V
+5V

Results

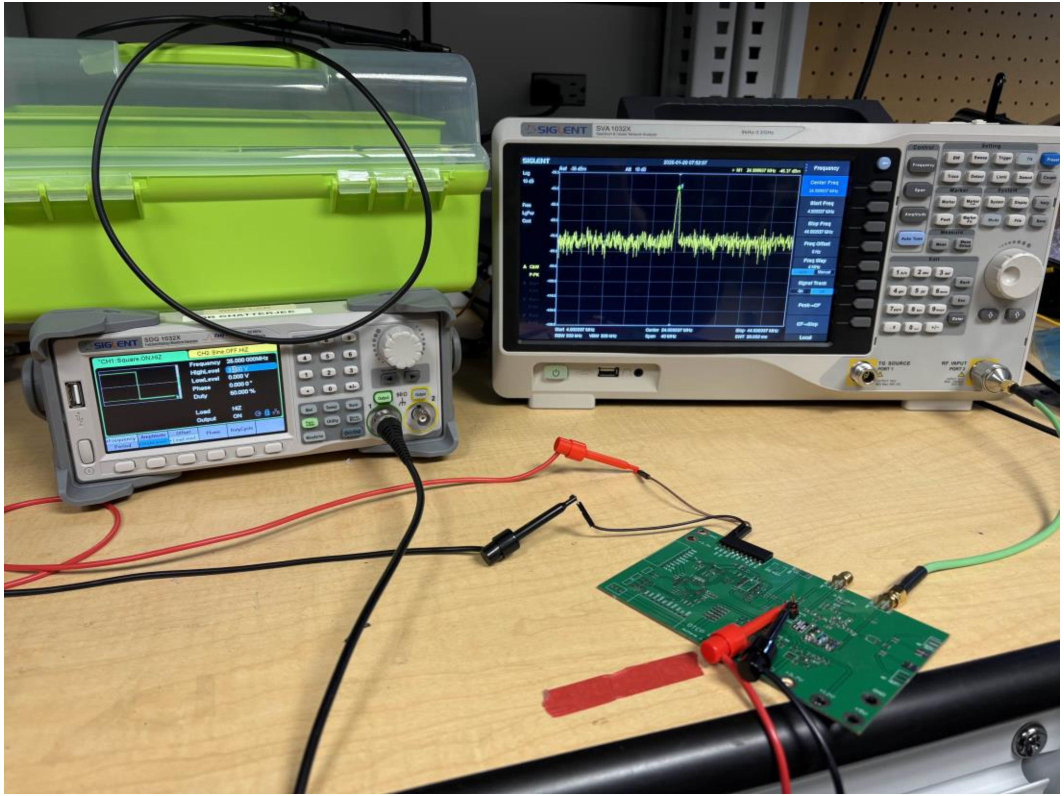




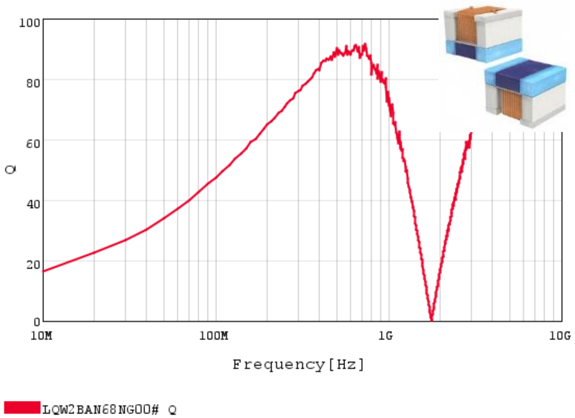
We made use of parts on hand to approximate the values for the capacitors and inductors used in simulation.

- Drain inductor used was 1.5uH.
- Approximated 120pF to 112pF using two 56pF capacitors in parallel.
- Approximated a 120nH and 85nH in series with two 68nH and two 39nH inductors in series.
- Used a single 470pF capacitor and a 330pF and 100pF capacitor in parallel to make a 430pF equivalent capacitor.

Results



Results (Checking Inductors Used)

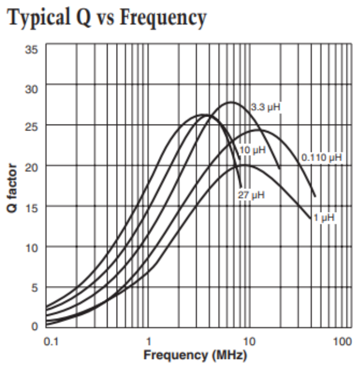


Datasheet for the 68NH 1.25A 200mOHM SMD

Electrical Properties:

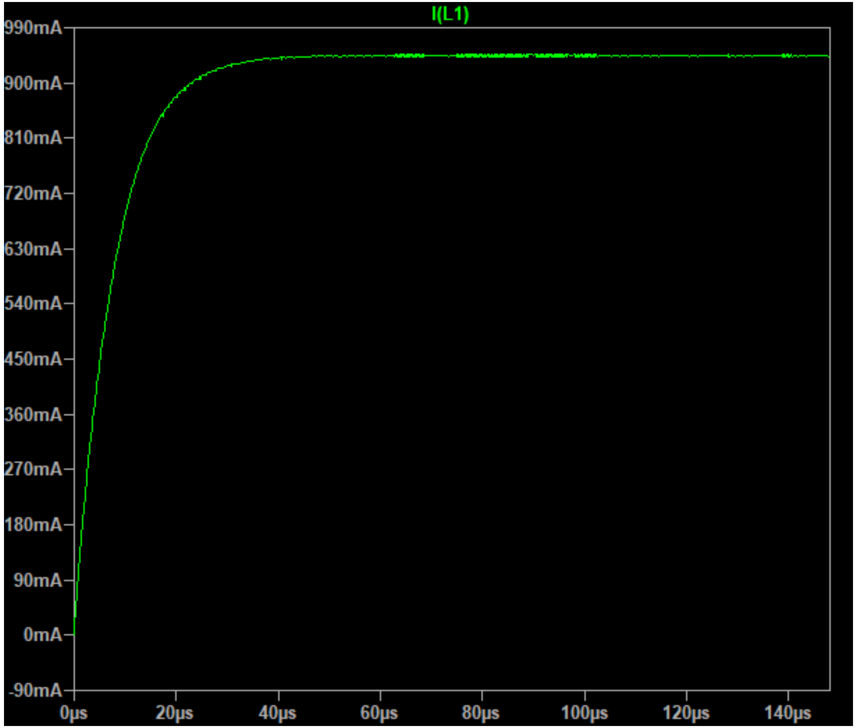
Properties	Test conditions		Value	Unit	Tol.
Inductance	250 MHz	L	39	nH	±5%
Q-Factor	250 MHz	$Q_{min.}$	36		min.
DC Resistance	@ 20 °C		0.21	Ω	max.
Rated Current	ΔT = 15 K	I_R	600	mA	max.
Self Resonant Frequency		f_{res}	2200	MHz	min.

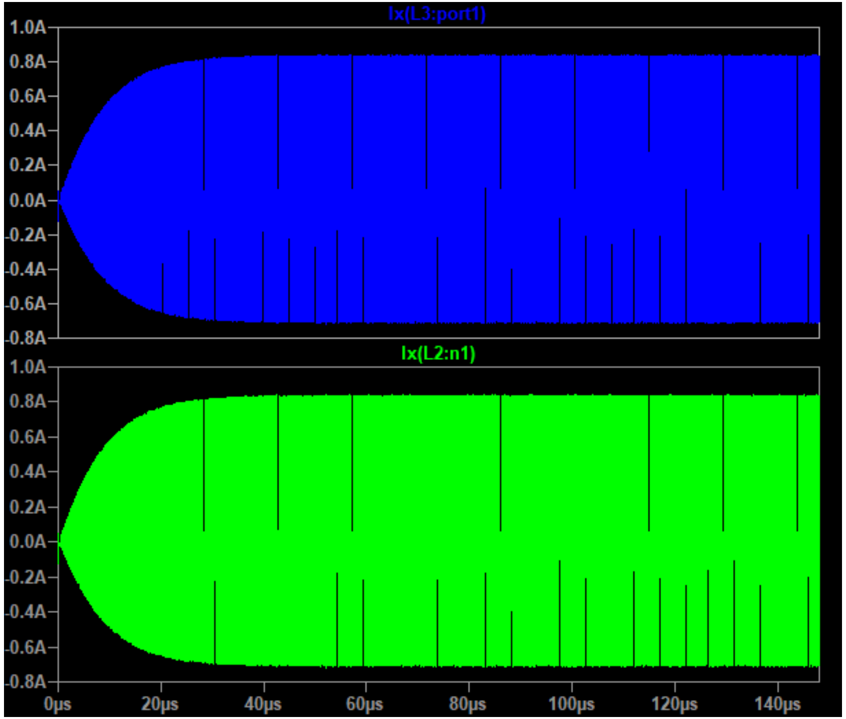
Datasheet for the 744761139C: 39NH 600MA 210MOHM SMD



Datasheet for the 0805LS-152XJRC: 1.5UH 490MA 210MOHM SMD

Results (Theoretical Currents)





Results (Checking Capacitors Used)

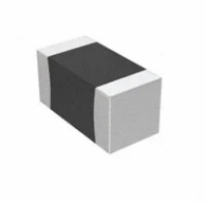


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

CC0402KRX7R9BB331

DigiKey Part Number	311-1027-2-ND - Tape & Reel (TR) 311-1027-1-ND - Cut Tape (CT) 311-1027-6-ND - Digi-Reel®
Manufacturer	YAGEO
Manufacturer Product Number	CC0402KRX7R9BB331
Description	CAP CER 330PF 50V X7R 0402
Manufacturer Standard Lead Time	16 Weeks
Customer Reference	
Detailed Description	330 pF ±10% 50V Ceramic Capacitor X7R 0402 (1005 Metric)
Datasheet	Datasheet
EDA/CAD Models	CC0402KRX7R9BB331 Models

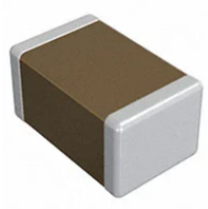


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

GRM1885C1H560JA01D

DigiKey Part Number	490-1421-2-ND - Tape & Reel (TR) 490-1421-1-ND - Cut Tape (CT) 490-1421-6-ND - Digi-Reel®
Manufacturer	Murata Electronics
Manufacturer Product Number	GRM1885C1H560JA01D
Description	CAP CER 56PF 50V C0G/NP0 0603
Customer Reference	
Detailed Description	56 pF ±5% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)
Datasheet	Datasheet
EDA/CAD Models	GRM1885C1H560JA01D Models

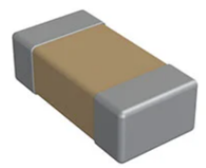


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

C0603C101K4GACTU

DigiKey Part Number	399-17390-2-ND - Tape & Reel (TR) 399-17390-1-ND - Cut Tape (CT) 399-17390-6-ND - Digi-Reel®
Manufacturer	KEMET
Manufacturer Product Number	C0603C101K4GACTU
Description	CAP CER 100PF 16V C0G/NP0 0603
Manufacturer Standard Lead Time	14 Weeks
Customer Reference	
Detailed Description	100 pF ±10% 16V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)
Datasheet	Datasheet
EDA/CAD Models	C0603C101K4GACTU Models

Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

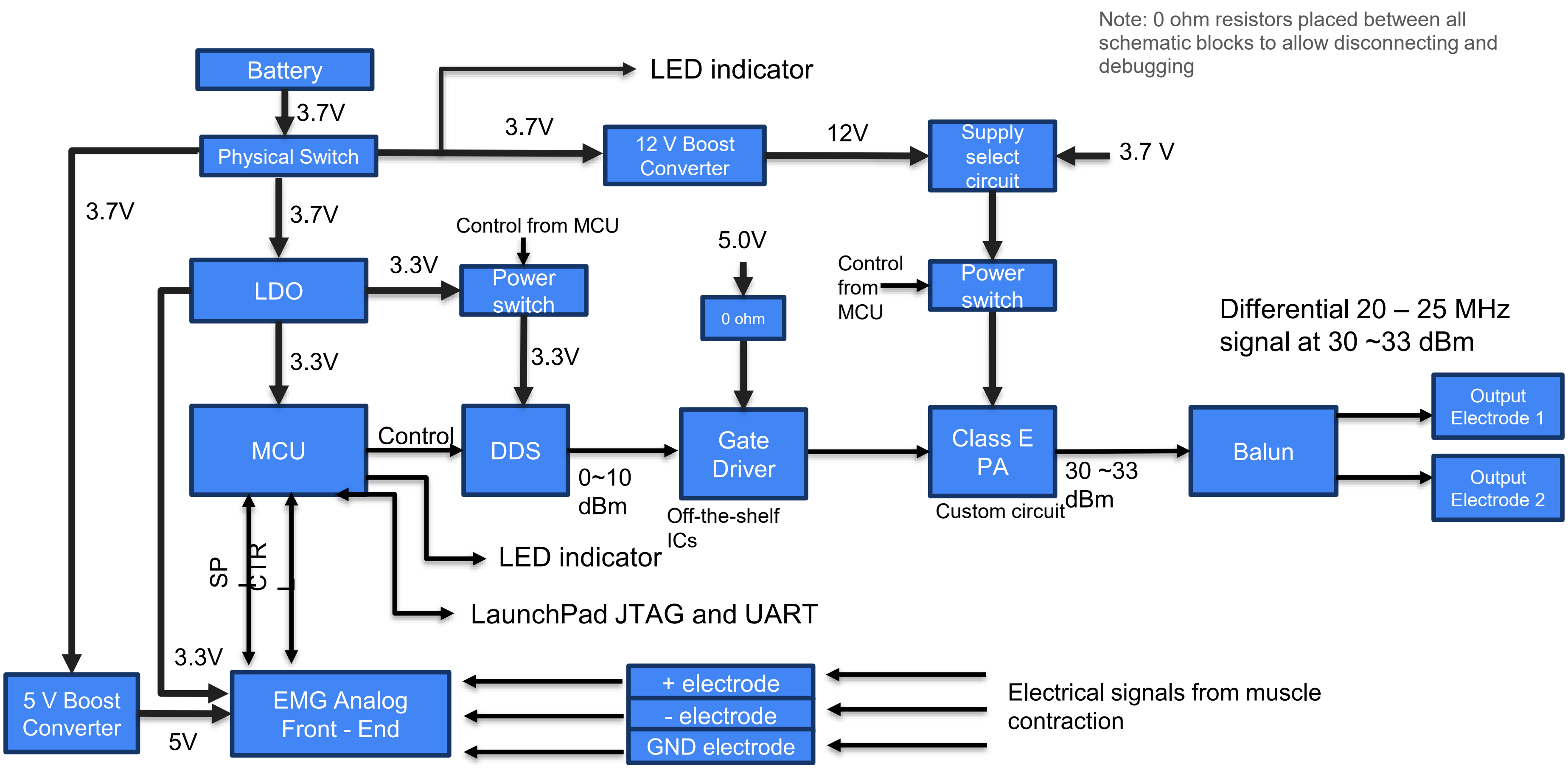
VJ0603D471KXXAJ

DigiKey Part Number	720-1335-2-ND - Tape & Reel (TR) 720-1335-1-ND - Cut Tape (CT) 720-1335-6-ND - Digi-Reel®
Manufacturer	Vishay Vitramon
Manufacturer Product Number	VJ0603D471KXXAJ
Description	CAP CER 470PF 25V C0G/NP0 0603
Manufacturer Standard Lead Time	15 Weeks
Customer Reference	
Detailed Description	470 pF ±10% 25V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)
Datasheet	Datasheet
EDA/CAD Models	VJ0603D471KXXAJ Models

Observation/Conclusion

- We need to weigh the cost of troubleshooting this design versus dropping it in favor of the Class-A PA design that already has results.
- If we do continue trying to get a result out of the Class-E PA we may need to buy new parts (capacitors and inductors) with sufficient Q factors at the frequencies we plan to use the device in to avoid the large loss in power we experienced in our initial tests.
- We double check current rating of SMDs used to make sure they can handle about 1A to be safe.
- We can modify the design to increase the voltage at drain to compensate for lower current rating of SMDs.

Fabricated system architecture



Testing Class-E PA

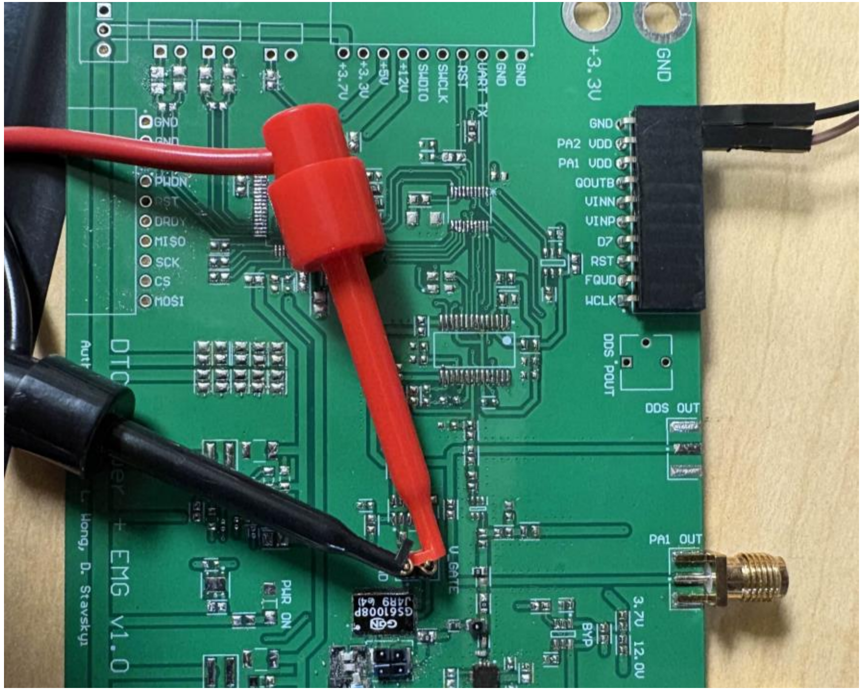
Luis Wong 01/28/2026

Objective

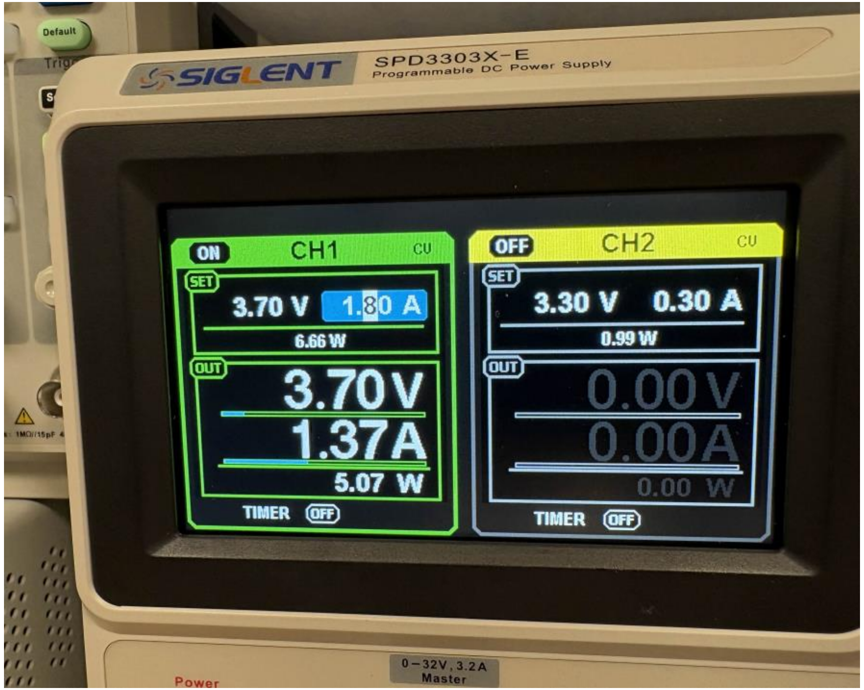
- Sweep frequencies of Class-E PA to check which frequencies the current design on the board works best with.

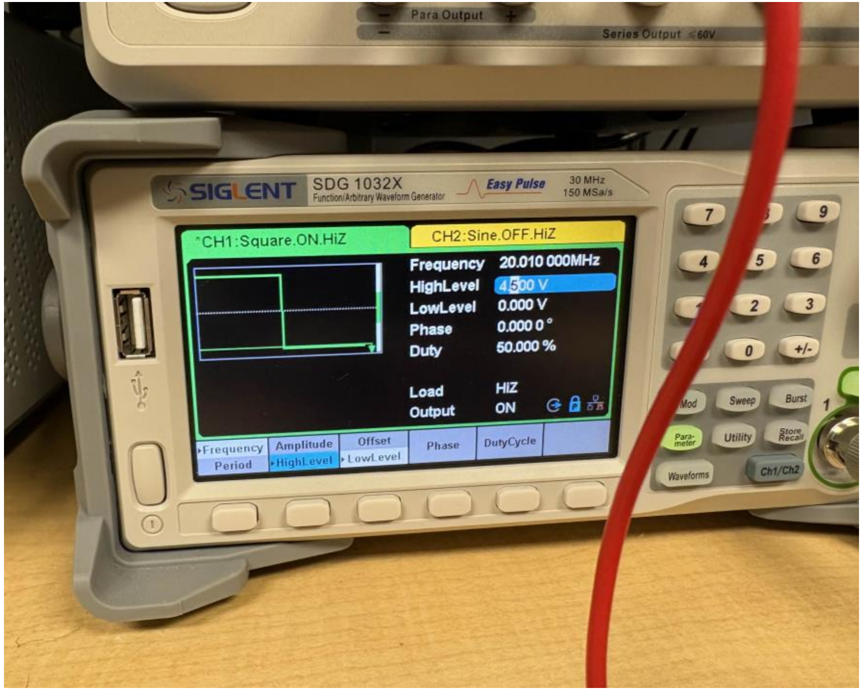
Methods

- Use spectrum analyzer, PSU, signal generator, and assembled PCB to test output power at different frequencies.



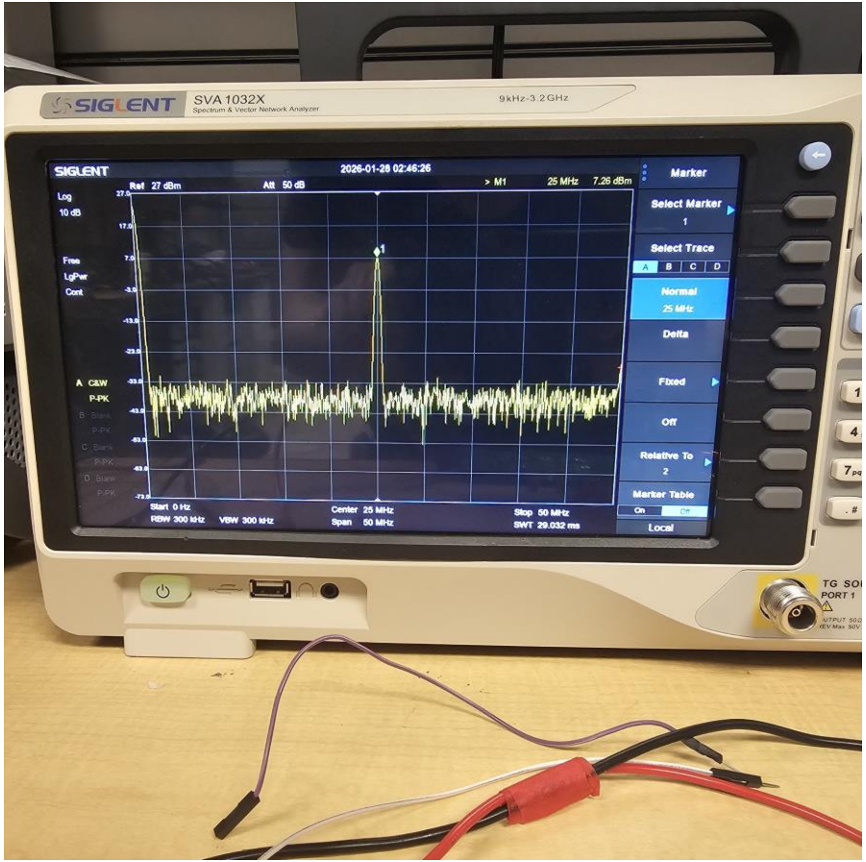
Results


SIGLENT
SPD3303X-E
Programmable DC Power Supply
Default
ON
CH1
CU
SET
3.70 V
1.80 A
6.66 W
OUT
3.70V
1.37A
5.07 W
TIMER
OFF
OFF
CH2
CU
SET
3.30 V
0.30 A
0.99 W
OUT
0.00V
0.00A
0.00 W
TIMER
OFF
0-32V, 3.2A
Master
Power


Para Output
Series Output ≤60V
SIGLENT
SDG 1032X
Function/Arbitrary Waveform Generator
Easy Pulse
30 MHz
150 MSa/s
CH1:Square.ON.HiZ
CH2:Sine.OFF.HiZ
Frequency
20.010 000MHz
HighLevel
4.500 V
LowLevel
0.000 V
Phase
0.000 0 °
Duty
50.000 %
Load
HiZ
Output
ON
Frequency
Period
Amplitude
HighLevel
Offset
LowLevel
Phase
DutyCycle
7
9
4
5
6
2
3
0
+/-
Mod
Sweep
Burst
Para-meter
Utility
Store Recall
Waveforms
Ch1/Ch2

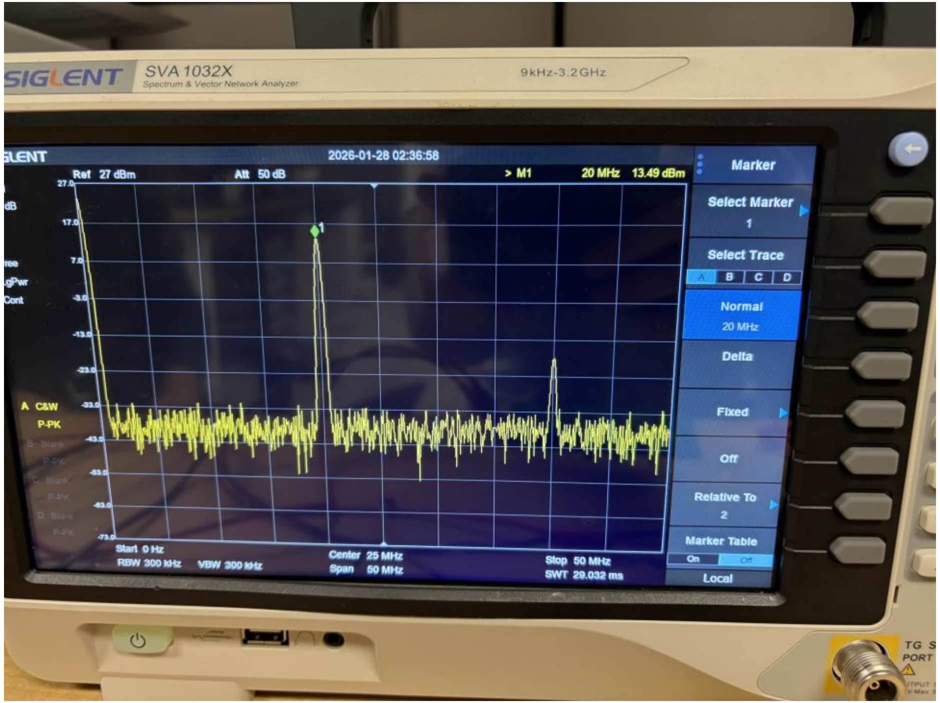
Results

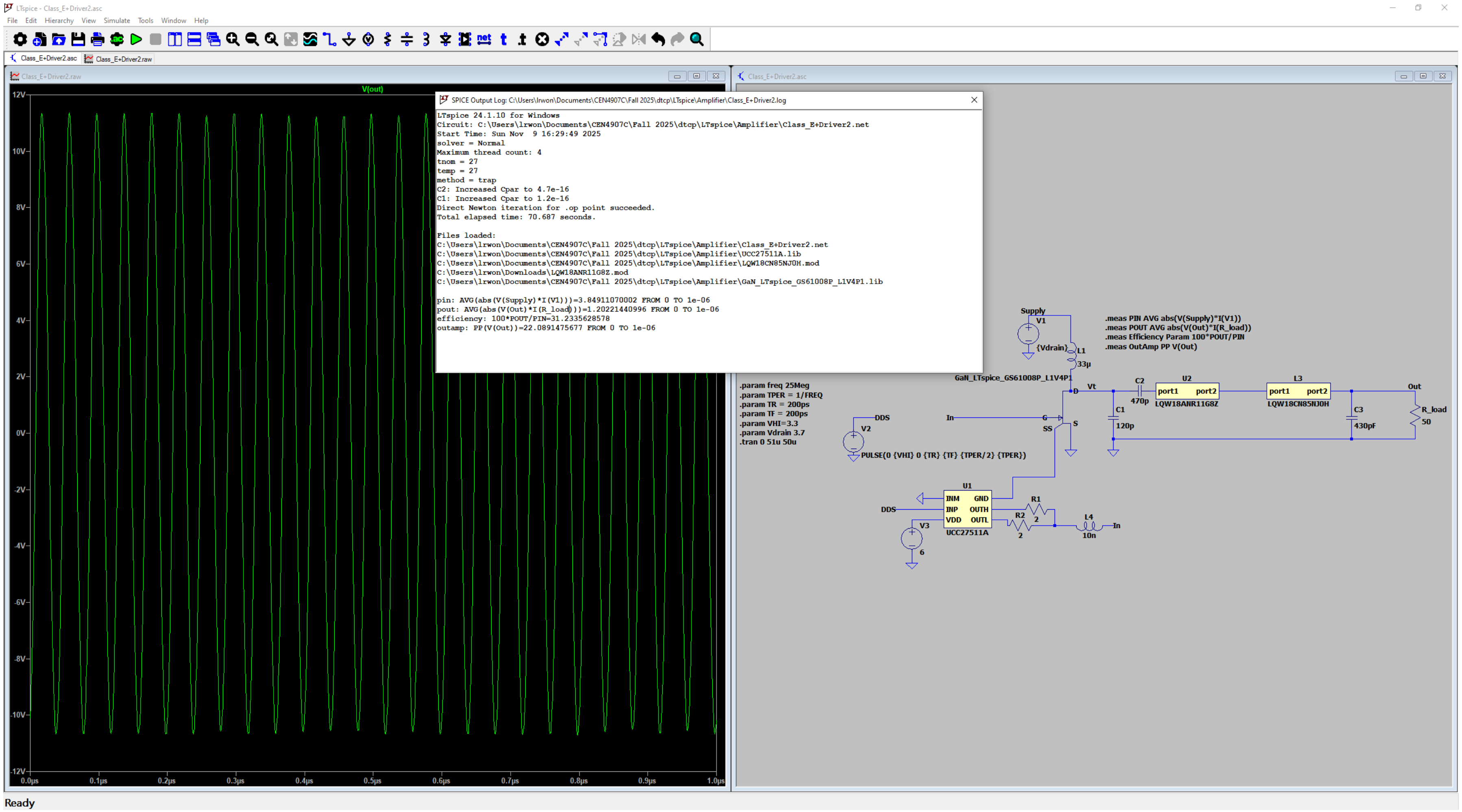
- At drain we have the same 1.5 uH inductor as before made use of four total with 2 pairs of them in series then connected in parallel.
- At 25 MHz after this change we are now getting 7.26 dBm instead of the negative 40 dBm we got before using a 0-4.5V square wave.
- When the signal at the gate is modified to a 0 to 4 V square wave with 3.3V or 5V at drain we get about negative 2 dBm. We seem to be getting the best results in terms of power output using a 0V - 4.5V square wave.



Results

- When sweeping frequencies, it becomes evident that the power output of the device peaks at around 18-20 MHz.
- Without changing the setup of using 3.7V at drain and a 0-4.5V square wave at the PA gave 13.5 dBm at 20 MHz.
- After increasing the drain voltage to 5V we saw that the signal amplified further to 15dBm but we had to stop there as we were worried we would damage the SMAs if we continued.




LTspice - Class_E+Driver2.asc
File Edit Hierarchy View Simulate Tools Window Help
Class_E+Driver2.asc
Class_E+Driver2.raw
V(out)
12V
10V
8V
6V
4V
2V
0V
-2V
-4V
-6V
-8V
-10V
-12V
0.0µs 0.1µs 0.2µs 0.3µs 0.4µs 0.5µs 0.6µs 0.7µs 0.8µs 0.9µs 1.0µs
SPICE Output Log: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.log
LTspice 24.1.10 for Windows
Circuit: C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
Start Time: Sun Nov 9 16:29:49 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = trap
C2: Increased Cpar to 4.7e-16
C1: Increased Cpar to 1.2e-16
Direct Newton iteration for .op point succeeded.
Total elapsed time: 70.687 seconds.
Files loaded:
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\Class_E+Driver2.net
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\UCC27511A.lib
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\LQW18CN85NJ0H.mod
C:\Users\lrwon\Downloads\LQW18ANR11G8Z.mod
C:\Users\lrwon\Documents\CEN4907C\Fall 2025\dtcp\LTspice\Amplifier\GaN_LTspice_GS61008P_L1V4P1.lib
pin: AVG(abs(V(Supply)*I(V1)))=3.84911070002 FROM 0 TO 1e-06
pout: AVG(abs(V(Out)*I(R_load)))=1.20221440996 FROM 0 TO 1e-06
efficiency: 100*POUT/PIN=31.2335628578
outamp: PP(V(Out))=22.0891475677 FROM 0 TO 1e-06
Supply
V1
{Vdrain}
L1
33µ
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
GaN_LTspice_GS61008P_L1V4P1
.param freq 25Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
D
Vt
C2
470p
U2
port1 port2
LQW18ANR11G8Z
L3
port1 port2
LQW18CN85NJ0H
Out
R_load
50
C1
120p
C3
430pF
DDS
V2
In
G
SS
S
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
U1
INM GND
INP OUTH
VDD OUTL
UCC27511A
DDS
V3
6
R1
2
R2
2
L4
10n
In
Ready

Observation/Conclusion

- With some modification to the design using 4 inductors at the drain instead of one we got better results. It seems we may just need to tune the rest of the circuit to get better results.
- Due to time constraints Jerimiah has opted to go with the two stage Class A PA as we already have it working and meeting specification requirements (30dBm).

Next Steps

- I have installed Code Composer Studio and tested its functionality with Dmytro.
- I will now be working on helping Dmytro with the Firmware and contributing to the second design with Jerimiah with Altium.

Simulating Class-E PA

Luis Wong 02/04/2026

Objective

- Simulate the design on the board using its parts and sweep frequencies.

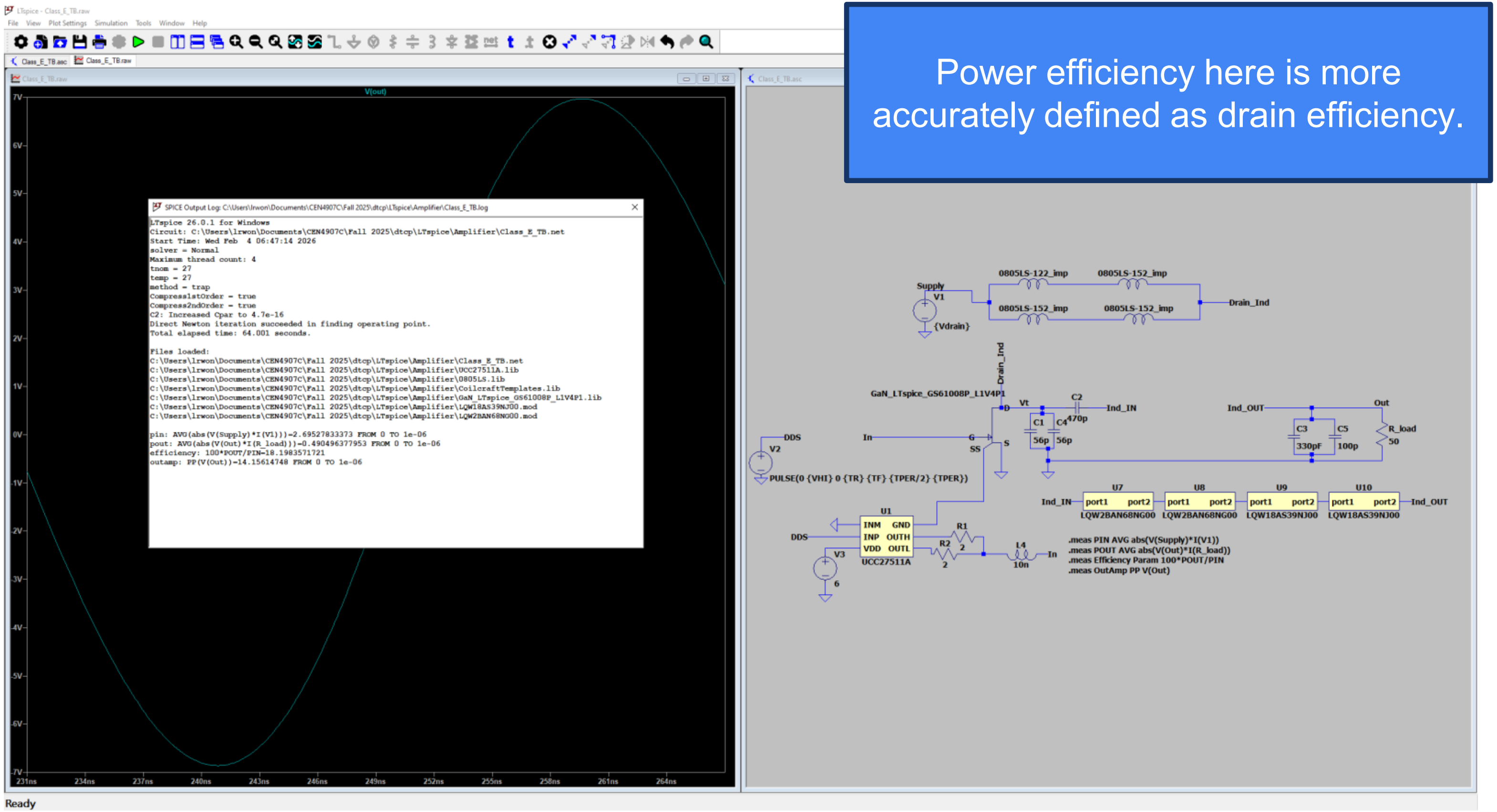
Methods

- LTspice

Target Specifications

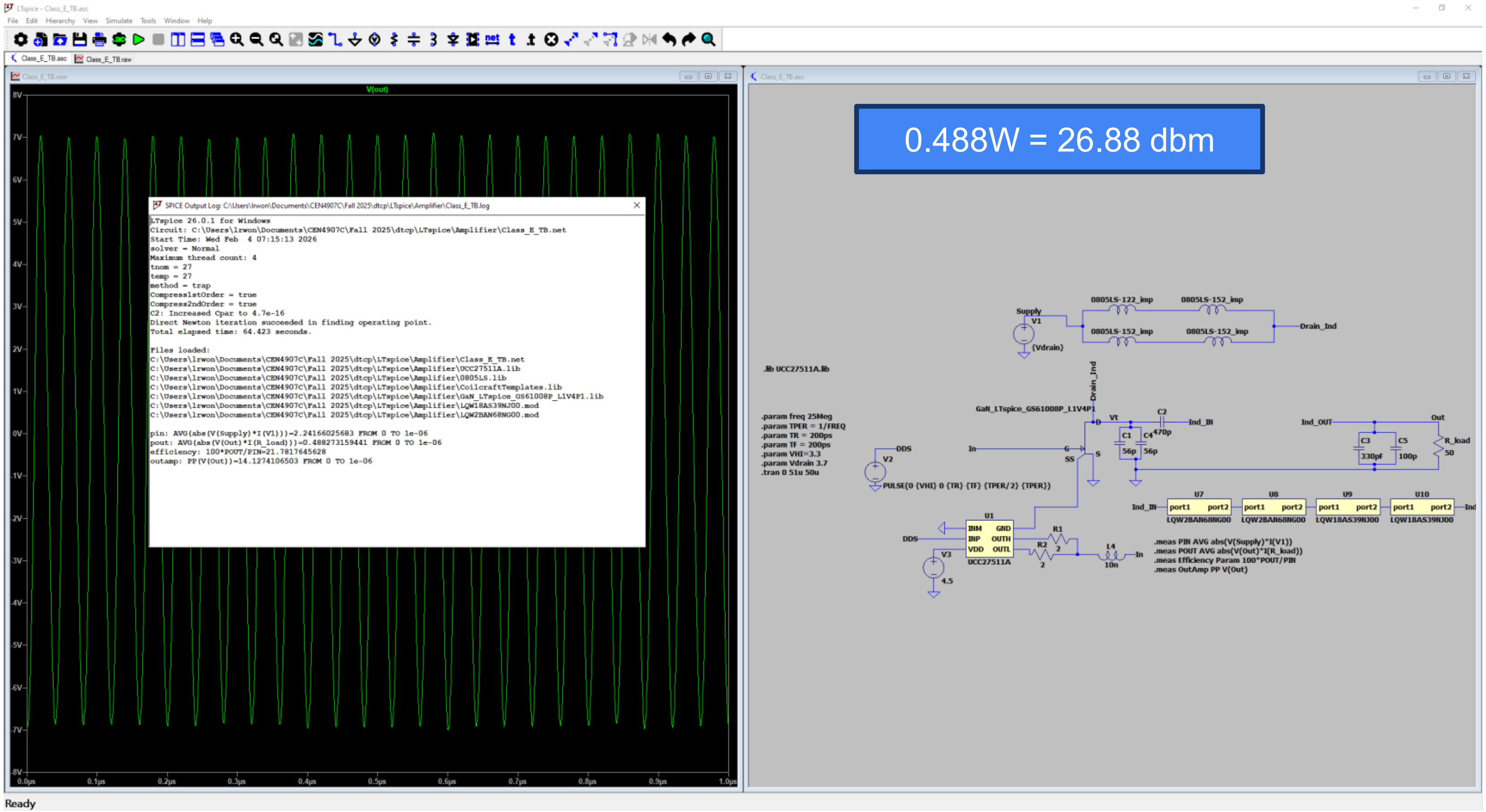
Performance metric	Target value	Measured value as of 1/28/2026	Explanation
Output frequency	25 MHz	25 MHz	Measured with an RF spectrum analyzer.
Output power	≥30 dBm differential (1W)	7.26 dBm (0.0053 W)	Measured with an RF spectrum analyzer.
Static current	<10 mA	0 A (without power switches or driver)	Current consumed when the RF signal is not being transmitted. Class E PAs drive the transistor fully on and off like a switch. If the switch is off no DC path from supply to ground.
Tx current	<500 mA	1.37 A	Current consumed when the RF signal is being transmitted.

Results (6V square wave)

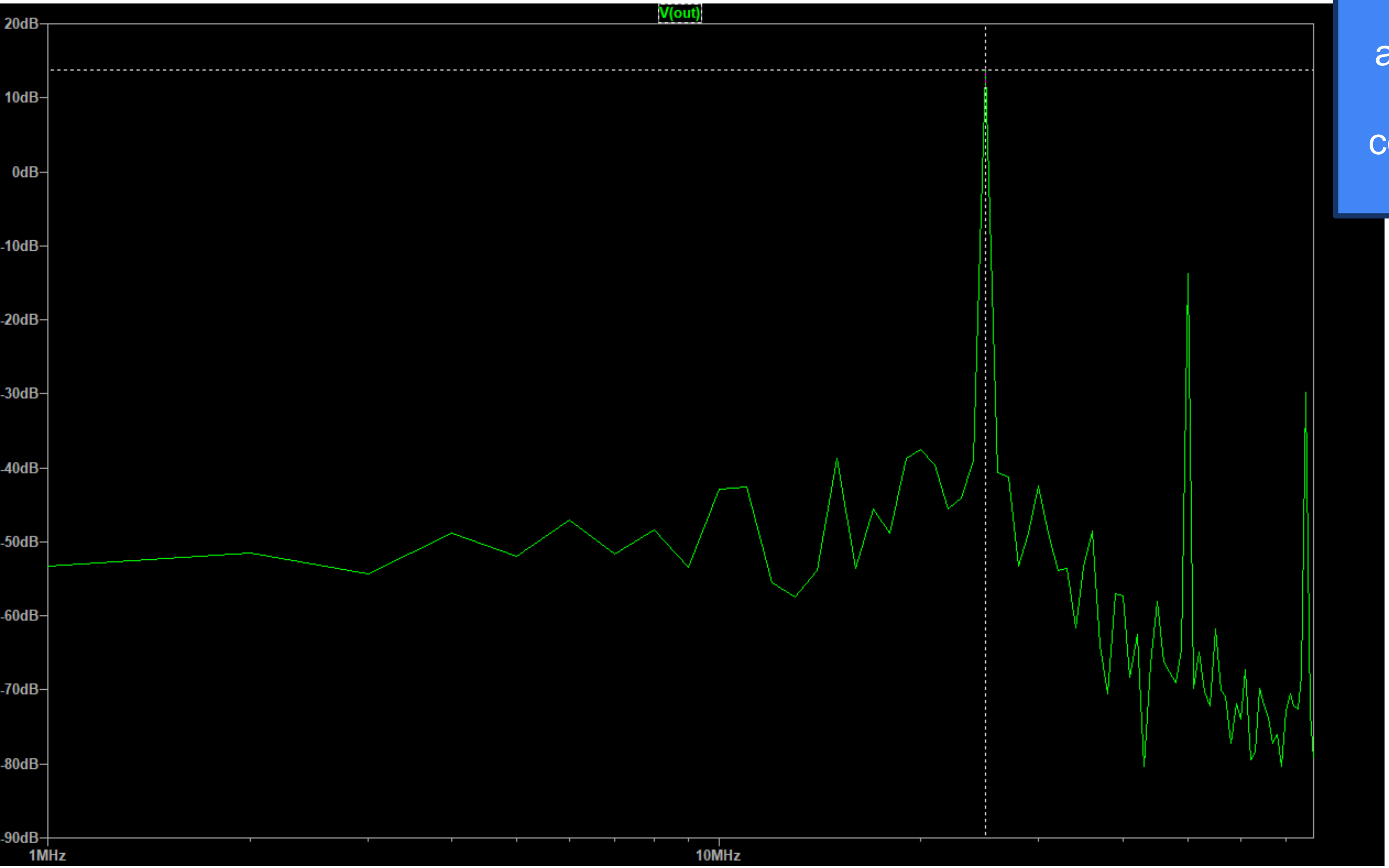


Power efficiency here is more accurately defined as drain efficiency.

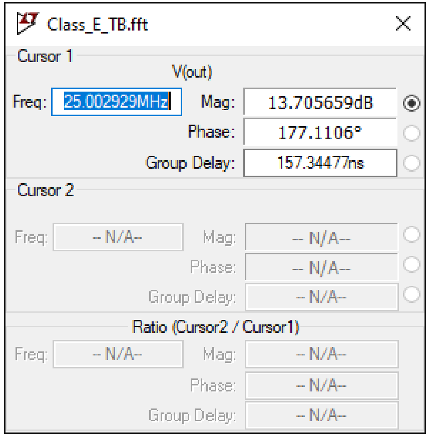
Results (4.5V square wave)

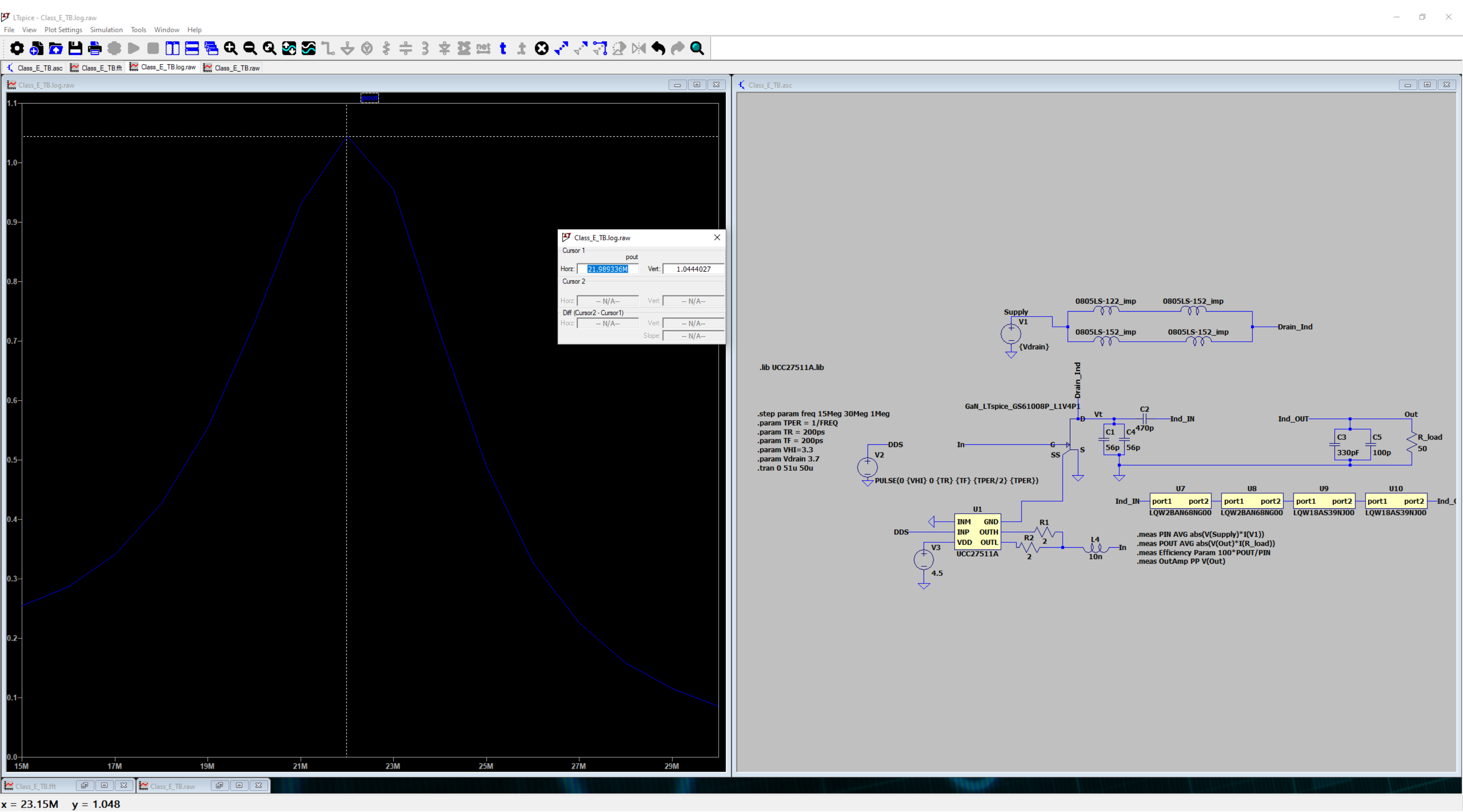


Results (4.5V square wave FFT)



Since I use transient analysis I calculated power using .meas POUT AVG abs(V(Out)*I(R_load)). This FFT done on the voltage confirms the signal is mostly 25 MHz.





LTspice - Class_E_TB.log.raw
File View Plot Settings Simulation Tools Window Help
Class_E_TB.asc
Class_E_TB.fft
Class_E_TB.log.raw
Class_E_TB.raw
pout
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
15M
17M
19M
21M
23M
25M
27M
29M
Cursor 1
Horz: 21.989336M
Vert: 1.0444027
Cursor 2
Diff (Cursor2 - Cursor1)
Slope:
-- N/A--
Supply
V1
{Vdrain}
0805LS-122_imp
0805LS-152_imp
0805LS-152_imp
0805LS-152_imp
Drain_Ind
.lib UCC27511A.lib
.step param freq 15Meg 30Meg 1Meg
.param TPER = 1/FREQ
.param TR = 200ps
.param TF = 200ps
.param VHI=3.3
.param Vdrain 3.7
.tran 0 51u 50u
GaN_LTspice_GS61008P_L1V4P1
Vt
C2
470p
Ind_IN
Ind_OUT
Out
DDS
V2
PULSE(0 {VHI} 0 {TR} {TF} {TPER/2} {TPER})
In
G
SS
S
D
C1
56p
C4
56p
C3
330pF
C5
100p
R_load
50
U7 U8 U9 U10
port1 port2
LQW2BAN68NG00
LQW2BAN68NG00
LQW18AS39NJ00
LQW18AS39NJ00
U1
INM GND
INP OUTH
VDD OUTL
UCC27511A
V3
4.5
R1
2
R2
2
L4
10n
.meas PIN AVG abs(V(Supply)*I(V1))
.meas POUT AVG abs(V(Out)*I(R_load))
.meas Efficiency Param 100*POUT/PIN
.meas OutAmp PP V(Out)
x = 23.15M y = 1.048

Results (Calculating Theoretical Gain @ 25 MHz)

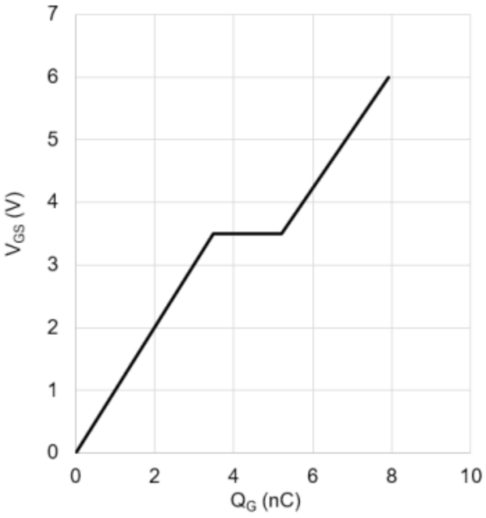
$$P_{in} = Q_G \cdot V_{GS} \cdot f$$

$$P_{in} = \left(6.3 \times 10^{-9}\,\mathrm{C}\right) \cdot (4.5\,\mathrm{V}) \cdot \left(25 \times 10^{6}\,\mathrm{Hz}\right) = 0.709\ \mathrm{Watts}$$

$$Gain = \frac{Pout}{Pin} \qquad Gain = \frac{0.488W}{0.709W} = 0.688$$

$$G_{db} = 10 * \log_{10} 0.688 = -0.162$$

Gate Charge, Q_G Characteristic



GaN Systems

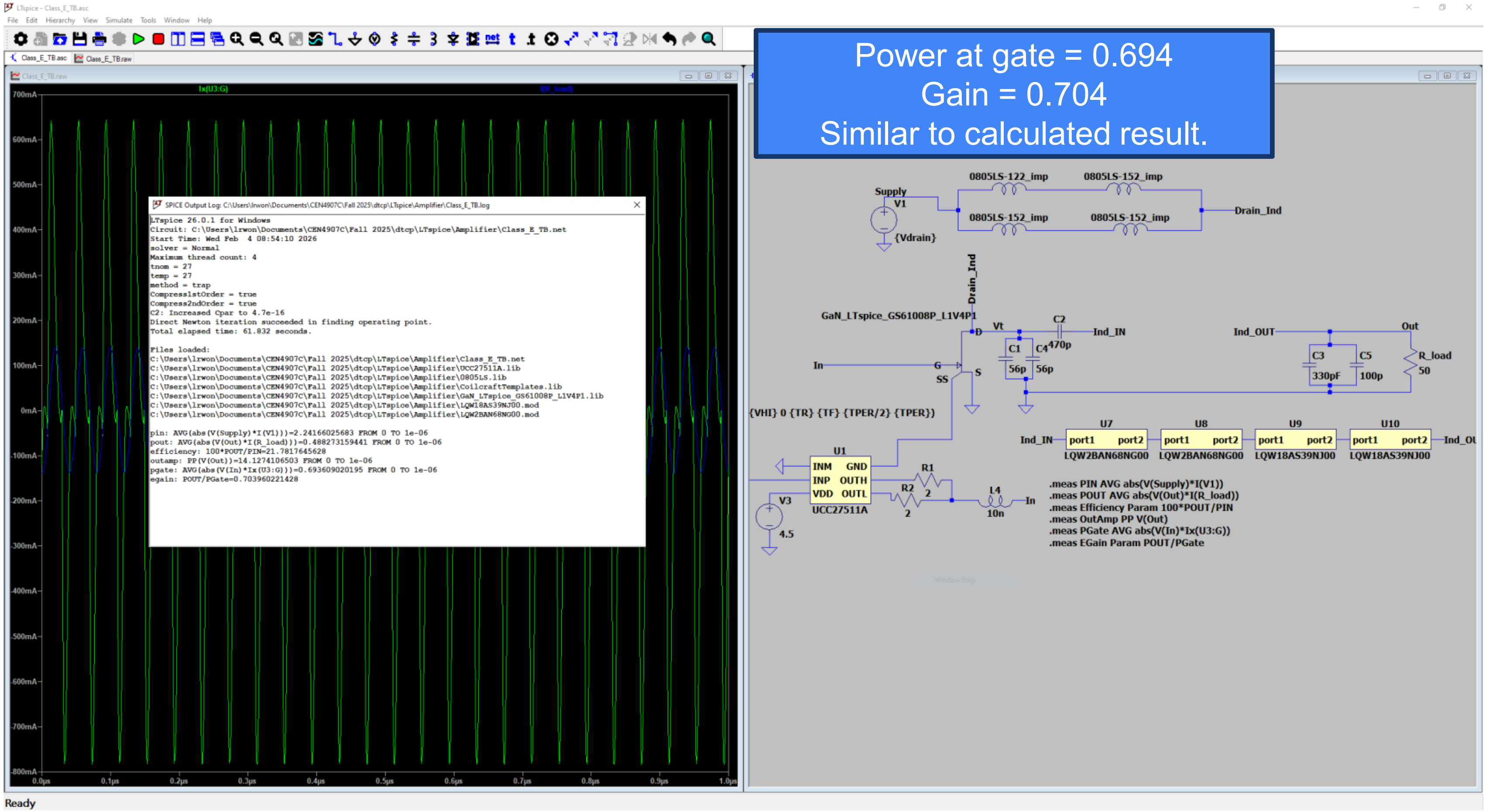
GS61008P
Bottom-side cooled 100 V E-mode GaN transistor
Datasheet

Electrical Characteristics (Typical values at T_J = 25 °C, V_{GS} = 6 V unless otherwise noted)

Parameters	Sym.	Min.	Typ.	Max.	Units	Conditions
Drain-to-Source Blocking Voltage	$V_{(BL)DSS}$	100			V	V_{GS} = 0 V, I_{DSS} = 50 µA
Drain-to-Source On Resistance	$R_{DS(on)}$		7	9.5	mΩ	V_{GS} = 6 V, T_J = 25 °C I_{DS} = 27 A
Drain-to-Source On Resistance	$R_{DS(on)}$		17.5		mΩ	V_{GS} = 6 V, T_J = 150 °C I_{DS} = 27 A
Gate-to-Source Threshold	$V_{GS(th)}$	1.1	1.7	2.6	V	V_{DS} = V_{GS}, I_{DS} = 7 mA
Gate-to-Source Current	I_{GS}		200		µA	V_{GS} = 6 V, V_{DS} = 0 V
Gate Plateau Voltage	V_{plat}		3.5		V	V_{DS} = 50 V, I_{DS} = 90 A
Drain-to-Source Leakage Current	I_{DSS}		0.5	50	µA	V_{DS} = 100 V, V_{GS} = 0 V T_J = 25 °C
Drain-to-Source Leakage Current	I_{DSS}		100		µA	V_{DS} = 100 V, V_{GS} = 0 V T_J = 150 °C
Internal Gate Resistance	R_G		0.8		Ω	f = 5 MHz, open drain
Input Capacitance	C_{ISS}		600		pF	V_{DS} = 50 V V_{GS} = 0 V f = 100 kHz
Output Capacitance	C_{OSS}		250		pF	
Reverse Transfer Capacitance	C_{RSS}		12		pF	
Effective Output Capacitance, Energy Related (Note 4)	$C_{O(ER)}$		302		pF	V_{GS} = 0 V V_{DS} = 0 to 50 V
Effective Output Capacitance, Time Related (Note 5)	$C_{O(TR)}$		385		pF	
Total Gate Charge	Q_G		8		nC	V_{GS} = 0 to 6 V V_{DS} = 50 V I_{DS} = 90 A
Gate-to-Source Charge	Q_{GS}		3.5		nC	
Gate threshold charge	$Q_{G(th)}$		1.9		nC	
Gate switching charge	$Q_{G(sw)}$		3.3		nC	
Gate-to-Drain Charge	Q_{GD}		1.7		nC	
Output Charge	Q_{OSS}		20		nC	V_{GS} = 0 V, V_{DS} = 50 V
Reverse Recovery Charge	Q_{RR}		0		nC	

Results (LTspice Gain Calc)

Power at gate = 0.694
Gain = 0.704
Similar to calculated result.



Observation/Conclusion

- In the simulation it show that at 25MHz we should expect a theoretical power output of 26.88 dBm or 0.488W.
- Power output changes to be about 30 dBm or 1W when we change the frequency at the gate to 22 MHz.
- Gain seems to be low which may disqualify the Class-E PA as a viable alternative with the current design found on the board.

Next Steps

- To be determined by Jerimiah and Han.

UART Firmware

Luis Wong 02/25/2026

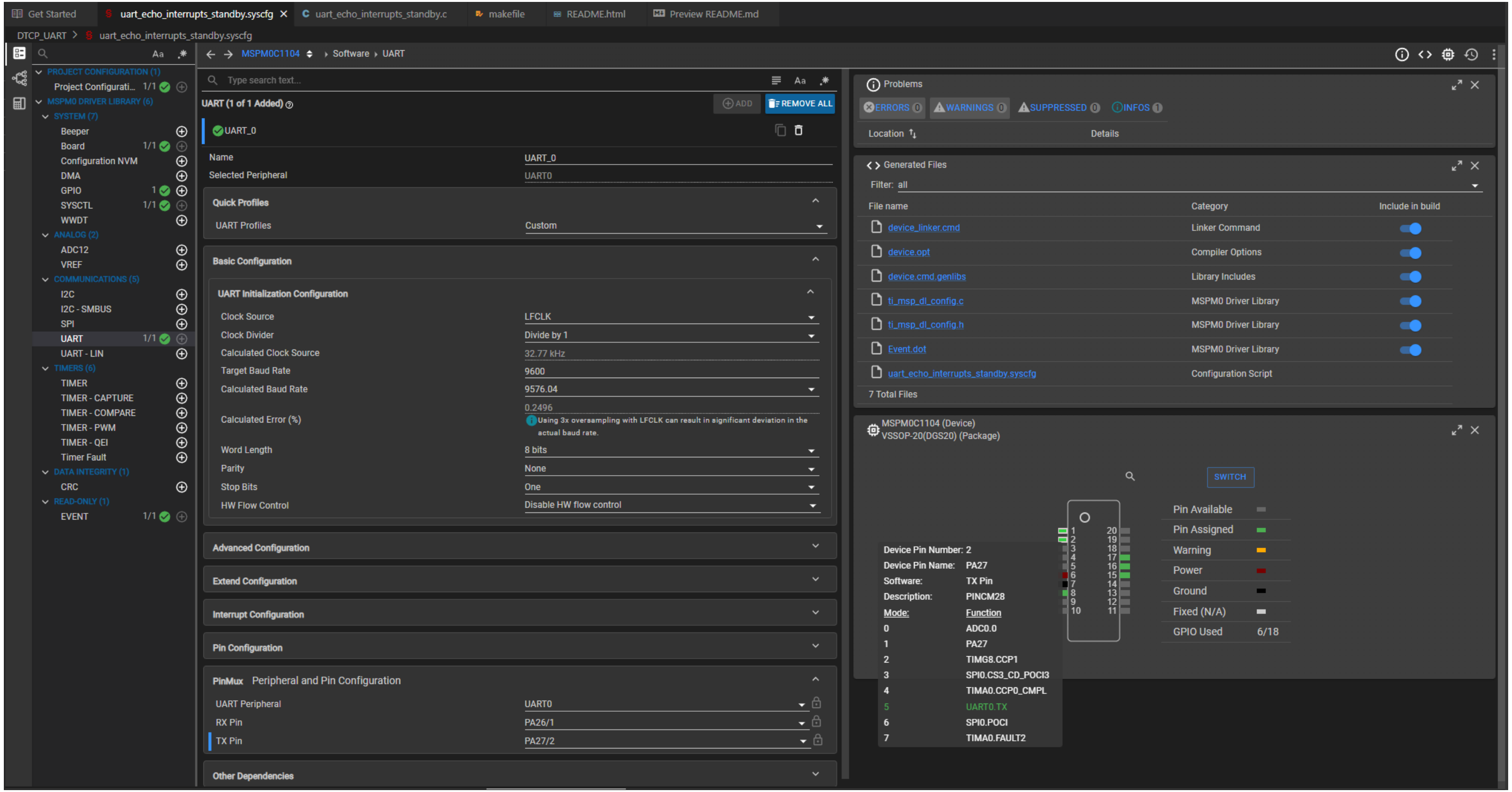
Objective

- Begin writing firmware to facilitate the transfer of data from the MCU to a PC via UART

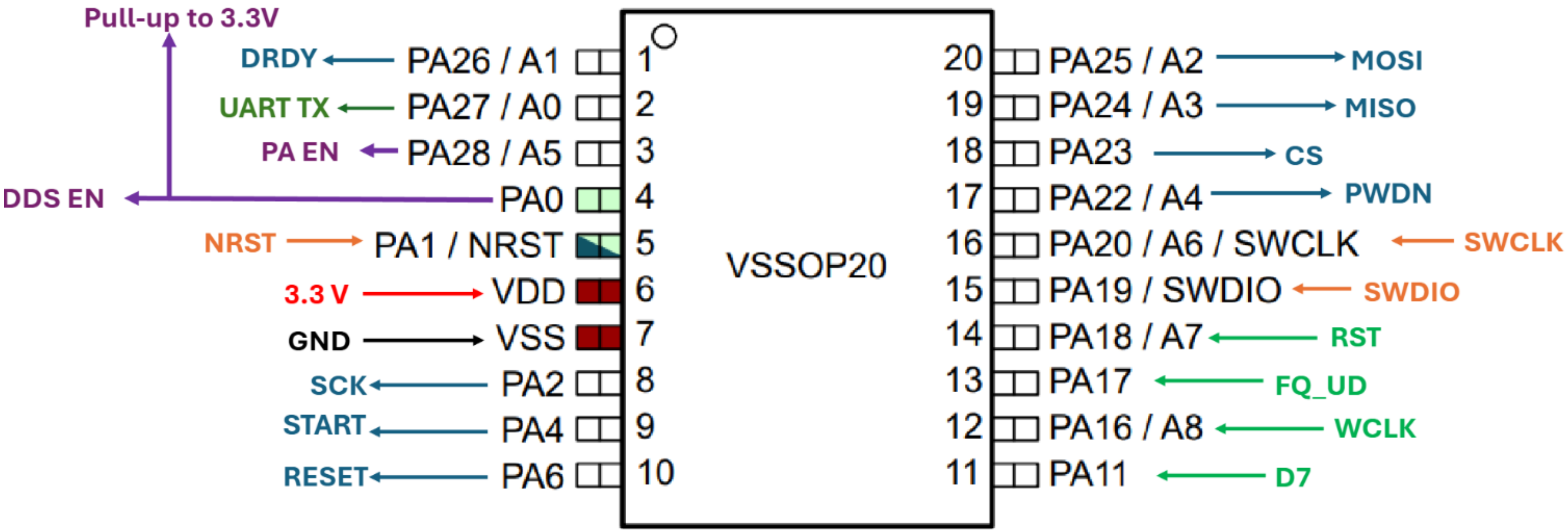
Methods

- CCS and Virtual COM Ports
- TI Launchpad

Results


Get Started
uart_echo_interrupts_standby.syscfg
uart_echo_interrupts_standby.c
makefile
README.html
Preview README.md
DTCP_UART > uart_echo_interrupts_standby.syscfg
MSPM0C1104 › Software › UART
PROJECT CONFIGURATION (1)
Project Configurati... 1/1
MSPM0 DRIVER LIBRARY (6)
SYSTEM (7)
Beeper
Board 1/1
Configuration NVM
DMA
GPIO 1
SYSCTL 1/1
WWDT
ANALOG (2)
ADC12
VREF
COMMUNICATIONS (5)
I2C
I2C - SMBUS
SPI
UART 1/1
UART - LIN
TIMERS (6)
TIMER
TIMER - CAPTURE
TIMER - COMPARE
TIMER - PWM
TIMER - QEI
Timer Fault
DATA INTEGRITY (1)
CRC
READ-ONLY (1)
EVENT 1/1
Type search text...
UART (1 of 1 Added)
ADD
REMOVE ALL
UART_0
Name UART_0
Selected Peripheral UART0
Quick Profiles
UART Profiles Custom
Basic Configuration
UART Initialization Configuration
Clock Source LFCLK
Clock Divider Divide by 1
Calculated Clock Source 32.77 kHz
Target Baud Rate 9600
Calculated Baud Rate 9576.04
Calculated Error (%) 0.2496
Using 3x oversampling with LFCLK can result in significant deviation in the actual baud rate.
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control
Advanced Configuration
Extend Configuration
Interrupt Configuration
Pin Configuration
PinMux Peripheral and Pin Configuration
UART Peripheral UART0
RX Pin PA26/1
TX Pin PA27/2
Other Dependencies
Problems
ERRORS 0
WARNINGS 0
SUPPRESSED 0
INFOS 0
Location
Details
Generated Files
Filter: all
File name
Category
Include in build
device_linker.cmd Linker Command
device.opt Compiler Options
device.cmd.genlibs Library Includes
ti_msp_dl_config.c MSPM0 Driver Library
ti_msp_dl_config.h MSPM0 Driver Library
Event.dot MSPM0 Driver Library
uart_echo_interrupts_standby.syscfg Configuration Script
7 Total Files
MSPM0C1104 (Device)
VSSOP-20(DGS20) (Package)
SWITCH
Device Pin Number: 2
Device Pin Name: PA27
Software: TX Pin
Description: PINCM28
Mode: Function
0 ADC0.0
1 PA27
2 TIMG8.CCP1
3 SPI0.CS3_CD_POCI3
4 TIMA0.CCP0_CMPL
5 UART0.TX
6 SPI0.POCI
7 TIMA0.FAULT2
Pin Available
Pin Assigned
Warning
Power
Ground
Fixed (N/A)
GPIO Used 6/18

Results



New wiring diagram of the MCU

Results

DTCP_UART > uart_echo_interrupts_standby.c > ...

```c
 * PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS;
 * OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
 * WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR
 * OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE,
 * EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 */

#include <stdio.h>
#include "ti_msp_dl_config.h"

// Array of floats
float dataArray[] = {12.34f, 56.78f, 90.12f, -5.67f};
#define ARRAY_SIZE (sizeof(dataArray) / sizeof(dataArray[0]))

void UART_transmitString(char* str) {
    while (*str) {
        // Wait for TX FIFO to have space, then send
        DL_UART_Main_transmitDataBlocking(UART_0_INST, *str++);
    }
}

void transmitFloatArray() {
    char buffer[32];
    for (int i = 0; i < ARRAY_SIZE; i++) {
        float val = dataArray[i];

        // Handle negative numbers
        if (val < 0) {
            UART_transmitString("-");
            val = -val;
        }

        int intPart = (int)val;
        int fracPart = (int)((val - intPart) * 100); // 2 decimal places

        // Format into buffer: "12.34\r\n"
        sprintf(buffer, "%d.%02d\r\n", intPart, fracPart);
        UART_transmitString(buffer);
    }
}

int main(void) {
    SYSCFG_DL_init();

    // Transmit the whole array once
    transmitFloatArray();

    while (1) {
        __WFI();
    }
}
```

Problems 4 | Output | GEL Output | Debug Output | Debug Console | Serial Console - COM3

```
12.34
56.77
90.12
-5.67
```

Observation/Conclusion

- I have confirmed using our configuration that the basic functionality of UART TX should work as intended.

Next Steps

- Expand on the code to make the array of data points larger to mimic how our device actually runs.
- Real-Time plotting of data using PUTTY, Take a Video.
- Add function to access specific data points within array.
- Integrate code into ADS firmware to test transferring data supplied by ADS to the PC.
- Read Papers and look for bias and what their signals looks like.
- Pic of their animal setup.

UART Firmware

Luis Wong 03/04/2026

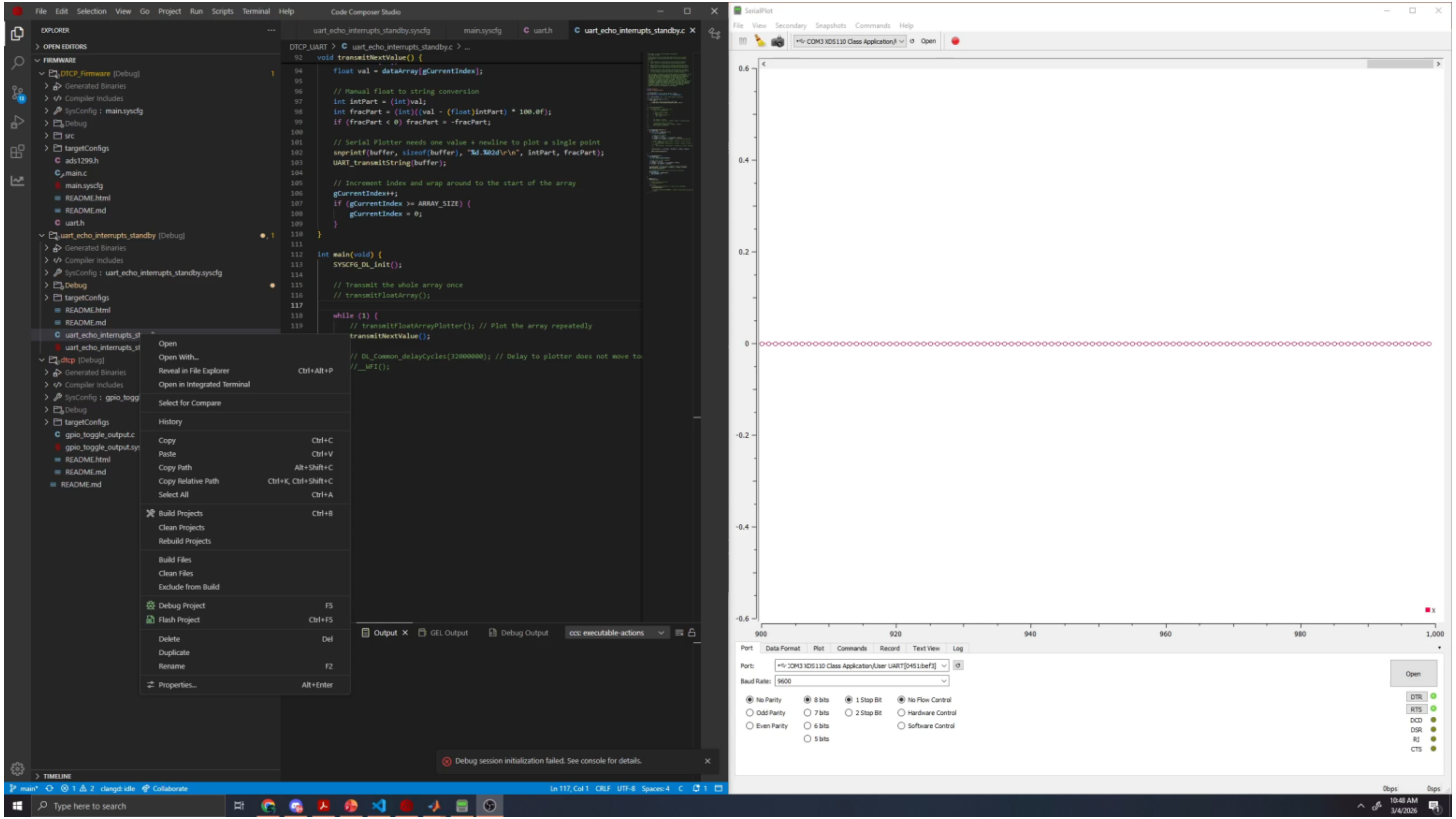
Objective

- Continue writing firmware to facilitate the transfer of data from the MCU to a PC via UART
- Look for info regarding ADS1299 used in EMG.

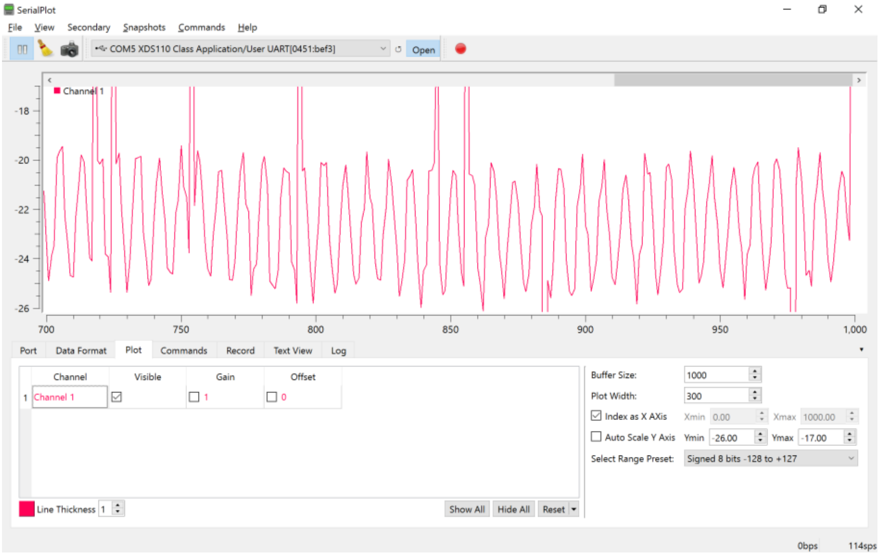
Methods

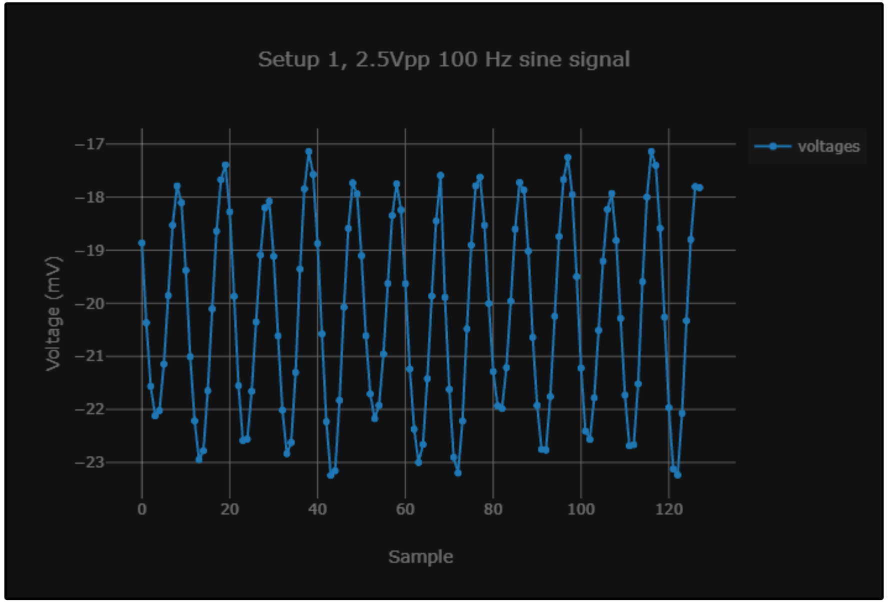
- CCS and Virtual COM Ports
- TI Launchpad
- Serial Plotter
- MATLAB

Results (Video of UART Plotting Working)



Results (UART Plotting vs CCS Debug Plot)





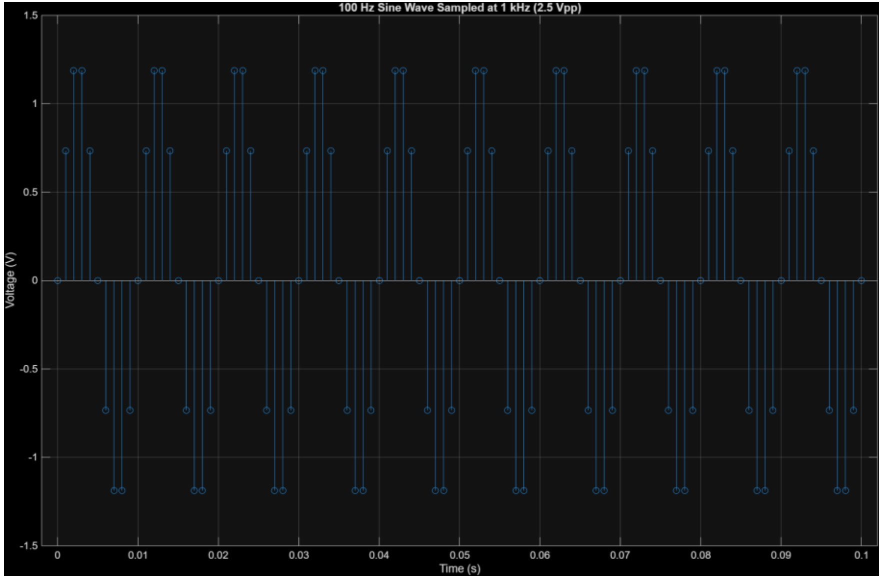
Try 15200 baud rate
- Video of setup with saline

Results

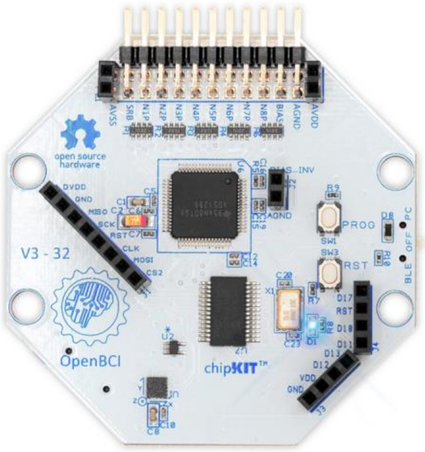
```matlab
% Parameters
f = 100;            % Signal Frequency (Hz)
Fs = 1000;          % Sampling Frequency (Hz)
Vpp = 2.5;          % Peak-to-peak voltage
duration = 0.05;    % Duration (seconds) - shows 5 cycles

% Time vector
t = 0:1/Fs:T;

% Amplitude calculation (Vp = Vpp / 2)
Amp = Vpp / 2;

% Generate sine wave
x = Amp * sin(2*pi*f*t);
```

```matlab
% Format the array as a comma-separated list
fprintf('%.4f, ', x(1:end)); % Print all but the last element with a comma
```

```matlab
% Plot
stem(t, x);
title(['100 Hz Sine Wave Sampled at 1 kHz (2.5 Vpp)']);
xlabel('Time (s)');
ylabel('Voltage (V)');
grid on;
```



Results (Cyton Board)



Cyton Board Overview

The OpenBCI Cyton Board is an Arduino-compatible, 8-channel bio-amplifier with a 32-bit processor. At its core, the OpenBCI Cyton Board implements the PIC32MX250F128B microcontroller, giving it lots of local memory and fast processing speeds. The board comes pre-flashed with the chipKIT™ bootloader, and the latest OpenBCI firmware. Data is sampled at 250Hz on each of the eight channels.

The OpenBCI Cyton Board can be used to sample brain activity (EEG), muscle activity (EMG), and heart activity (ECG), and much more. The board communicates wirelessly to a computer via the the OpenBCI USB dongle using RFDuino radio modules.

If you are only interested in using the board for research and don't care about the hardware, don't fret! The board and dongle come pre-loaded with the latest firmware, making OpenBCI accessible to researchers and developers with little to no hardware experience. OpenBCI boards have a growing list of data output formats, making them compatible with an expanding collection of existing biofeedback applications and tools.

Cyton Board Technical Specifications

OpenBCI Cyton 8-channel Board:

- 8 differential, high gain, low noise input channels
- Compatible with active and passive electrodes
- Texas Instruments ADS1299 ADC (datasheet)
- PIC32MX250F128B microcontroller w/chipKIT™ bootloader
- RFduino™ Low Power Bluetooth™ radio (datasheet)
- 24-bit channel data resolution
- Programmable gain: 1, 2, 4, 6, 8, 12, 24
- 3.3V digital operating voltage
- +/-2.5V analog operating voltage
- ~3.3-12V input voltage
- LIS3DH accelerometer (datasheet)
- Micro SD card slot
- 5 GPIO pins, 3 of which can be Analog

Results (OpenBCI Citations)

OpenBCI Documentation Main Site Shop Forum Documentation Github Citations Search Ctrl K

OpenBCI Citation List

Here is a list of research papers and articles that have used OpenBCI hardware and software for their research.

Are you using OpenBCI for research?

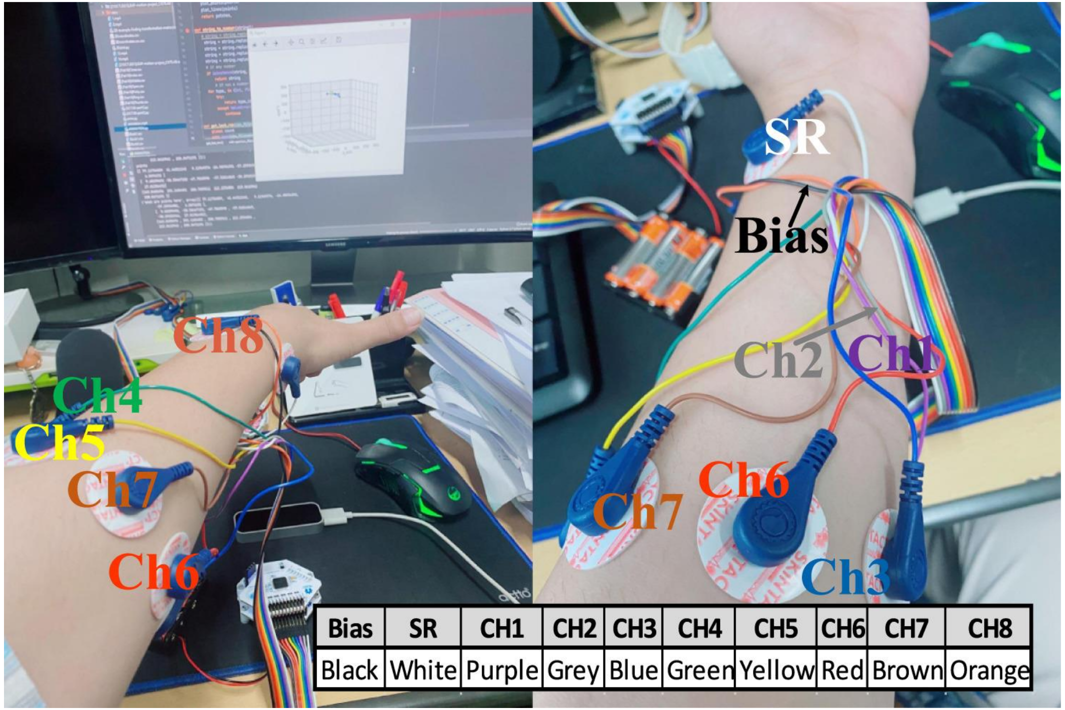
OpenBCI Research Collection (see openbci.com/citations)

g	OpenBCI Product Used	Electrode / Oth	Author(s)	Title	Website	Journal/Conference	DO
2026	Cyton+Daisy Board		M. Tairab, Alaeldin; Liu, Weiqun; Zhuo, Junchao;	A centrifugal mechanism hybrid energy harvester with AI-enabled self-sensing and self-powere	https://iopscience.iop.org	Smart Materials and Structures	10.
2026	Cyton+Daisy Board		Saglam, Abdullah Yigit; Tezel, Tugce Gulseren; K	Video game-driven EEG signal classification for rehabilitation applications using LSTM network	https://www.spiedigitallib	Eighteenth International Conference on Machine Vision (ICM	10.
2026	Cyton+Daisy Board		Ghadah Alosaimi, Maha Alsayyari, Yixin Sun, Sta	EEG-Driven Intention Decoding: Offline Deep Learning Benchmarking on a Robotic Rover	https://arxiv.org/abs/2602		10.
2026	Cyton Board	Ultracortex Ma	N, Keerthika.; V, Kiruthika.	Comparison of preprocessing techniques for effective cognitive analysis using electroencephal	https://linkinghub.elsevie	MethodsX	10.
2026	Cyton+Daisy Board		İbili, Aysel Burcu; Fidan, Ugur; Ibılı, Emin; Çavuş	Çocuklarda Migren Tespiti İçin Işıklı Uyarım Frekansı ve EEG Kanallarının Belirlenmesi: Makin	https://dergipark.org.tr/tr/	Afyon Kocatepe University Journal of Sciences and Enginee	10.
2026	Cyton+Daisy Board		Xiong, Qinqin; Gui, Longjin; Shu, Chuan	Support vector machine algorithm-based wearable device in sports rehabilitation training for pe	https://www.nature.com/	Scientific Reports	10.
2026			Abdollahzadeh, Hamed; Montaldi, Elisa; Olivieri,	Peak-Independent Cuffless Blood Pressure Monitoring Using a Smart Sock: The Role of Temp	https://www.mdpi.com/14	Sensors	10.
2026	Cyton Board		D'Amico, Alessandro; de Sa, Virginia R.	Comparing Simultaneous Scalp EEG Recordings from the OpenBCI Cyton and Brain Products	https://www.mdpi.com/14	Sensors	10.
2026	Cyton Board		Ramu, Vadivelan; Lakshminarayanan, Kishor	IoT-Enabled Framework for OpenViBE-Based Online BCI with an OSC-Driven Somatosensory	https://onlinelibrary.wiley	Artificial Intelligence in Instrumentation, Control and Automat	10.
2026	Cyton Board		Bahramsari, Parsa; Behzadipour, Saeed	Classification of the lower limb motor imagery and execution using high vs. low channel EEG d	http://biorxiv.org/lookup/		10.
2026	Cyton Board		Gornale, Shivanand S.; Palaiahnakote, Shivakun	A Novel EEG-ECG Based Fusion Model for Multimodal Emotional Classification	https://www.worldscientif	International Journal of Pattern Recognition and Artificial Inte	10.
2026	Cyton Board		Palmisciano, Andrea Costanzo; Farabbi, Andrea;	Form factor meets function: Anatomy-dependent electrode-skin coupling and signal content in	https://iopscience.iop.org	Journal of Neural Engineering	10.
2026	Cyton Board	Ultracortex Ma	Muhammad, Riaz; Nettey-Oppong, Ezekiel Edwa	Neural Efficiency and Attentional Instability in Gaming Disorder: A Task-Based Occipital EEG a	https://www.mdpi.com/23	Bioengineering	10.
2026	Cyton+Daisy Board		Alawee, Wissam H.; Al-Haddad, Luttfi A.; Basem,	Enhancing a task recognition model for real-time control of artificial limbs using data-driven boo	https://link.springer.com/	Multimedia Tools and Applications	10.
2026	Cyton Board		Fathur Rahman, Imam; Melinda, Melinda; Yunida	MEWT-Enhanced EEGNet for ASD EEG Classification: Performance Evaluation with k-Fold Cr	https://ijeeemi.org/index.	Indonesian Journal of Electronics, Electromedical Engineerir	10.

https://docs.openbci.com/citations/

Results (Example Setup found in Literature)

EMG-based 3D hand gesture prediction using transformer–encoder classification



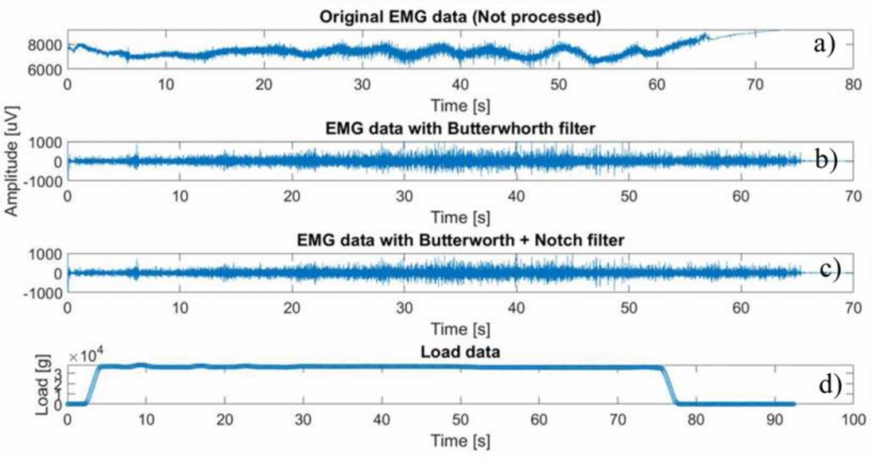
https://doi.org/10.1016/j.icte.2023.04.005

- “The gel electrodes for recording sEMG channel information are attached around the arm and the SR and Bias are attached near the wrist.”

Results (Example Signal + Setup)

A comprehensive system for the acquisition of EMG signals and muscle force in lower limb

DOI: 10.1109/ICECCME55909.2022.9988114



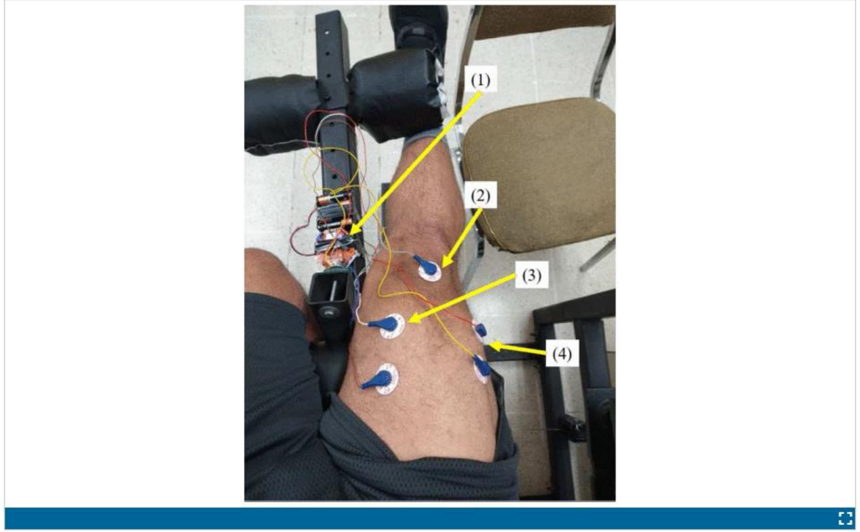


Fig. 3.
Use of test equipment: (1) Cyton board; (2) reference electrode; (3) EMG surface electrodes in rectus femoris; (4) EMG surface electrodes on vastus medialis

Observation/Conclusion

- I have confirmed using our configuration UART can plot in real-time what is being recorded by the ADS. There may be some issues present as some spikes in voltage were present in the UART measurement not found in the debug graphs.

Next Steps

- Add function to access specific data points within array. *
  - We can export data in serial plot as CSV so maybe this isn’t needed?
- Import generated array sampled at 1kHz to CCS to test if the spikes in UART still occur.
  - Could be due to improper float to string conversion, may want to test sending as raw binary instead.
- Retest with Dmytro ADS + UART plotting.
- Show Jerimiah schematic for AFE and compare. (Discuss need for additional power LED).
- After PCB design is sent to manufacturer.
  - Read Papers again and look for bias and what their signals looks like.
  - Look for pics of their animal setup.
  - Data processing for AFE in code. Manually generate signal we want to detect and test. Artifact (LARGE) → Signal